

Important Information

Thank you for choosing Freenove products!

Getting Started

If you have not yet downloaded the zip file, associated with this kit, please do so now and unzip it.

https://github.com/Freenove/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi

Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

support@freenove.com

Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. **Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.**
- When the product is turned ON, activated or tested, some parts will move or rotate. **To avoid injuries to hands and fingers keep them away from any moving parts!**
- It is possible that an improperly connected or shorted circuit may cause overheating. **Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

<http://www.freenove.com>

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Preface

Welcome to use the Freenove Computer Case Kit Pro for Raspberry Pi



This product is exclusively designed for the RPi 5, a popular single-board computer (SBC). Follow this tutorial, you can install a sleek, multi-functional computer case kit for your Raspberry Pi 5 (RPi 5).

The RPi 5 delivers highly powerful performance with a CPU frequency of up to 2.4GHz. However, this increased power consumption also leads to significant heat generation. Without an appropriate heatsink, thermal throttling may severely limit the Raspberry Pi 5's performance. Additionally, the removal of the 3.5mm audio jack in the RPi 5 has caused inconvenience for many users.

In this context, our case kit came into being. It not only addresses these issues, but also introduces advanced features.

If you encounter any issues, feel free to contact us for prompt and free technical support.

support@freenove.com

Key Features

This product integrates a wide range of features and supports multiple optional expansions, as detailed below:

- **Standard Features:**
 - Built-in dual 4Ω 3W speakers for audio output
 - Equipped with a 3.5mm JACK audio interface
 - Applying a tower cooler for more efficient heat dissipation
 - Built-in five ARGB 4010 cooling fans for lower overall case temperature and enhanced lighting effects
 - Classic case design
 - Integrated 2x6 ARGB lights, supporting multiple lighting modes and customizable effects
 - Power button with indicator light
 - Low standby power consumption design; all lights and power can be turned off after shutdown
 - Integrated a 0.96-inch OLED display for real-time system information monitoring
 - Built-in case temperature sensor for real-time internal temperature monitoring
 - Featuring a 1220 RTC battery holder
 - Ports are uniformly arranged on the back of the case, including USB, Ethernet, UART, HDMI, 3.5mm JACK, SD card slot, and power interface for easy connection and management
- **Optional Features:**
 - Optional 4.3-inch IPS display with true-to-life colors and responsive touch control, suitable for real-time monitoring and displaying comprehensive system operation information
 - Supports M2Me M.2 slot expansion in 2230/2242/2260/2280 specifications; optional configurations of 1/2/4 slots to meet different storage needs
 - Optional 128GB SSD for basic storage requirements

Product Variants

This product is currently offered in multiple variants. The primary distinction lies in the display functionality, divided into two series: **one without a display** and **one equipped with a 4.3-inch IPS display**.

Beyond this, further model differentiation is available based on the presence and number of SSD slots.

The table below details the specific configurations of the **non-display series**.

non-display series	NVMe M.2 SSD x1	Number of NVMe Slot
FNK0107A	✗	1
FNK0107B	✗	2
FNK0107C	✗	4
FNK0107H	✓	1
FNK0107K	✓	2
FNK0107L	✓	4

The table below shows the detailed configurations of the variants **equipped with the 4.3-inch IPS display**.

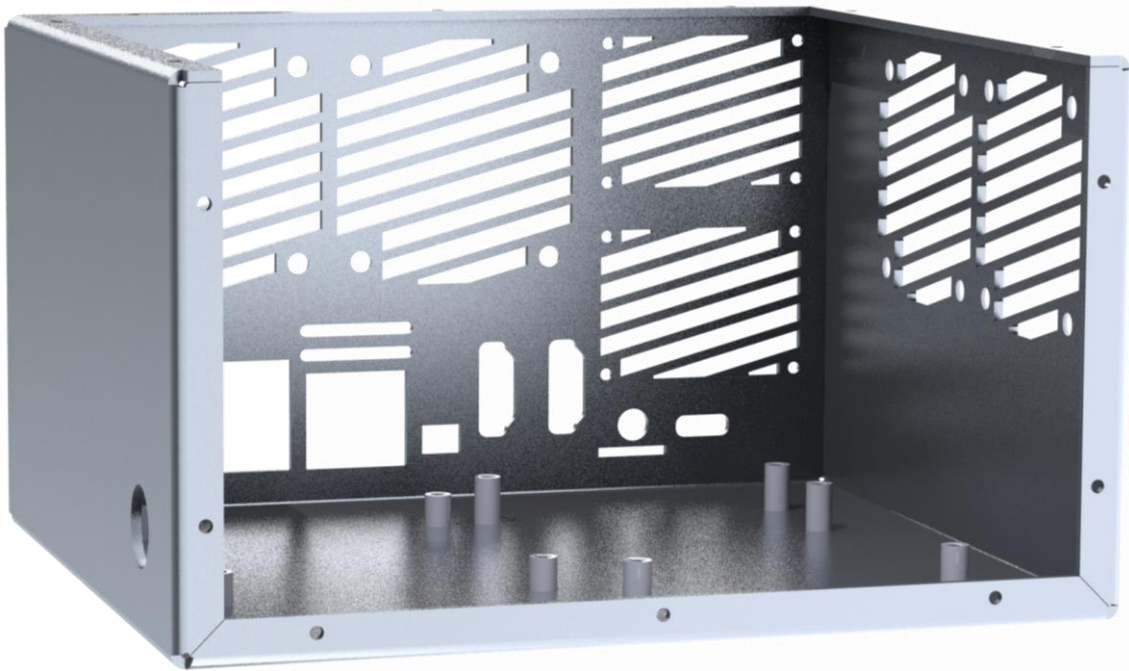
with 4.3inch IPS display	NVMe M.2 SSD x1	Number of NVMe Slot
FNK0107P	✗	1
FNK0107Q	✗	2
FNK0107R	✗	4
FNK0107U	✓	1
FNK0107V	✓	2
FNK0107W	✓	4

List

Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

Metal Parts

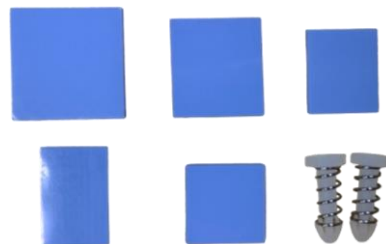
Metal case x1



Tower Cooler x1

Thermal Pad x5

Nylon Standoff x2



Acrylic Parts

Note: Please tear off the protective films from the acrylic parts before use.

Acrylic Top Plate x 1

Acrylic Top Plate × 1

Landscape Orientation Screen Bracket × 2

Portrait Orientation Screen Bracket × 2

(Only for FNK0107P\Q\R\U\V\W)

Acrylic Side Plate x 1

Acrylic Side Plate(Only for FNK0107P\Q\R\U\V\W)

LED Light Diffusion Plate x 1

Acrylic for OLED x1

Speaker Acrylic Pad x2

Machinery Parts

All fasteners come in a larger bag, please open it and check whether they are complete.

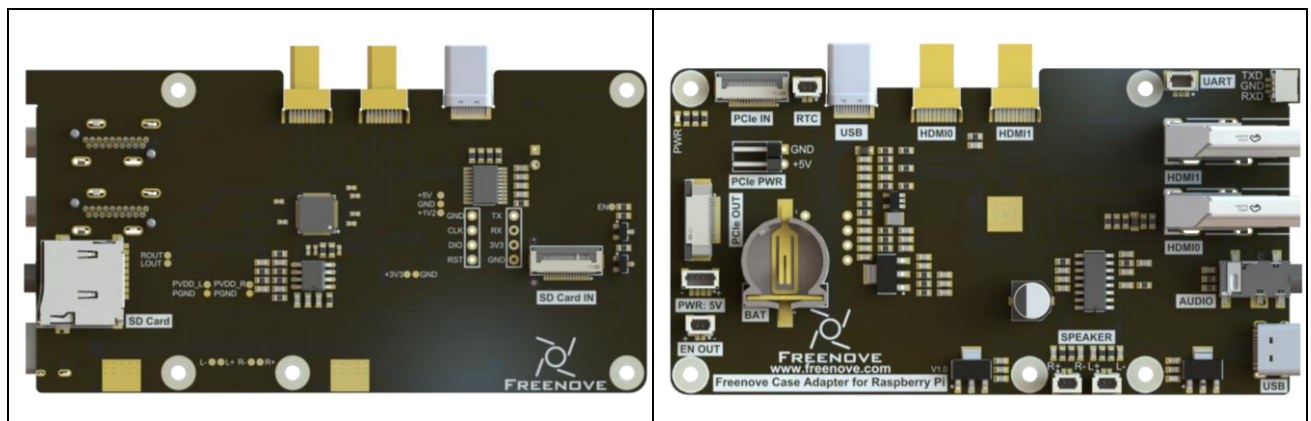
M2.5*5 Countersunk Head Screw  x50 Freenove	M3.7*10 Countersunk Head Screw  x18 Freenove	M2*6 Countersunk Head Screw  x10 Freenove
M2.5*12 Brass Standoff  x2 Freenove	M2.5*12+4 Brass Standoff  x4 Freenove	M2.5*25+4 Brass Standoff  x3 Freenove

The fasteners listed below are not packaged in the same bag as those above. They are only provided with the **FNK0107P/Q/R/U/V/W** versions.

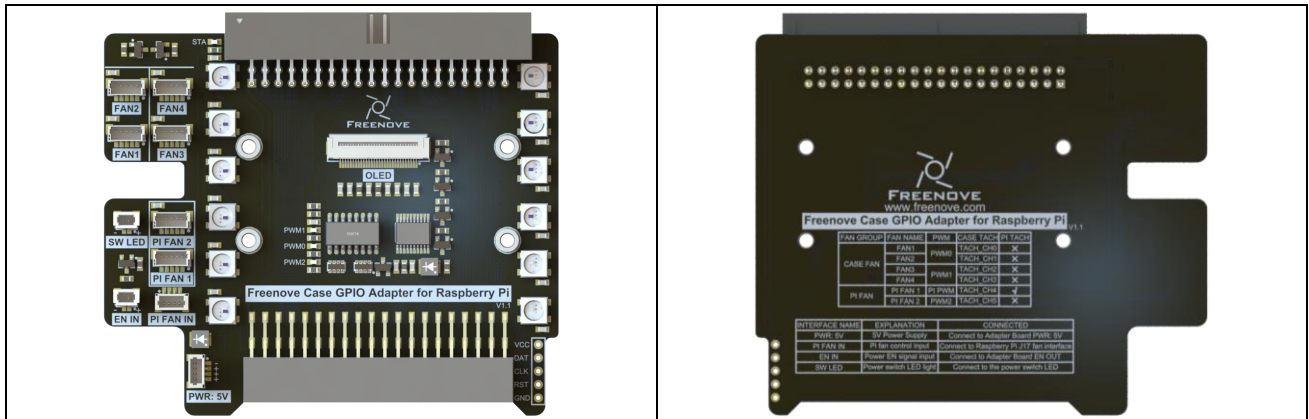
M2.5*9 Brass Standoff  x5 Freenove

Electronic Parts

Freenove Case Adapter for Raspberry Pi



Freenove Case GPIO Adapter for Raspberry Pi



Freenove Power Button Board for Raspberry Pi



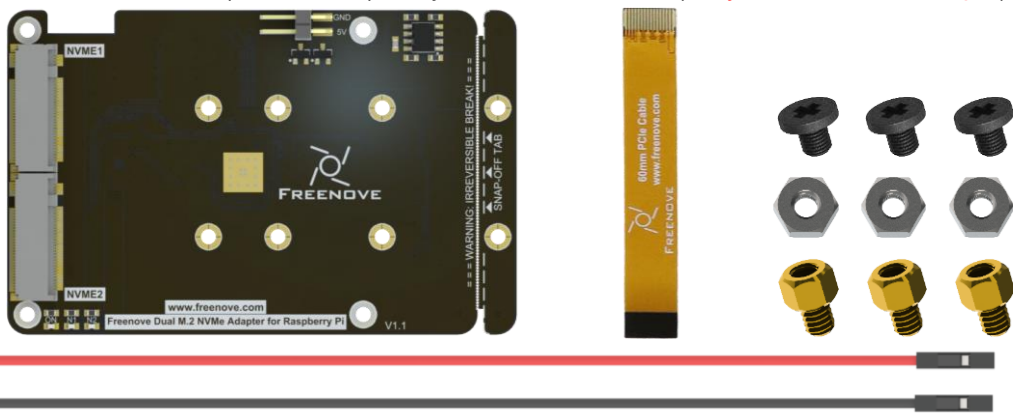
Freenove M.2 NVMe Adapter Series for Raspberry Pi

Note: The components included in the NVMe Adapter Combo Pack vary by product version. Please verify that the contents match your model before installation.

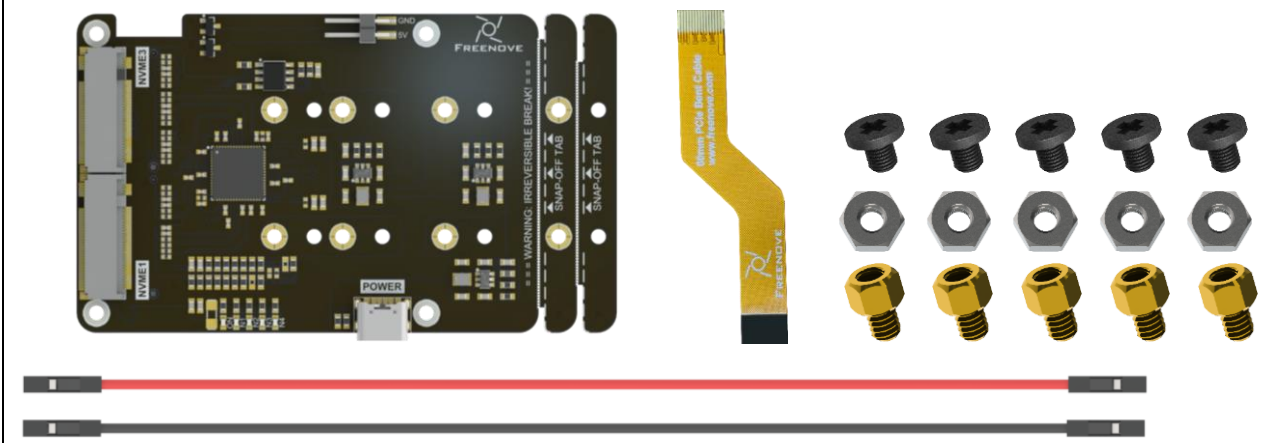
Freenove M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0107A/H/P/U)



Freenove Dual M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0107B/K/Q/V)

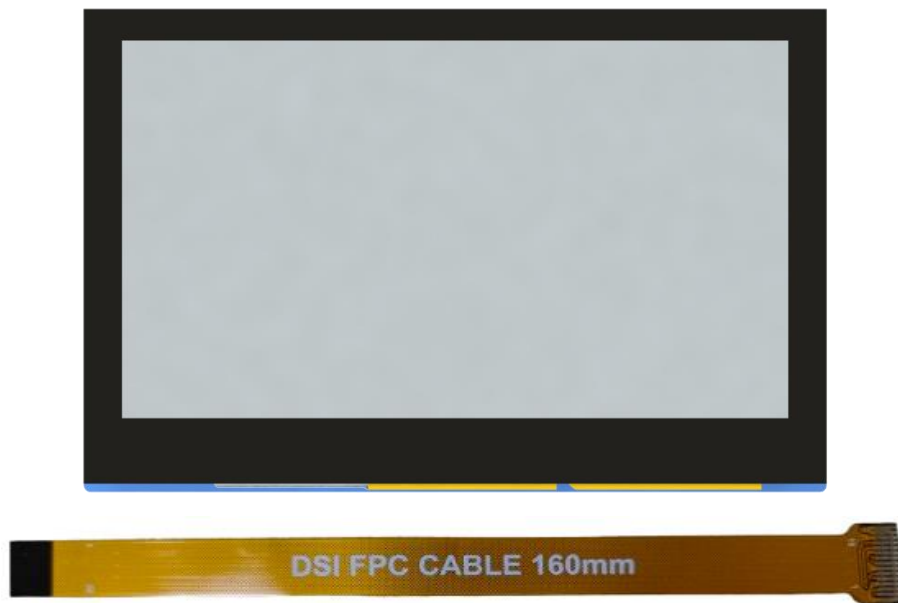


Freenove Quad M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0107C/L/R/W)



Electronic Modules

4.3-inch Screen x1, DSI FPC CABLE 160mm x1 (Only for FNK0107P/Q/R/U/V/W)



0.96 inch OLED display x1



Important: The display glass is fragile – please exercise extreme caution during installation.

ARGB Fan x4

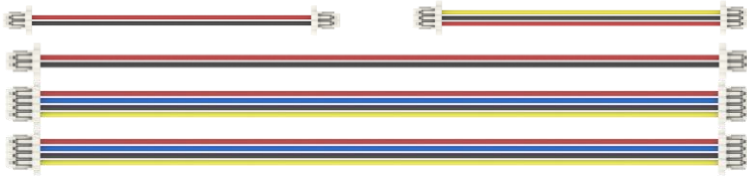


Speaker x2



Wires

SH1.0mm_2P Same-Direction Cable 5cm x1
 SH1.0mm_3P Same-Direction Cable 5cm x1
 SH1.0mm_2P Same-Direction Cable 15cm x1
 SH1.0mm_4P Same-Direction Cable 15cm x2



SH1.0mm 2-Pin to 2.8mm Quick-Disconnect Terminal Cable (Red-Black), 7cm, × 1
 1.25mm 2-Pin to 2.8mm Quick-Disconnect Terminal Cable (Yellow-Yellow), 7cm, × 1



SD Card to 0.5mm-16P FPC cable



reverse straight cable x1



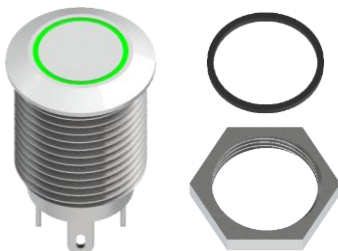
Tools

Screwdriver Bit Holder × 1
 Hex Shank Phillips #2 Bit × 1
 Hex Shank Phillips #0 Bit × 1

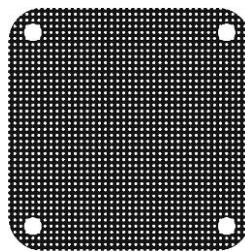


Others

12mm LED Power Button x1
 Black Sealing Gasket x1
 M12 Nut x1



Fan Dust Filter x4



Round Black Non-Slip Foot Pad x5

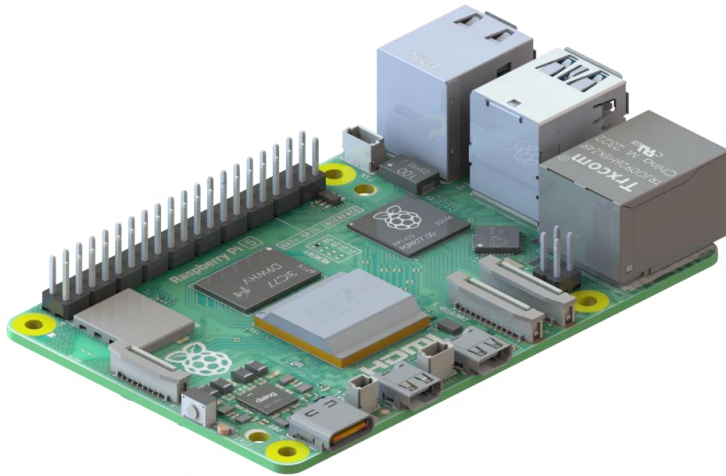


Masking Tape x 1



Required but NOT Contained Parts

Raspberry Pi 5 x 1



27W Power Adapter x 1 (or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A)



Micro SD Card (TF Card), Card Reader x 1



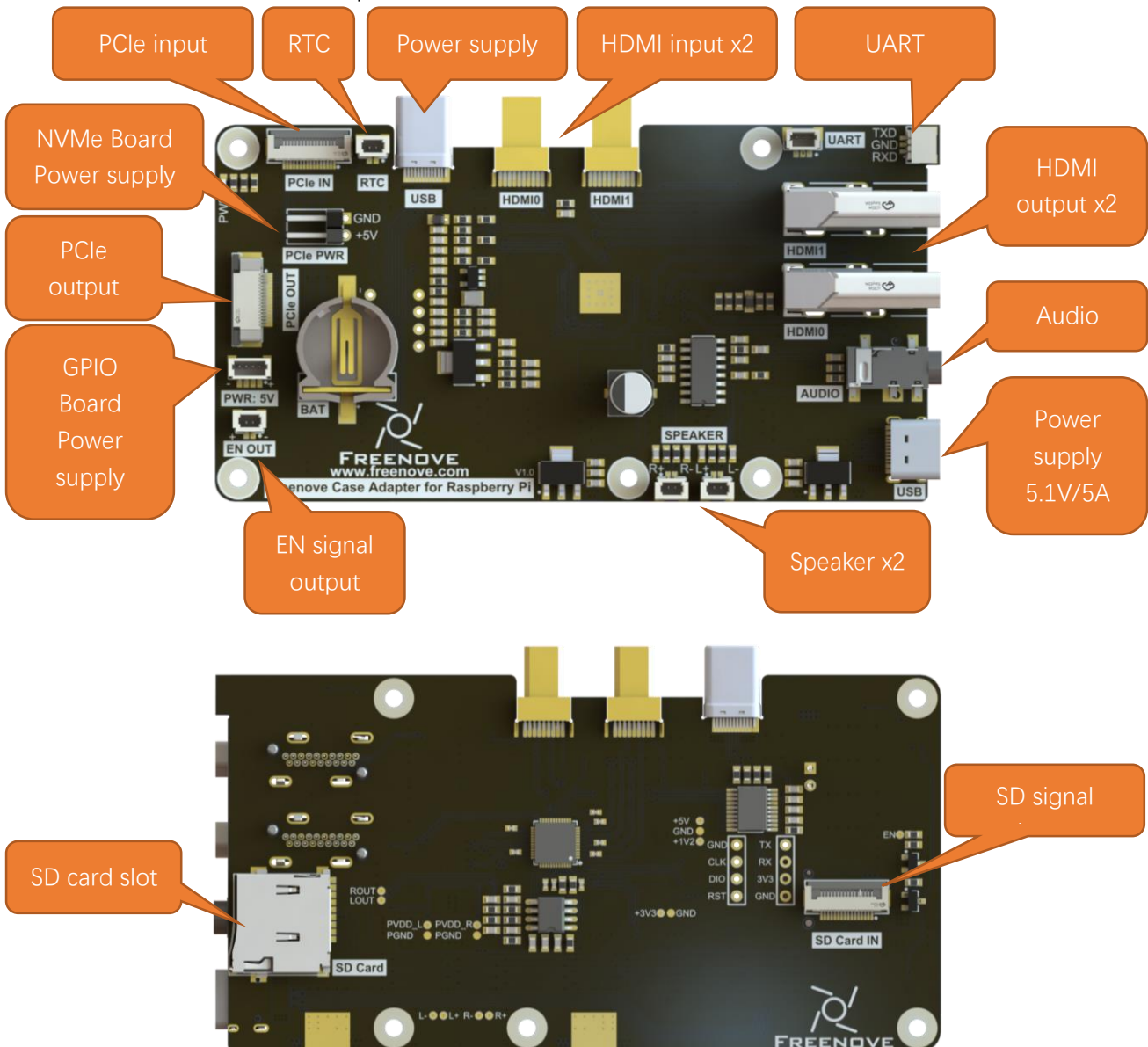
Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

Chapter 1 Introduction to Main Components

In this chapter, we will mainly introduce the main components of this case and their functionalities.

1.1 Introduction to Freenove Case Adapter for Raspberry Pi

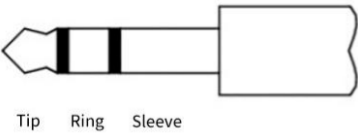
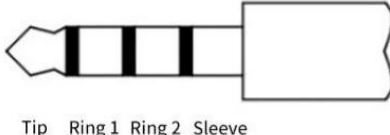
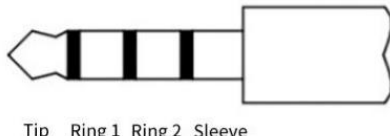
In this tutorial, we name this component as **Audio-Video Board**.



The Audio-Video Board is specifically designed for the Raspberry Pi 5, aiming to transform it into a fully functional home computer. This board integrates a wide range of interfaces, significantly expanding its native capabilities. Key features includes a PCIe adapter port, an RTC interface, a Type-C power supply interface, dual HDMI adapter ports, a UART adapter port, a 3.5mm JACK audio interface, dual speaker connectors, an EN (Enable) signal interface, a power interface for the GPIO Board, a power interface for the NVMe adapter board, and an SD card slot.

3.5mm JACK Audio Interface

In the RPi 5 design, the traditional 3.5mm audio output interface has been removed, and audio signals are now solely output through HDMI. To address this, we have developed an audio separation circuit on the Case Adapter Board. This circuit enables audio to be output via the 3.5mm audio interface on the board. You can use the external 3.5mm TRS headphone jack on the Raspberry Pi 5 to listen to audio output. For proper functionality, ensure your headphone connector is compatible. Below is an explanation of common jack types:

TRS JACK  Tip Ring Sleeve		Tip	Left Channel
		Ring	Right Channel
		Sleeve	GND
OMTP	TRRS JACK  Tip Ring 1 Ring 2 Sleeve	Tip	Left Channel
		Ring1	Right Channel
		Ring2	MIC
		Sleeve	GND
CTIA	TRRS JACK  Tip Ring 1 Ring 2 Sleeve	Tip	Left Channel
		Ring1	Right Channel
		Ring2	GND
		Sleeve	MIC

Speaker Interface

Additionally, we have designed an audio amplification circuit. With this circuit, you can control the two speakers, each with an impedance of 4Ω and a power rating of 3W.

PCIe Interface

This interface adapts the PCIe interface from the RPi 5 to the Audio-Video Board, enabling convenient connection to 1/2/4-slot NVMe adapter boards. To ensure stability during high-power SSD operation or when multiple SSDs are running simultaneously, an additional 2-pin independent power supply interface has been integrated onto the board. This design effectively prevents potential device damage caused by power insufficiency and guarantees the stable operation of your NVMe storage system.

EN OUT

This signal is used to control the power to both the Audio-Video Board and the GPIO Board. When the Raspberry Pi 5 is shut down, it automatically cuts off power and lighting, achieving high energy efficiency and complete silence.

SD Card Slot

This interface adapts the Raspberry Pi 5's SD card slot to the case interface area, achieving a unified layout where all ports are located on the same side. The design strictly complies with the SDR104 transmission

protocol, ensuring full compatibility with official Raspberry Pi SD cards and delivering bus speeds of up to 104 MB/s.

<https://www.raspberrypi.com/documentation/accessories/sd-cards.html#content>

<https://www.sdcard.org/developers/sd-standard-overview/bus-speed-default-speed-high-speed-uhs-sd-express/>

RTC Interface

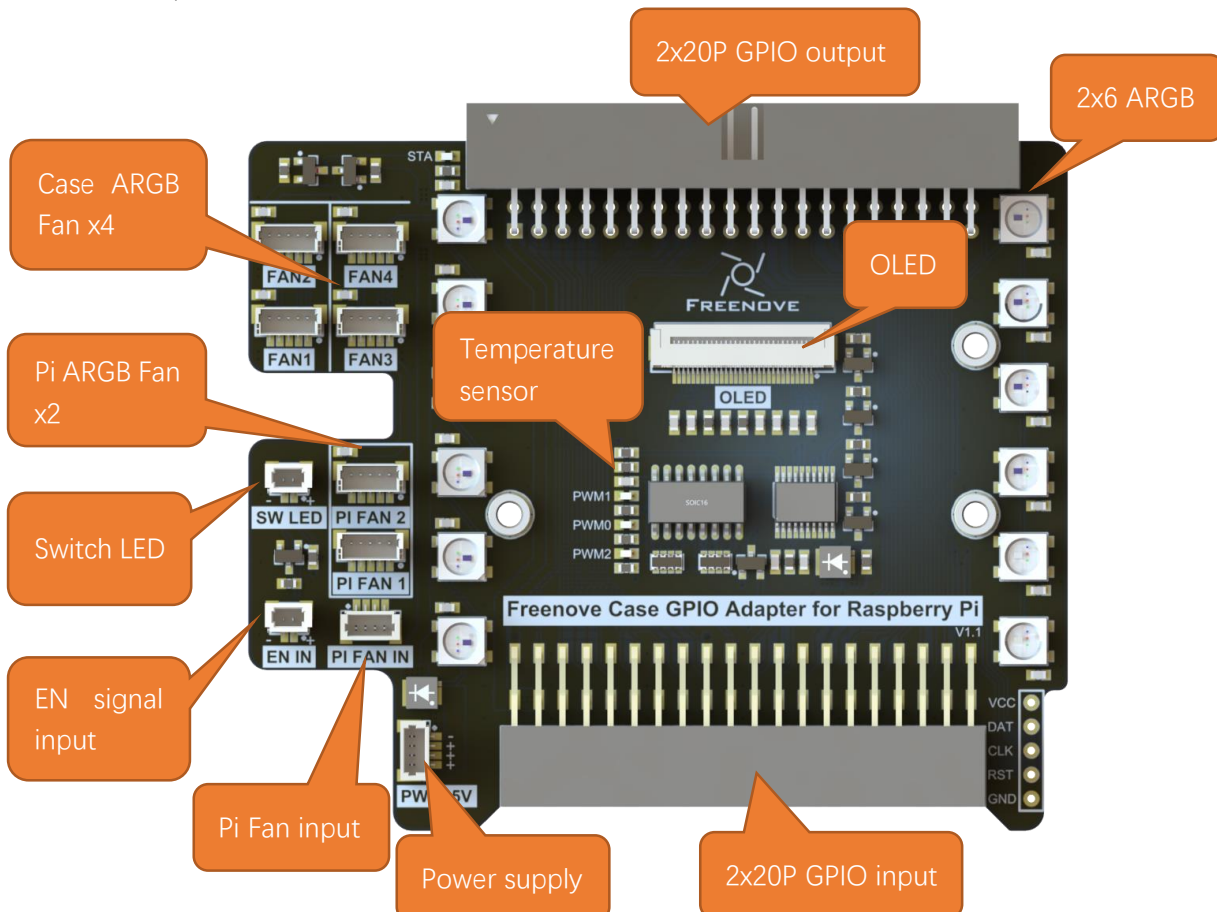
To enhance the accuracy of the Raspberry Pi's clock, it is advisable to add a battery to the RTC. For this purpose, we have installed a battery holder on the board. If you decide to add a battery, we recommend either purchasing a rechargeable lithium-manganese button-cell battery (with dimensions of 1220 and a voltage of 3V) or an official battery.

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#real-time-clock-rtc>

Due to the addition of audio processing capabilities, multiple lighting effects, an enhanced cooling system, and multiple NVMe interfaces (supporting additional SSDs), the overall power consumption during operation has increased. Therefore, only the official 5.1V/5A power adapter (or compatible models with equivalent specifications) should be used to ensure stable system performance.

1.2 Introduction to Freenove Case GPIO Adapter for Raspberry Pi

In this tutorial, we name this model as **GPIO Board**.



The GPIO Board extends all GPIO pins of the Raspberry Pi 5 to the exterior of the case, significantly simplifying the connection and debugging of external devices. Additionally, the board is highly integrated with multiple control functions, including interfaces for 2×6 ARGB LEDs and 6 ARGB fans, supporting customizable lighting modes. It also features onboard connectors for an OLED display and an EN (Enable) power control interface, further enhancing system expandability and power management capabilities.

ARGB

The board features 2×6 onboard ARGB LEDs, which operate synchronously in two channels. Via I²C communication, users can switch between multiple preset lighting modes and individually control each LED, enabling highly customizable dynamic lighting effects.

Case ARGB Fans

The case is equipped with four ARGB fan headers, whose lighting effects are synchronized with the onboard ARGB LEDs. The system supports two-channel independent PWM control, allowing users to precisely adjust fan speeds via I²C communication and monitor real-time RPM data for each fan. This enables efficient thermal management while maintaining unified visual aesthetics.

Pi ARGB Fan

Pi FAN 1: It is controlled directly by Raspberry Pi 5. For details, see:

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#fan-cases>

Pi FAN 2: It is controlled by GPIO board. The way to control it is the same as the aforementioned Case ARGB Fans.

EN IN (Enable Input)

This interface connects to the EN_OUT port on the Audio-Video Board to synchronize the power state of the entire system. When the Raspberry Pi is shut down, it automatically cuts off all power to the GPIO Board, effectively eliminating standby power consumption and ensuring that all connected devices (such as lighting) are completely turned off.

Temperature sensor

Equipped with a temperature sensor, the system monitors the internal temperature in real-time and linearly adjusts the fan speed based on the temperature data, achieving automatic closed-loop control of the cooling system. At the same time, users can view precise temperature readings at any time to clearly monitor the internal operating status of the case.

1.3 Introduction to Freenove Power Button Board for Raspberry Pi

In this tutorial, we name this model as **Switch Board**.



The Switch Board connects to the J2 interface of the Raspberry Pi 5 via pogo pins, enabling the use of a custom power button. A 12mm power button with an indicator light is included, offering identical functionality to the onboard power button of the Raspberry Pi 5. For detailed specifications and usage of the power button, please refer to the Raspberry Pi official documentation:

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#power-button>

1.4 Introduction to Freenove M.2 NVMe Adapter Series for Raspberry Pi

The Raspberry Pi 5 includes a PCIe x1 slot that is certified for PCIe Gen 2.0, providing a theoretical maximum throughput of 5GT/sec, which roughly translates to 500MB/sec for read and write operations. Although this slot is not officially certified for PCIe Gen 3.0, it is possible to force the use of Gen 3.0 for potentially higher speeds.

The PCIe consortium states that the speed of PCIe Gen 3.0 x1 is up to 8GT/sec, which translates to approximately 985MB/sec.

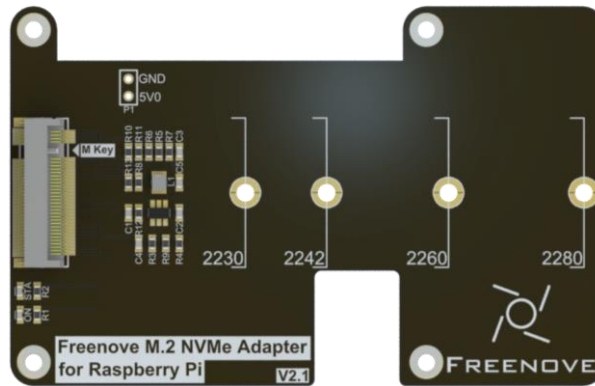
https://en.wikipedia.org/wiki/PCI_Express#Comparison_table

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#pcie-gen-3-0>

SSDs generally provide significantly faster read and write speeds compared to SD cards and USB drives, which can notably elevate the user experience when operating the Raspberry Pi 5.

1.4.1 Freenove M.2 NVMe Adapter for Raspberry Pi

In this tutorial, we name this model as **NVMe Adapter Board**.

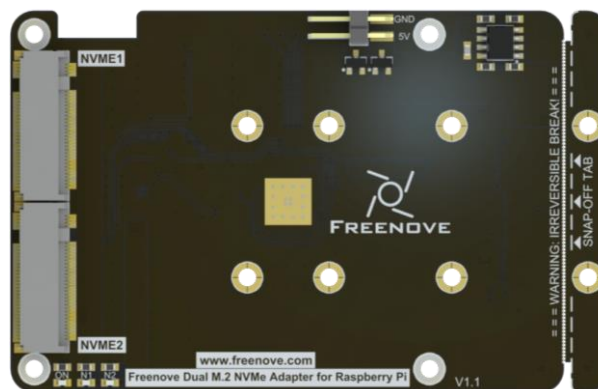


Here are its key features:

- **Interface Type:** M.2 with M-Key
- **Supported Protocol:** NVMe
- **PCIe Channel:** PCIe 3.0 x1(Compatible with PCIe 2.0)
- **Compatible Sizes:** 2230, 2242, 2260, 2280
- **Power Supply:** 3.3V, up to 3A (maximum)
- **Indicator Lights:** Includes both power and SSD status LEDs.

1.4.2 Freenove Dual M.2 Nvme Adapter for Raspberry Pi

This model has four NVMe SSD interface, supporting up to 2 NVMe SSDs to run simultaneously. In this tutorial, we name this model as **Dual-NVMe Adapter Board**.

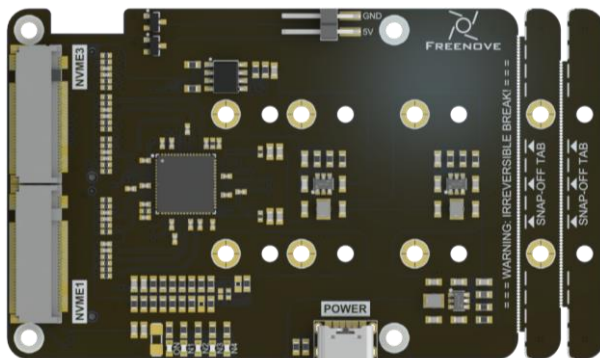


Here are its key features:

- **Interface Type:** 2x M.2 with M-Key
- **Supported Protocol:** NVMe
- **PCIe Channel:** PCIe 2.0 x2
- **Compatible Sizes:** 2230, 2242, 2260, 2280
- **Power Supply:** 3.3V, up to 3A (maximum)
- **Indicator Lights:** Includes both power and x2 SSD status LEDs.

1.4.3 Freenove Quad M.2 Nvme Adapter for Raspberry Pi

This model has four NVMe SSD interface, supporting up to 4 NVMe SSDs to run simultaneously. In this tutorial, we name this model as **Quad-NVMe Adapter Board**.



Here are its key features:

- **Interface Type:** 4x M.2 with M-Key
- **Supported Protocol:** NVMe
- **PCIe Channel:** PCIe 2.0 x4
- **Compatible Sizes:** 2230, 2242, 2260, 2280
- **Power Supply:** 3.3V, up to 3A (maximum)
- **Indicator Lights:** Includes both power and x4 SSD status LEDs.

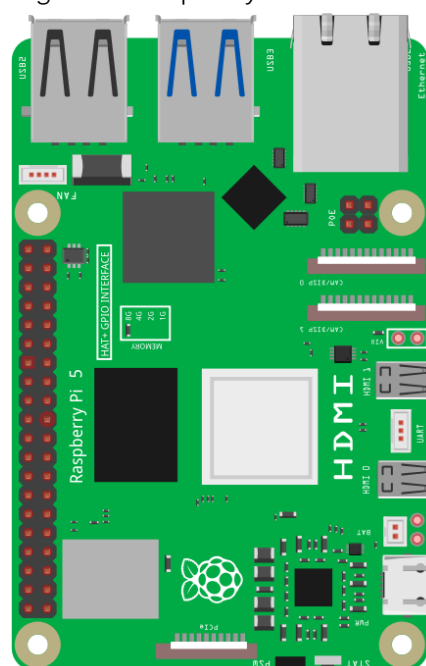
1.5 Introduction to Raspberry Pi 5(RPi 5)

At the time of this writing, this product only supports RPi5. The following shows the physical and model figures of an RPi 5.

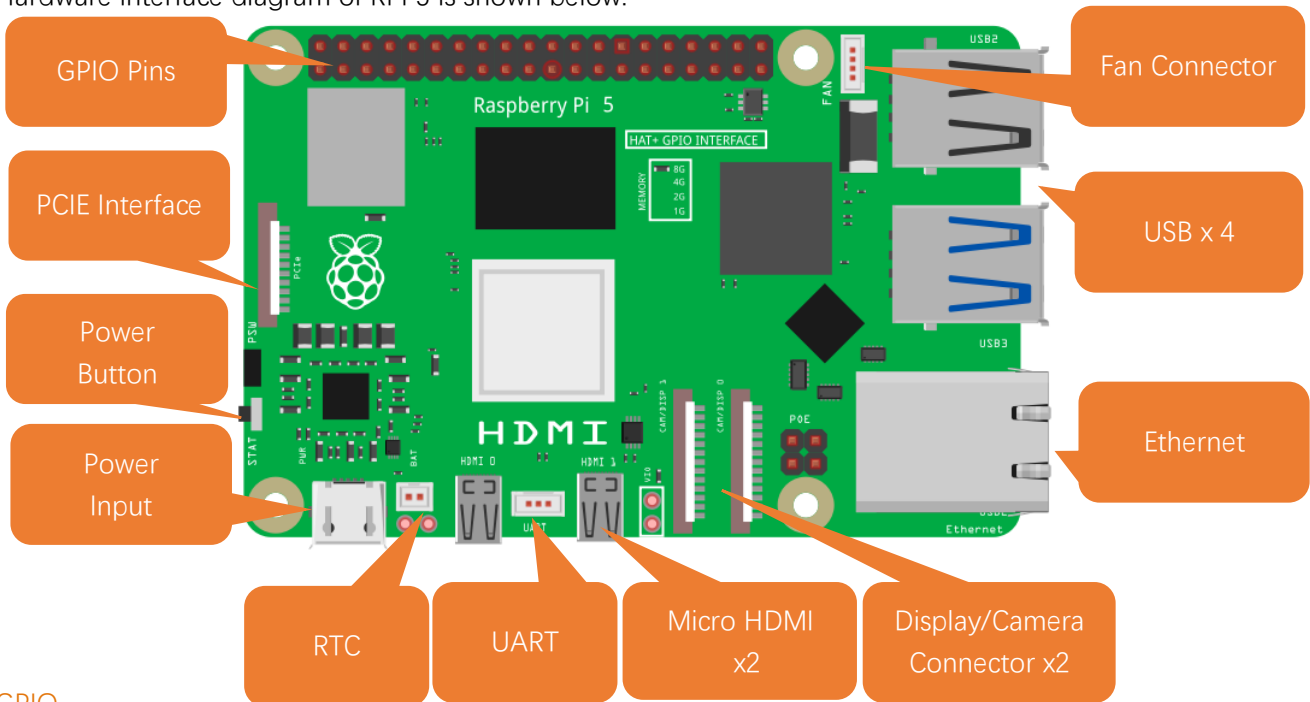
Practicality picture of Raspberry Pi 5:



Model diagram of Raspberry Pi 5:



Hardware interface diagram of RPi 5 is shown below:



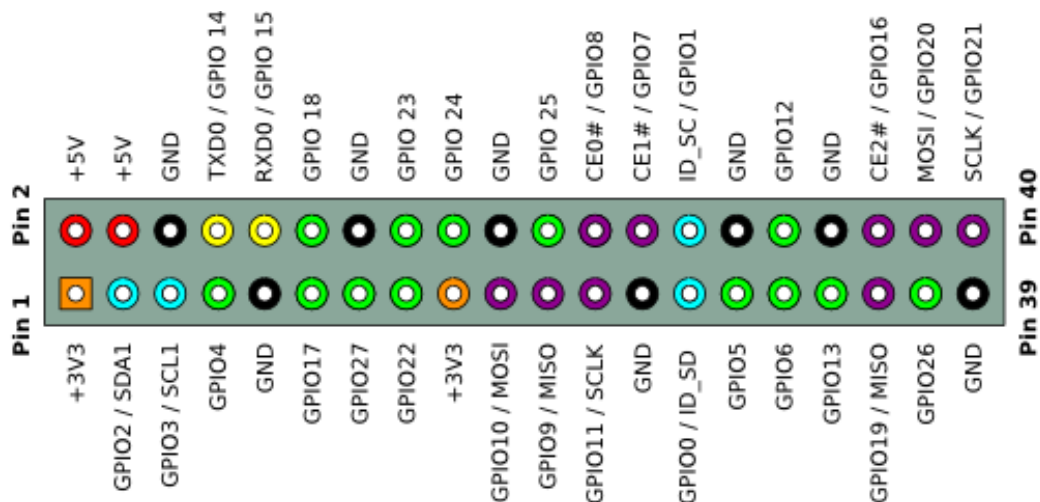
GPIO

GPIO: General purpose input/output. We will introduce the specific feature of the pins on the Raspberry Pi and how you can utilize them in all sorts of ways in your projects. Most RPi Module pins can be used as either an input or output, depending on your program and its functions. When programming the GPIO pins, there are three different ways to reference them: GPIO numbering, physical numbering, WiringPi GPIO Numbering.

BCM GPIO Numbering

The Raspberry Pi CPU uses Broadcom (BCM) processing chips BCM2835, BCM2836 or BCM2837. GPIO pin numbers are assigned by the processing chip manufacturer and are how the computer recognizes each pin. The pin numbers themselves do not make sense or have meaning, as they are only a form of identification. Since their numeric values and physical locations have no specific order, there is no way to remember them, so you will need to have a printed reference or a reference board that fits over the pins.

Each pin is defined as below:



For more details about pin definition of GPIO, please refer to <https://pinout.xyz/>

Power requirements of various versions of Raspberry Pi are shown in following table:

Product	Recommended PSU current capacity	Maximum total USB peripheral current draw	Typical bare-board active current consumption
Raspberry Pi Model A	700mA	500mA	200mA
Raspberry Pi Model B	1.2A	500mA	500mA
Raspberry Pi Model A+	700mA	500mA	180mA
Raspberry Pi Model B+	1.8A	600mA/1.2A (switchable)	330mA
Raspberry Pi 2 Model B	1.8A	600mA/1.2A (switchable)	350mA
Raspberry Pi 3 Model B	2.5A	1.2A	400mA
Raspberry Pi 3 Model A+	2.5A	Limited by PSU, board, and connector ratings only.	350mA
Raspberry Pi 3 Model B+	2.5A	1.2A	500mA
Raspberry Pi 4 Model B	3.0A	1.2A	600mA
Raspberry Pi 5 Model B	5.0A	1.6A (600mA if using a 3A power supply)	800mA
Raspberry Pi Zero W	1.2A	Limited by PSU, board, and connector ratings only.	150mA
Raspberry Pi Zero	1.2A	Limited by PSU, board, and connector ratings only	100mA

For more details, please refer to

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#power-supply>

In this product, the Raspberry Pi 5 is used and it must be powered by a 5.1V/5A power supply. Insufficient power may cause various functions to operate abnormally, or even permanently damage your Raspberry Pi 5. Therefore, we strongly recommend using a 5.1V/5A power supply to ensure optimal performance and avoid potential hardware failure.

Chapter 2 Case Assembly

Video: <https://youtu.be/RXRW5KxHWVQ>

It is recommended to assemble and use the Freenove Computer Case Kit Pro for Raspberry Pi according to this tutorial. Otherwise, it may lead to incorrect device installation or damage. Please check all the parts again. Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

Ensure the device is powered off before assembly.

Use of the Screwdriver Bit Holder

When installing the brass standoff, simply place it directly into the bit holder.



When installing the M3.7x10 Countersunk Head Screws, please use the PH2 hexagonal cross-bit.

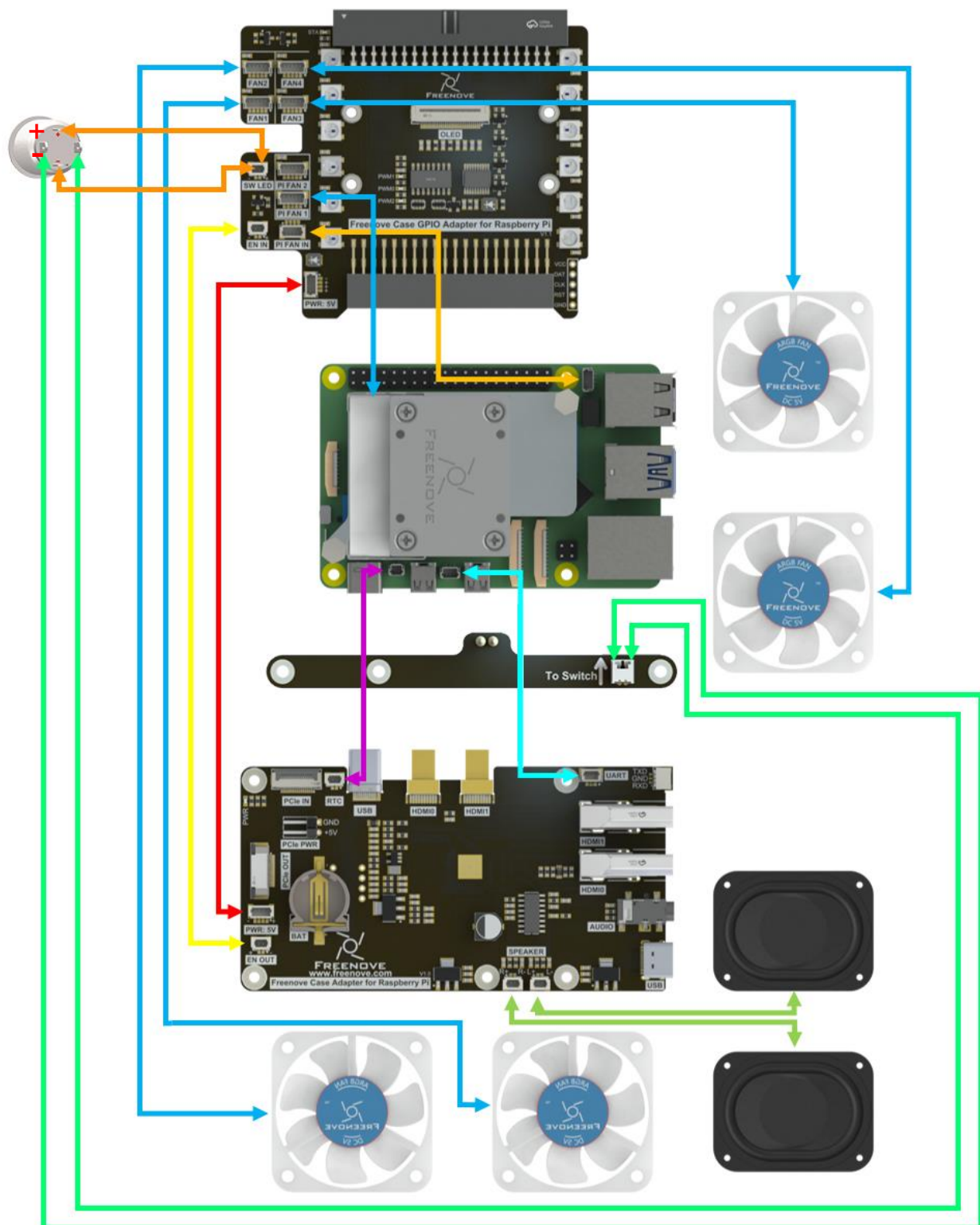


When installing other screws, please use the PH0 hexagonal cross-bit



Overall Wiring Diagram

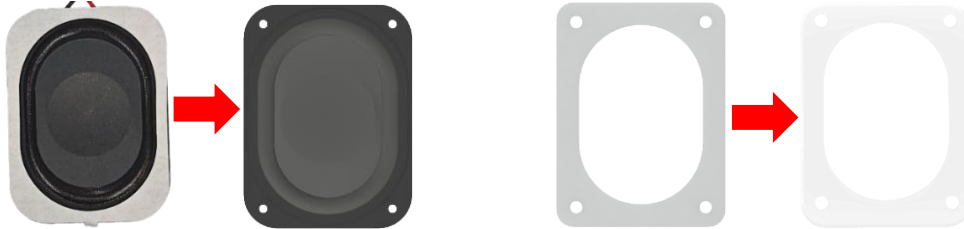
Note: The figure below is for reference only. Please assemble the Raspberry Pi case strictly according subsequent assembly steps.



2.1 Assmby of Speakers, Fans, and the Power Button

2.1.1 Assembling Speakers

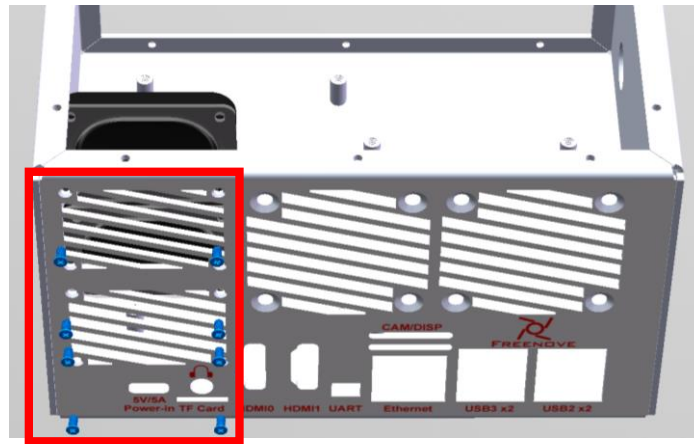
Step 1: Remove the white protective film from the speaker and the speaker's acrylic pad.



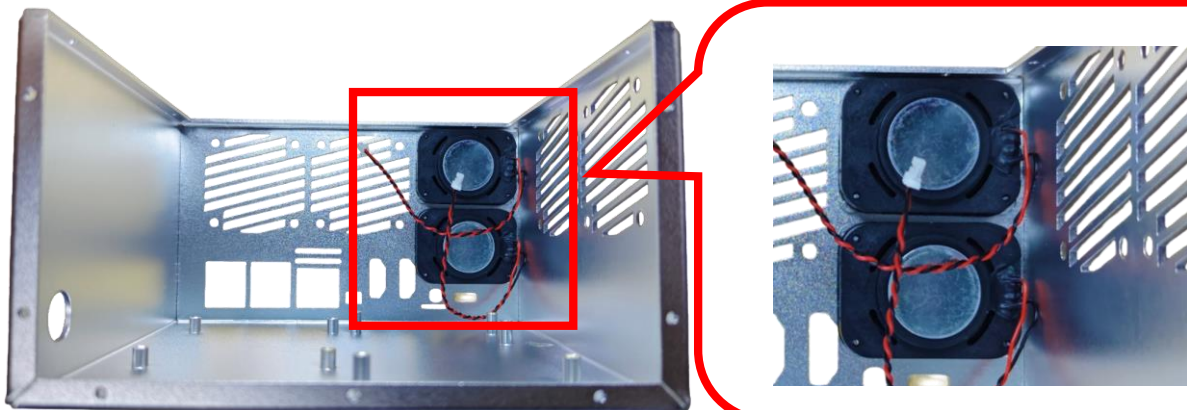
Step 2: Align the speaker acrylic pad with the speaker and attach them together.



Step 3: Mount the two speakers to the case from the outside with M2x6 Countersunk Head Screws.



Note: When installing, ensure the speaker orientation matches the diagram below, with the twisted pair located near the case housing.

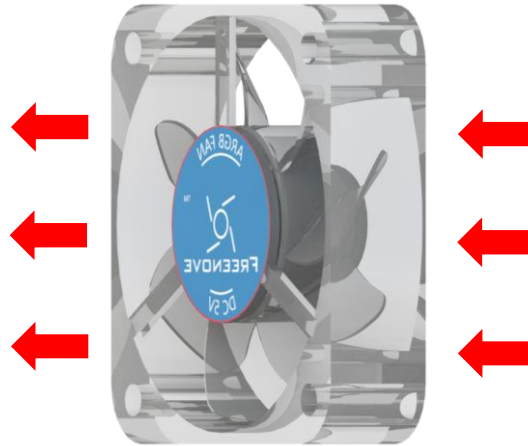


2.1.2 Assembling Fans

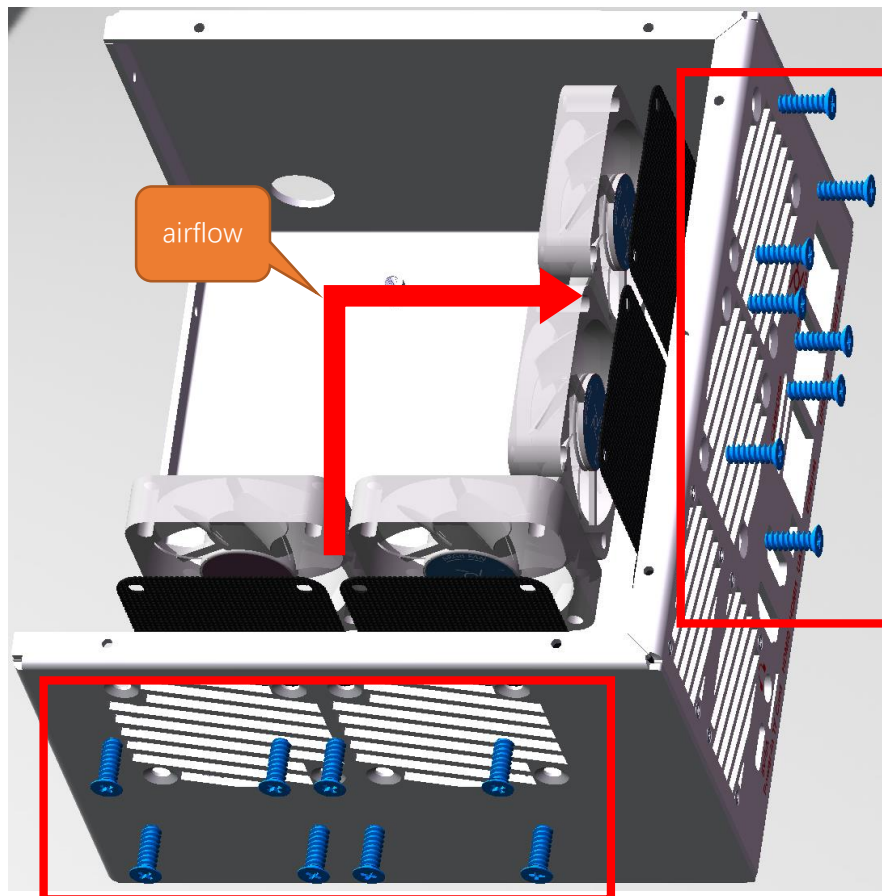
Airflow Direction Description:

The full fan blades are visible, is the **air intake**.

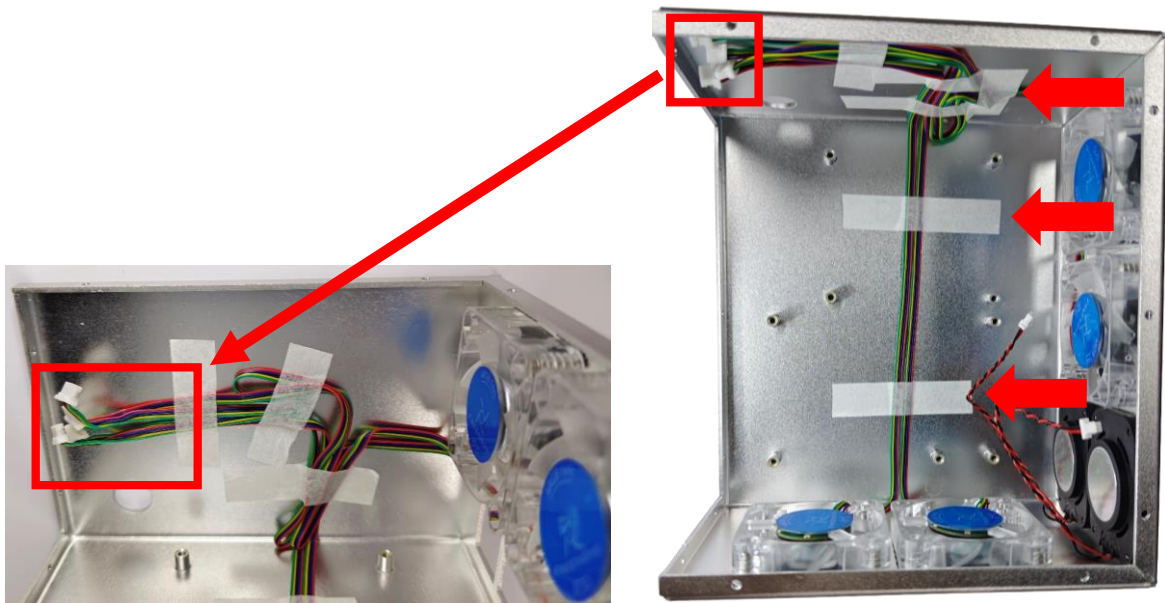
The side with the motor frame and protective grille is the **air outlet**.



Step 1: Install the case fan. Place the dust filter between the fan and the case. Secure the fan using four M3.7x10 countersunk head screws from the outside of the case. Note: During installation, ensure the fan's airflow direction is correct – it should draw cool air in from the bottom of the case and exhaust hot air from the right side. This creates a consistent airflow, effectively lowering the internal operating temperature of the case. **Note: For a cleaner cable layout, the fan cable side should be positioned to the bottom of the case.**



Step 2: Secure the fan cable to the case frame using masking tape. Route the connector to the center of the case for easy connection to the GPIO board, thus maintaining a tidy interior. (Please mark the fan connector for identification during wiring.)

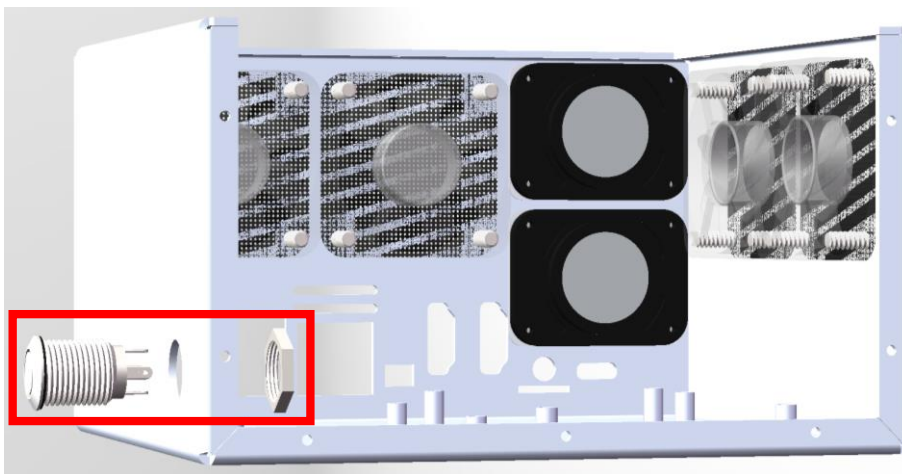


2.1.3 Installing the Power Button

Step 1: Insert the Black Sealing Gasket into the 12mm LED Power Button.



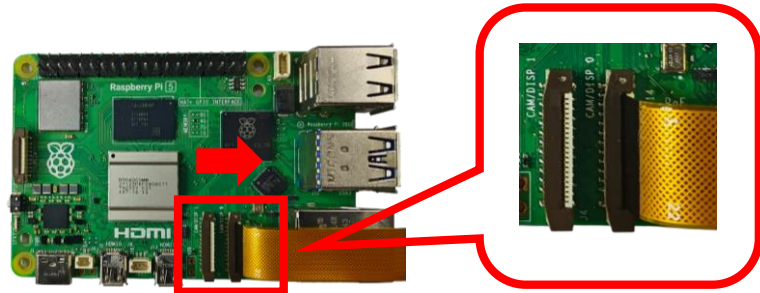
Step 2: Install the 12mm LED Power Button into the circular hole on the top of the case, and secure it from the inside with an M12 nut.



2.2 Installing Electronic Parts and Connecting Cables

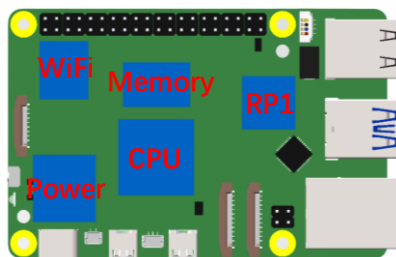
2.2.1 Installing the Tower Cooler

Step 1: For users buying the variants equipped with a **4.3-inch touchscreen (FNK0107P\Q\R\U\V\W)** and those who want to use the camera with this kit, please connect the DSI cable and the camera cable to the Raspberry Pi's **CAM/DISP** interface. (**Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection. The red arrow below indicates the orientation of the contacts**)

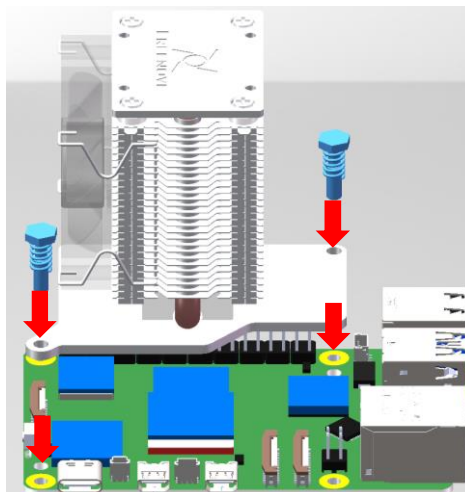


Note: The installation of a tower cooler may make it difficult to access the CAM/DISP interface. If your model includes a 4.3-inch screen (FNK0107P/Q/R/U/V/W) or if you need to connect a device (such as a camera) to this interface, you must connect the cable before installing the cooler. Otherwise, you may skip this step.

Step 2: Identify the five thermal pads by size and apply each one to its corresponding chip on the RPi 5. (Important: Ensure you remove the protective film from both sides of each pad to guarantee optimal thermal conductivity.)

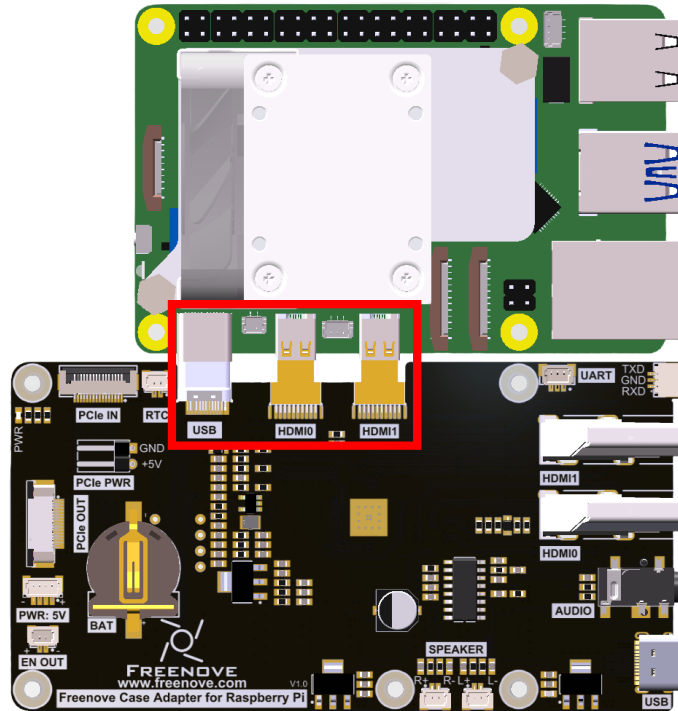


Steps 3: Fix the Tower Cooler to RPi 5 with the Nylon standoffs.

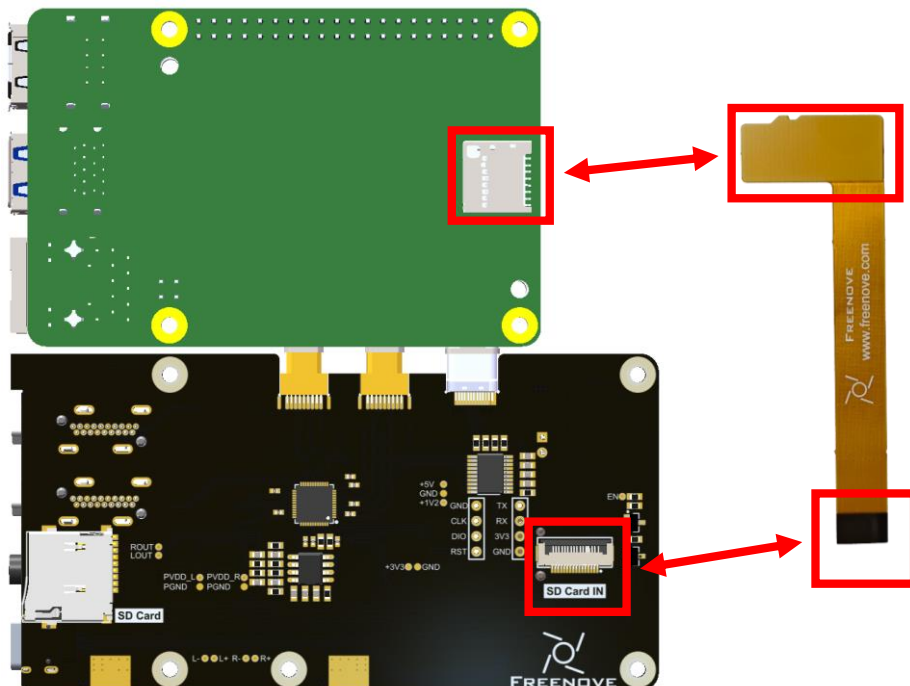


2.2.2 Installing Audio-Video Board

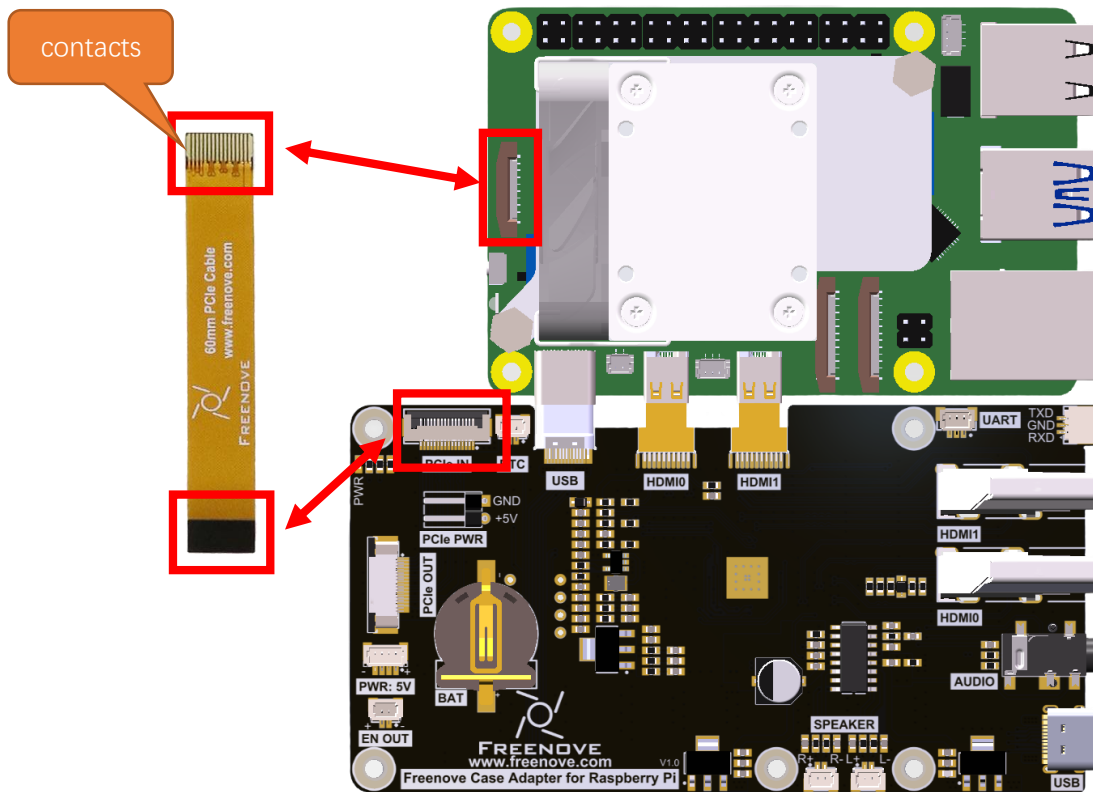
Step 1: Align the Type-C and HDMI ports of the Raspberry Pi 5 and the Audio-Video Board, then insert the Audio-Video Board into the Raspberry Pi 5.



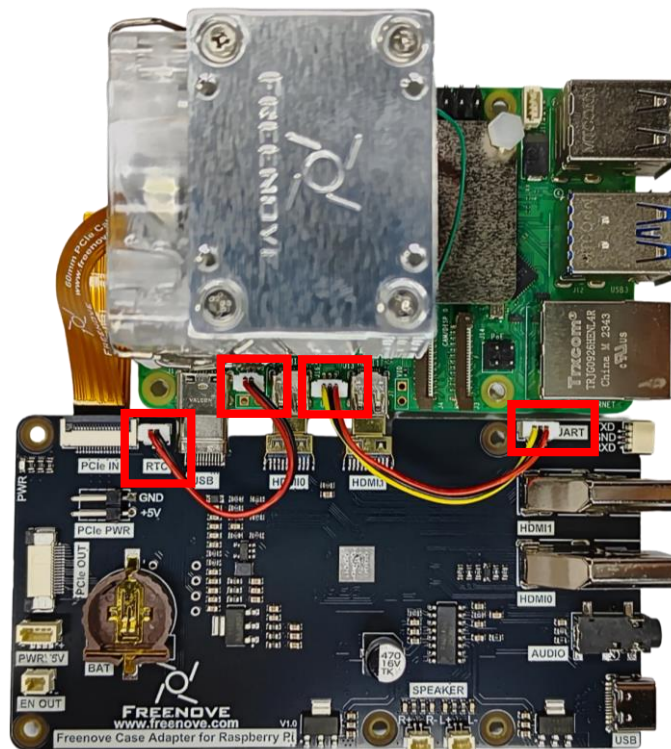
Step 2: Insert the **To Pi** end of the SD Card to 0.5mm-16P FPC cable to the **RPi 5's SD card slot**, then connect the **To Adapter** end to the **SD Card IN** interface on the Audio-Video Board. (**Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection. Contacts bottom.**)



Step 3: Insert one end of the reverse straight cable to the RPi 5's **PCIe interface** (The contacts should face the active cooler.), then connect the other end to the **PCIe IN interface** on the Audio-Video Board (The contacts should face the bottom. **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**)

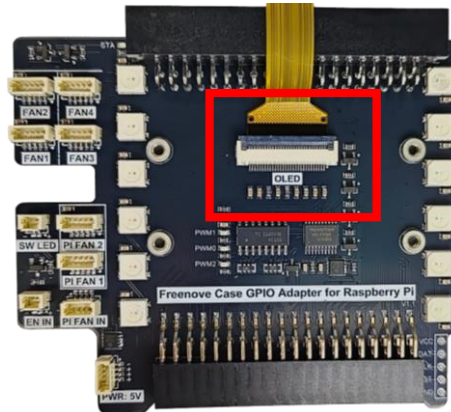


Step 4: Connect the **RTC interface** between the RPi 5 and the Audio-Video Board using a 5cm SH1.0mm 2-pin cable. Then, connect their **UART interface** using a 5cm SH1.0mm 3-pin cable.



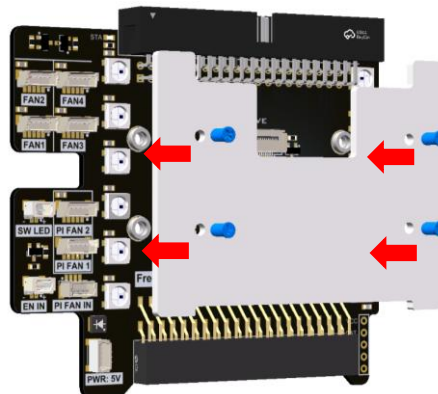
2.2.3 Installing the GPIO Board

Step 1: Connect the flex cable of the 0.96-inch OLED display to the OLED interface on the GPIO Board, with the contacts facing the bottom

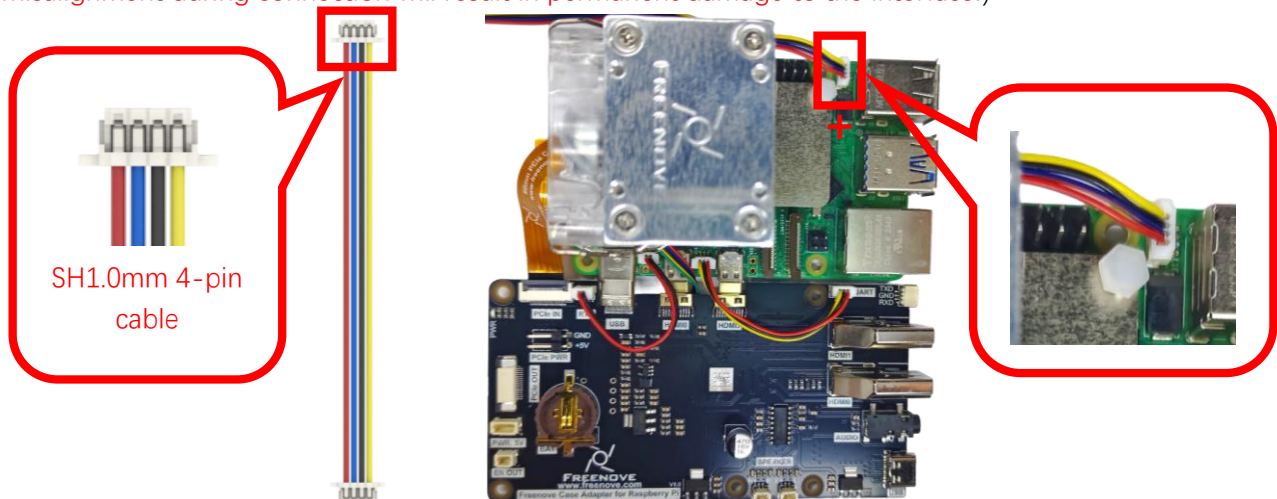


Important: The display glass is fragile – please exercise extreme caution during installation. Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.

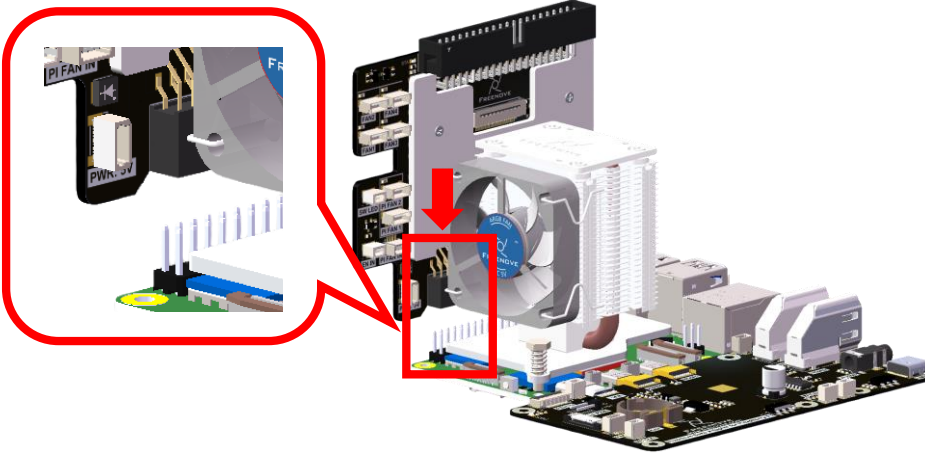
Step 2: Secure the LED light diffusion plate to the GPIO board's threaded insert with M2.5x5 countersunk head screws.



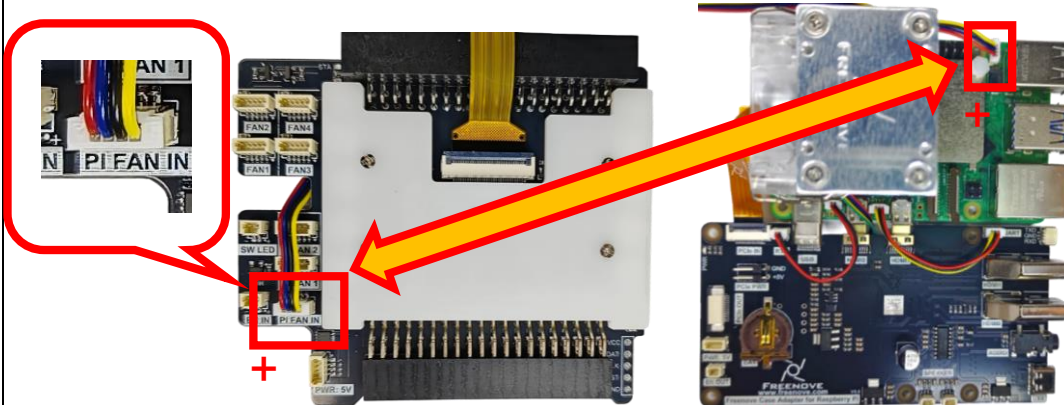
Step 3: Connect the 15cm SH1.0mm 4-pin cable to the FAN (J17) interface on the RPi 5. (Ensure the red wire connects to the terminal marked '+'. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



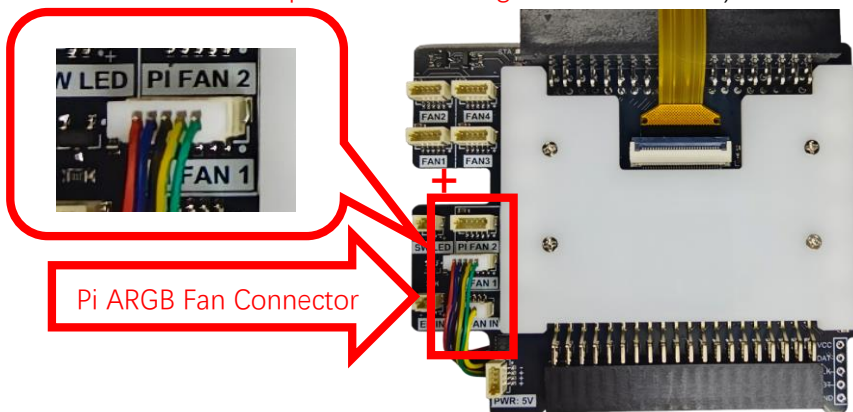
Step 4: Align the **female header** of the GPIO Board with the **GPIO header** on the RPi 5, then press the GPIO Board down to insert it. (**Caution: Proper alignment is imperative for installation. Failure to do so could lead to a short circuit and permanent burnout of the unit.**)



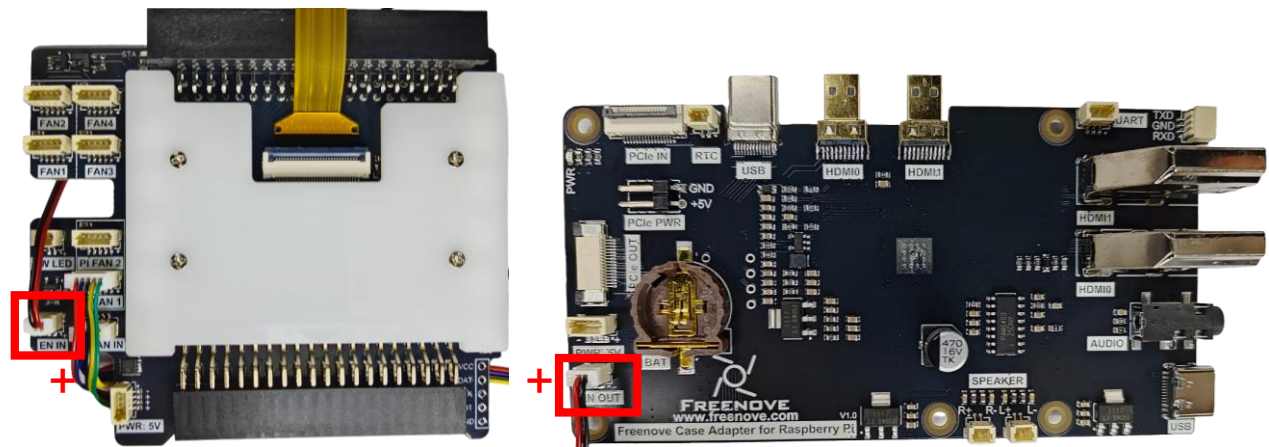
Step 5: Connect the other end of the 15cm SH1.0mm 4-pin cable to the **"PI FAN IN"** interface on the GPIO Board. (**Caution: Do not connect it to the "PWR:5V" interface, as this may damage the components. The red wire must go to the terminal marked '+'. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.**)



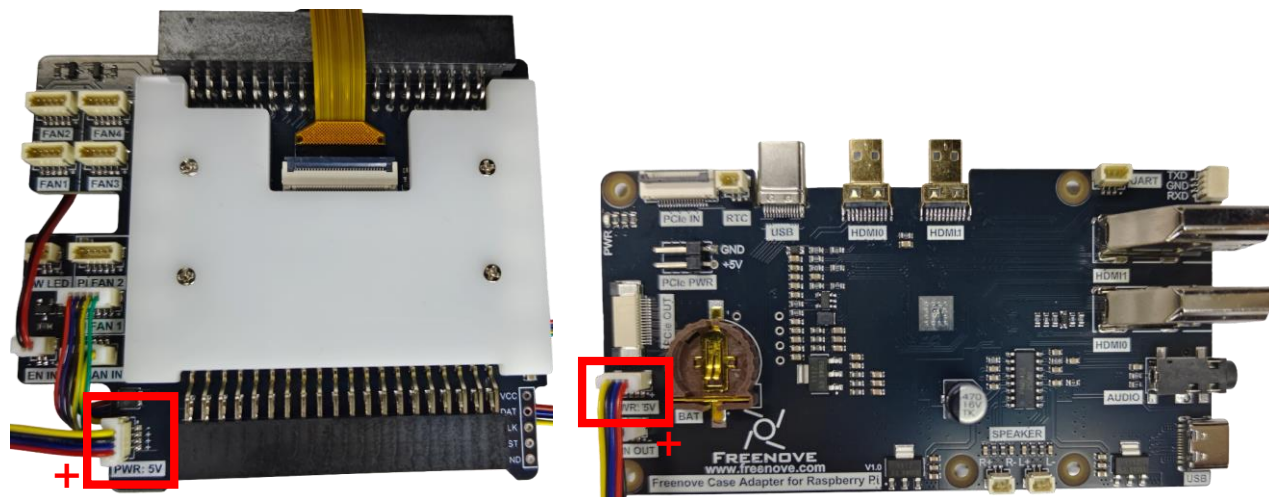
Step 6: Route the fan cable of the tower cooler around the bottom of the RPi 5 to the back of the GPIO Board, then insert it into either the **PI FAN 1 or PI FAN 2 interface**. PI FAN 1 is directly controlled by RPi 5 and PI FAN 2 is controlled by the GPIO board. We recommend connecting it to the PI FAN 1 interface. (**Note: Align the red wire with the '+' mark and insert carefully to avoid bending the pins. Use tweezers to gently straighten any bent pins. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.**)



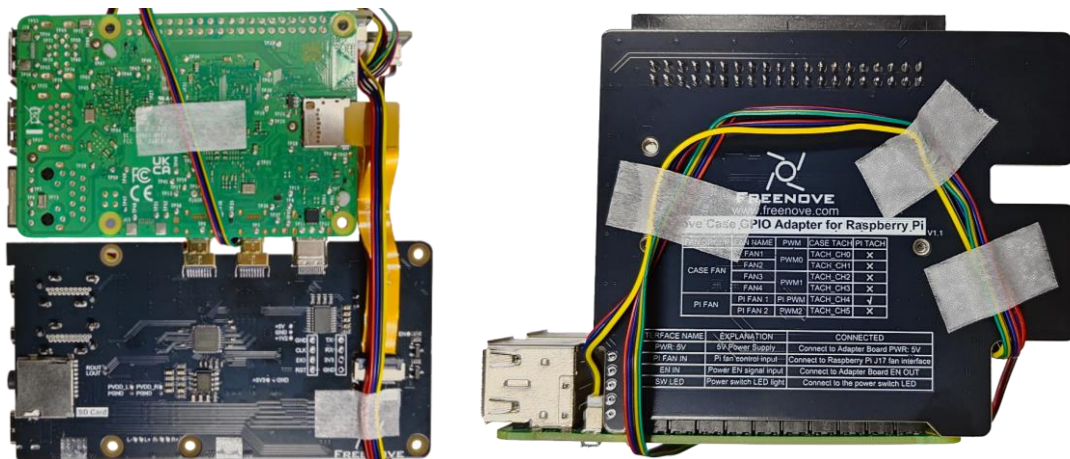
Step 7: Use the 15cm SH1.0mm 2-pin cable. Connect one end to the **EN IN** interface on the GPIO Board and the other end to the **EN OUT** interface on the Audio-Video Board. (Note: The red wire must be aligned with the terminal marked '+'. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



Step 8: Use the 15cm SH1.0mm 4-pin cable. Connect one end to the **PWR: 5V** interface on the GPIO Board and the other end to the **PWR: 5V** interface on the Audio-Video Board. (Note: The red wire must be aligned with the terminal marked '+'. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)

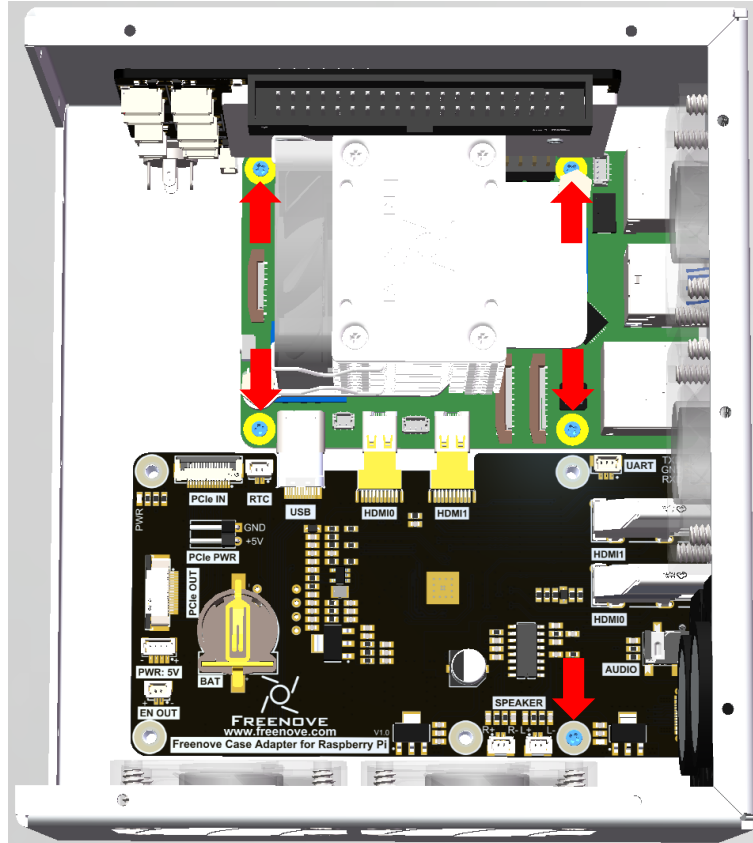


Step 9: For a clean installation, use masking tape to manage loose cables by securing them to the back or bottom of the PCB.

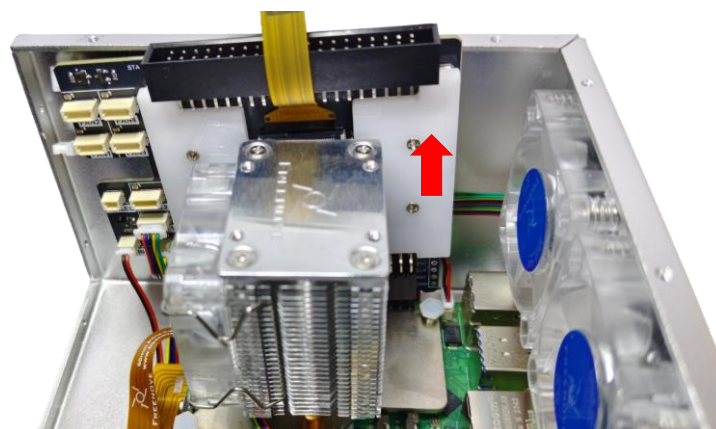


2.2.4 Installing to the Case Shell

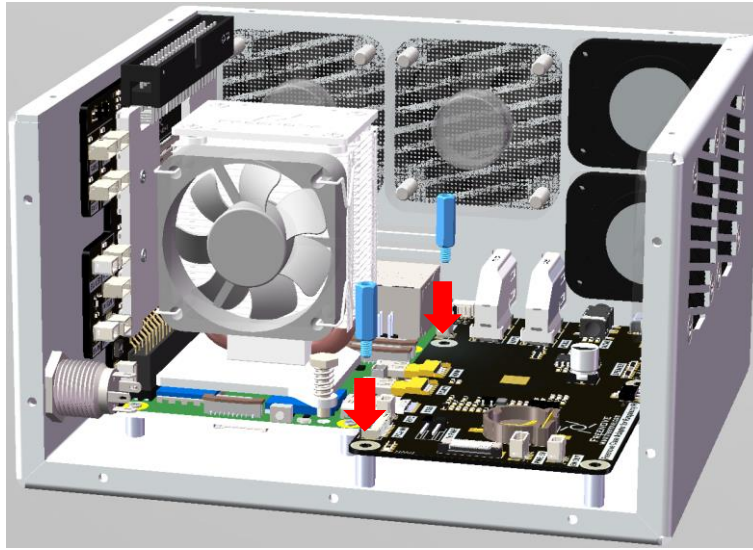
Step 1: Place the pre-assembled modules (RPi 5, Audio-Video Board, and GPIO Board) into the case. Align it with all the mounting posts, and then use M2.5x5 countersunk head screws to secure them to the case.



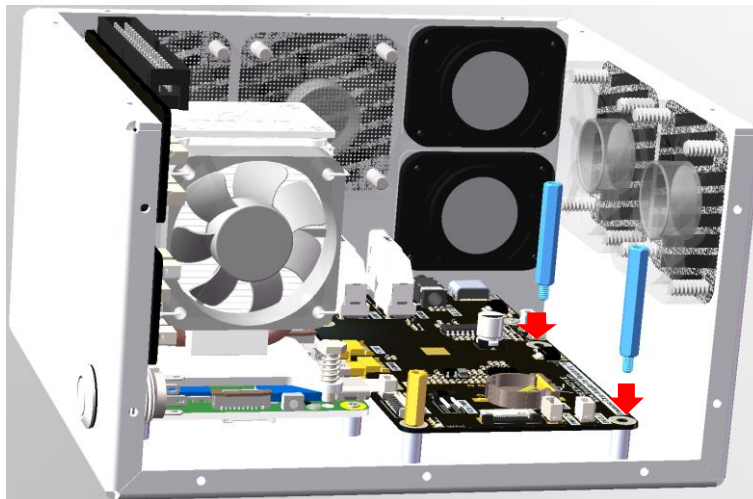
If there is insufficient operating space while installing the screw near the **RPi 5's FAN (J17) interface**, you can gently lift the GPIO Board on this side slightly upward. After the screw installation is complete, press it back into its original position.



Use the M2.5x12+4 brass standoffs to secure the side of the Audio-Video Board closest to the RPi 5 onto the mounting post below.

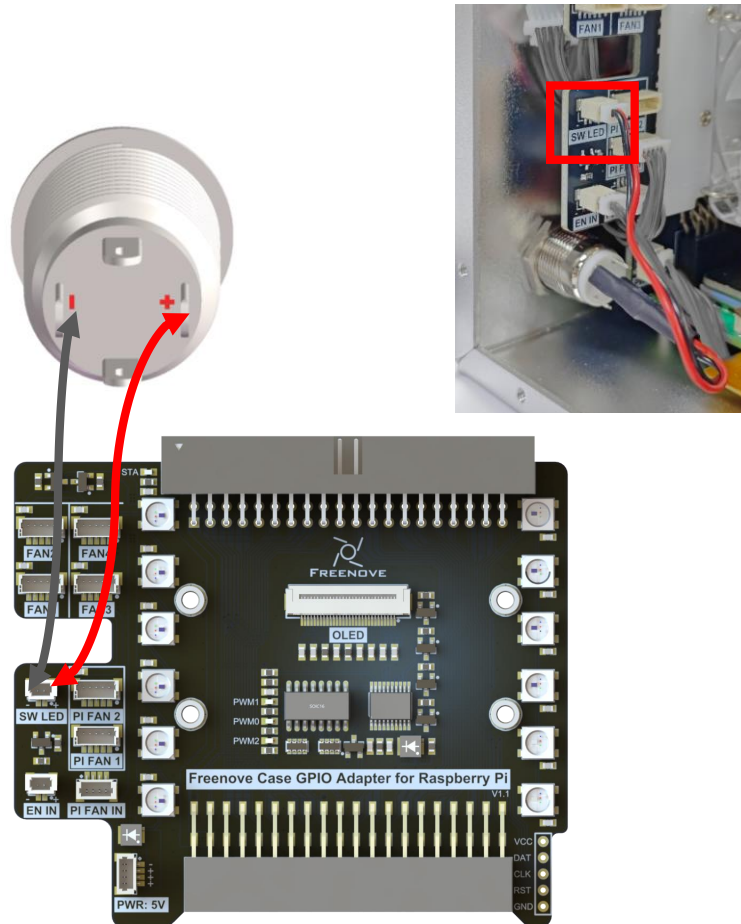


Use the M2.5x25+4 brass standoffs to secure the opposite side of the Audio-Video Board onto the mounting post.

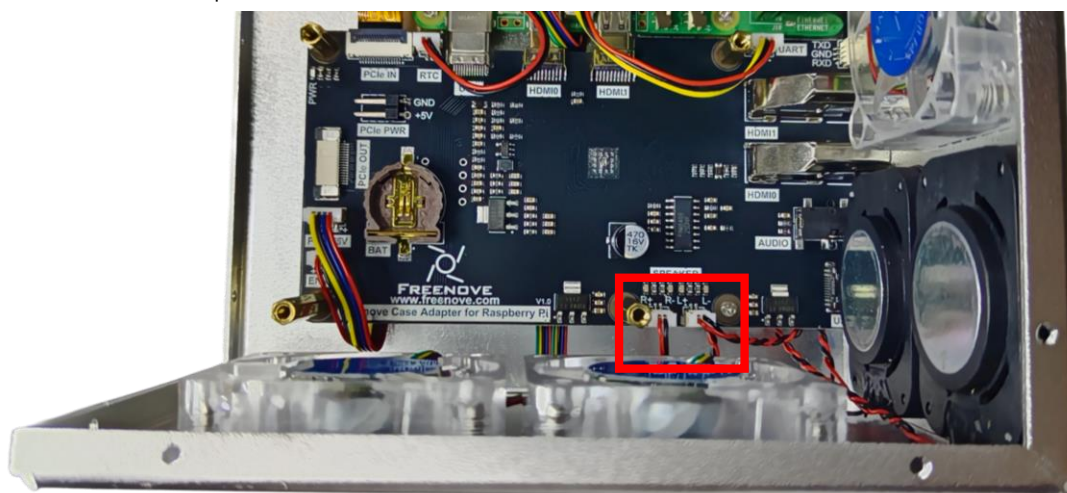


2.2.5 Connecting Cables

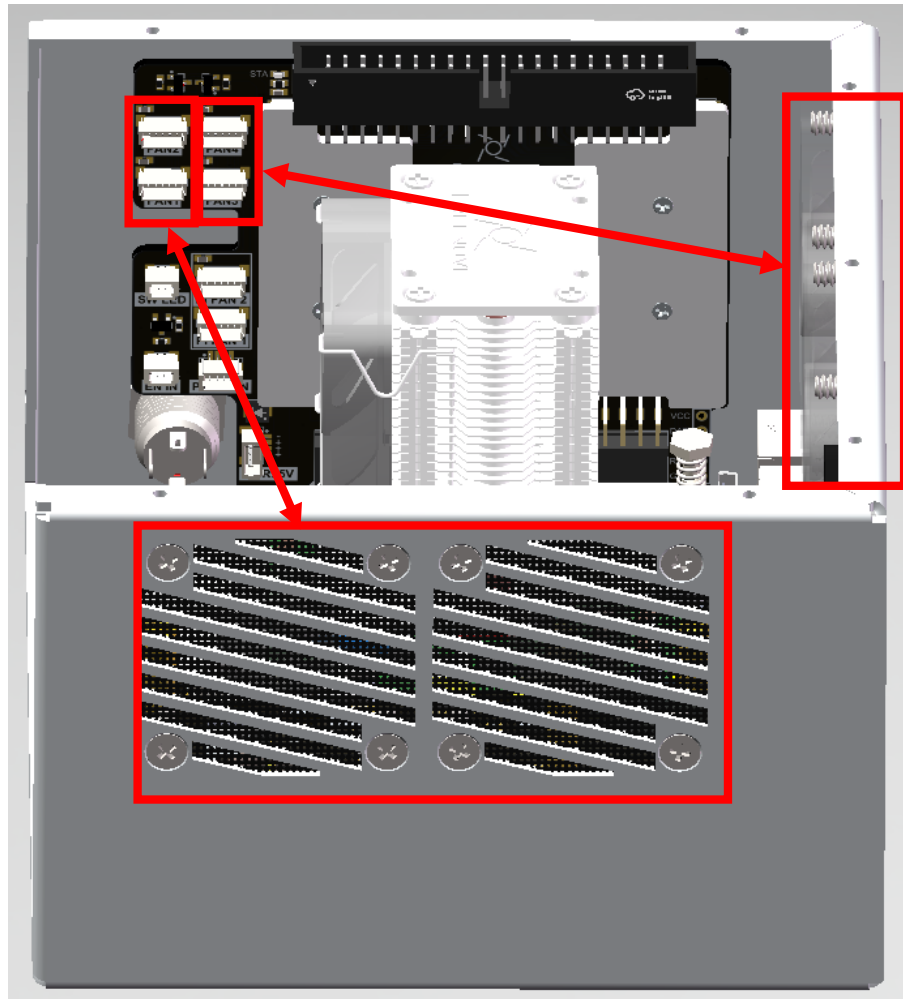
Step 1: Connect the quick-disconnect terminal end of the 7cm SH1.0mm to quick-disconnect terminal cable (red/black) to the 12mm LED Power Switch. (Note: The red wire connects to the switch's '+' terminal, and the black wire connects to the '-' terminal.) Then, connect the other end to the **SW LED interface** on the GPIO Board. (For a physical wiring reference, please see the actual installation photo below.)



Step 2: Connect the two speakers to the **SPEAKER interface** of the Audio-Video Board.

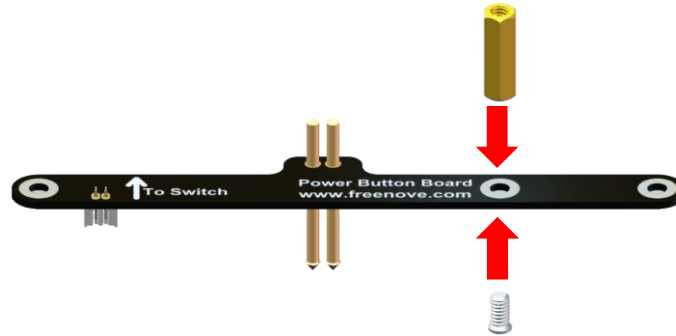


Step 3: Divide the four fans into two groups: FAN1/2 as one group, and FAN3/4 as the other. Connect the **rear fans** to the **FAN3/4 interface**, and connect the **bottom fans** to the **FAN1/2 interface**. (Note: This is a non-reversible connector; Please insert carefully to avoid bending the pins. Use tweezers to gently straighten any bent pins.)

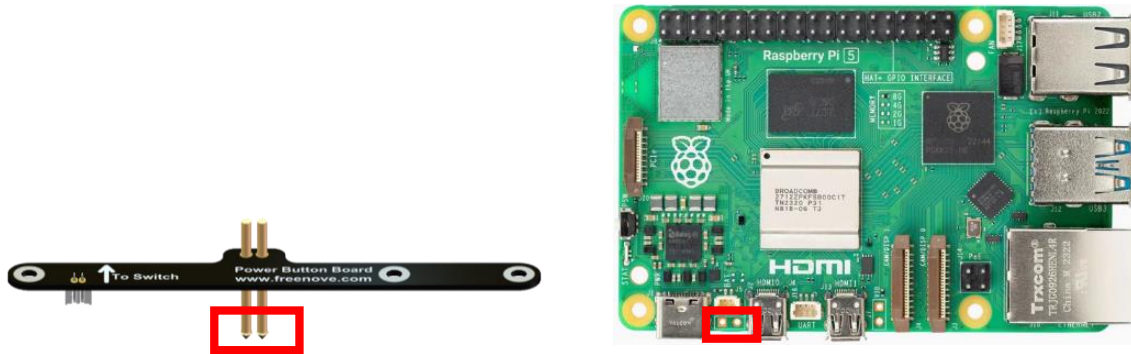


2.2.6 Connecting Switch Board

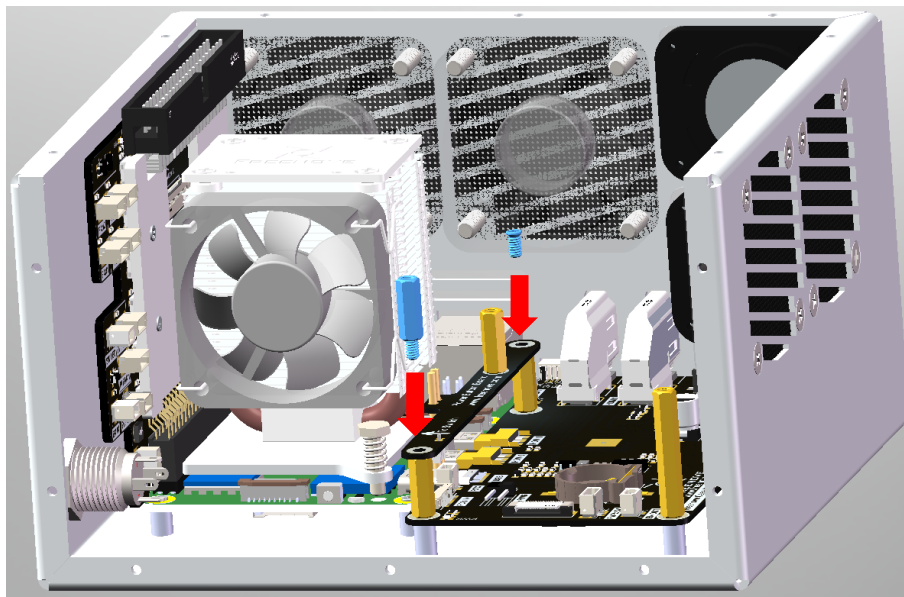
Step 1: Install the M2.5x12 brass standoff to the middle positioning location on the Switch Board, noting the orientation.



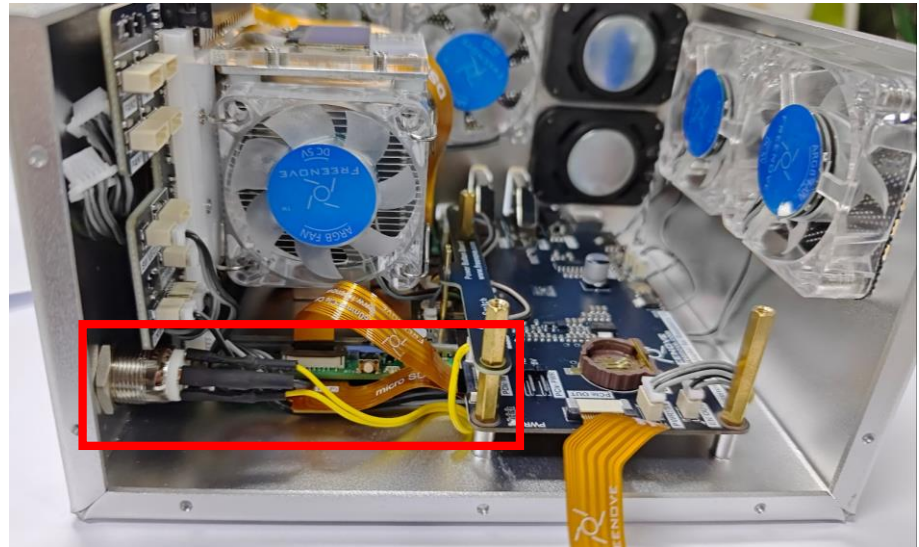
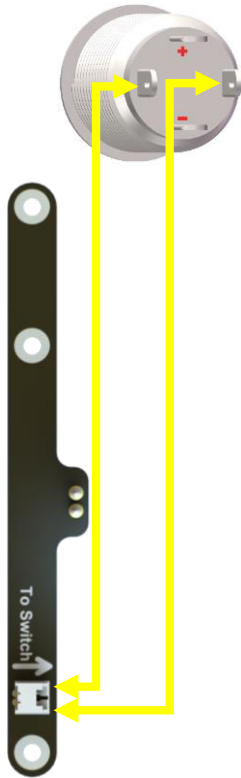
Step 2 : Align the pogo pin with the **J2 pad** located between the RPi 5's Type-C and HDMI0 ports, and then gently press the Switch Board into place.



Secure the inner side of the Switch Board with M2.5x5 Countersunk Head Screws. Then, fasten the outer side with the M2.5x12+4 Brass Standoffs.



Step 3: Insert the **terminal end** of the 7cm 1.25mm-to-2.8 quick-disconnect terminal cable (yellow-yellow) into the 12mm LED Power Switch. Then, insert the other end into the interface on the Switch Board.



2.2.7 Installing the NVMe Adapter Board

Please select your product model below to view the corresponding installation instructions for the NVMe Adapter Board.

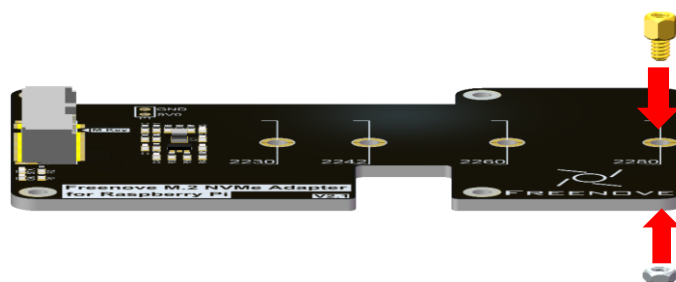
For **FNK0107A/H/P/U**: [NVMe Adapter Board](#)

For **FNK0107B/K/Q/V**: [Dual-NVMe Adapter Board](#)

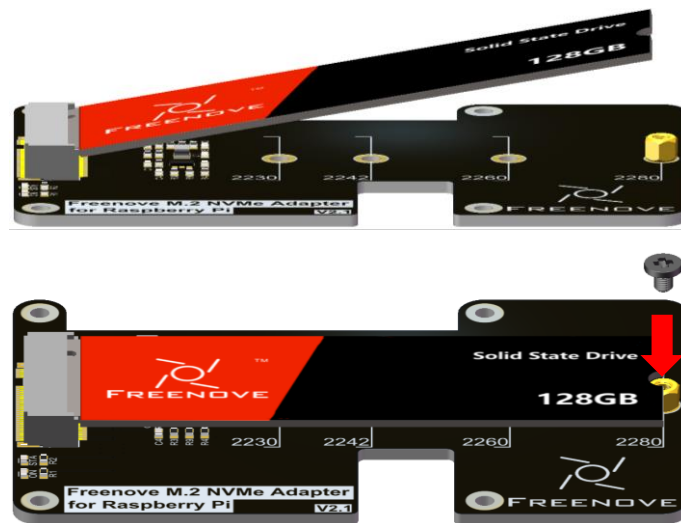
For **FNK0107C/L/R/W**: [Quad-NVMe Adapter Board](#)

Installing NVMe Adapter Board

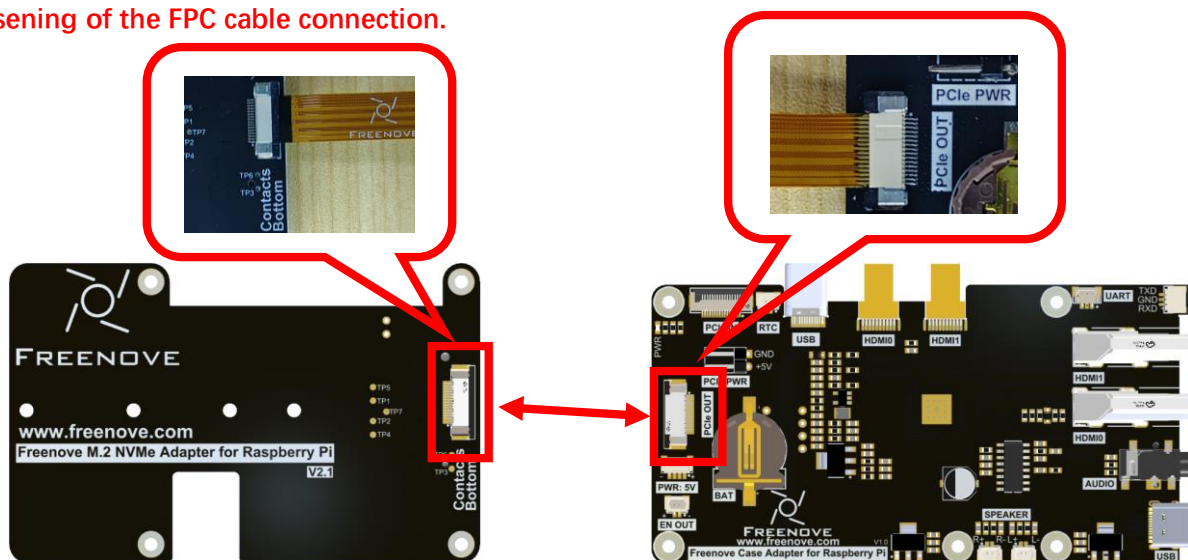
Step 1: Based on the physical dimensions of your SSD, install the M2.5x3+3 brass standoff into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure it from the reverse side using an M2.5 nut.



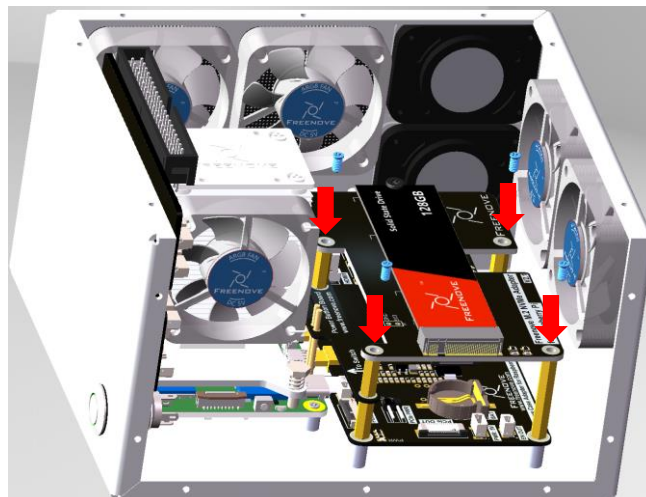
Step 2: Tilt the SSD to insert it into the NVMe slot, and then secure it using an M2.5x2.5x5 flat-head screw.



Step 3: Connect one end of the NVMe cable to the PCIe OUT interface on the Audio-Video Board (**contacts facing up**), then connect the other end to the FPC interface on the reverse side of the NVMe Adapter Board (**contacts facing down**). **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**



Step 4: Secure the NVMe Adapter Board with M2.5x5 Countersunk Head Screws



Installing Dual-NVMe Adapter Board

Step 1: Based on the physical dimensions of your SSD, install two M2.5x3+3 brass standoffs into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure them from the reverse side with M2.5 nuts.

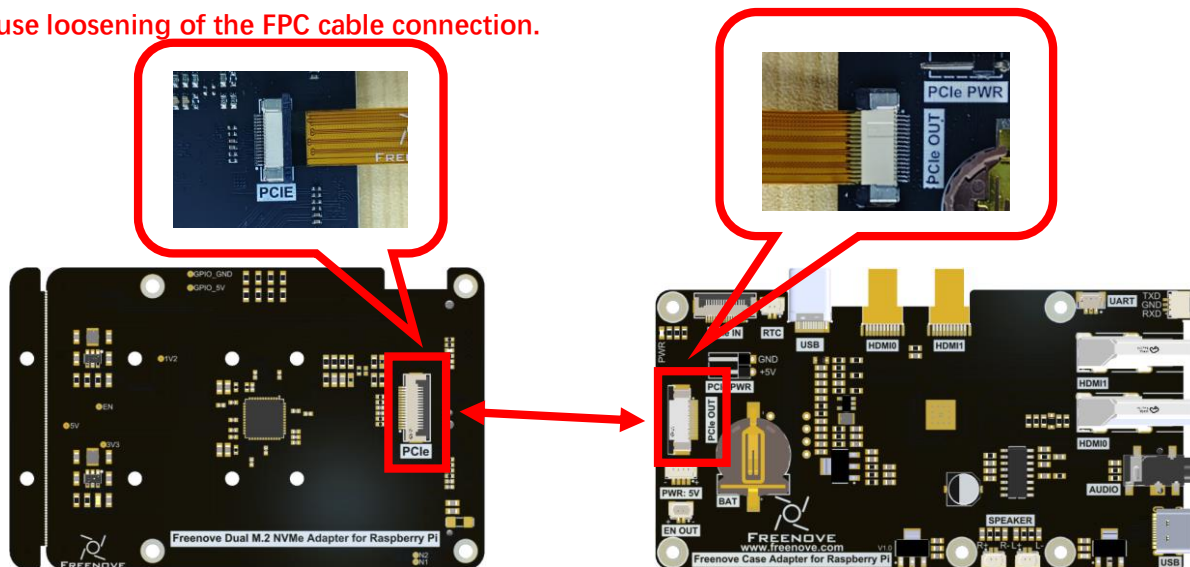


Note: This Adapter Board supports up to two SSDs. Install 0 to 2 SSDs based on your preference.

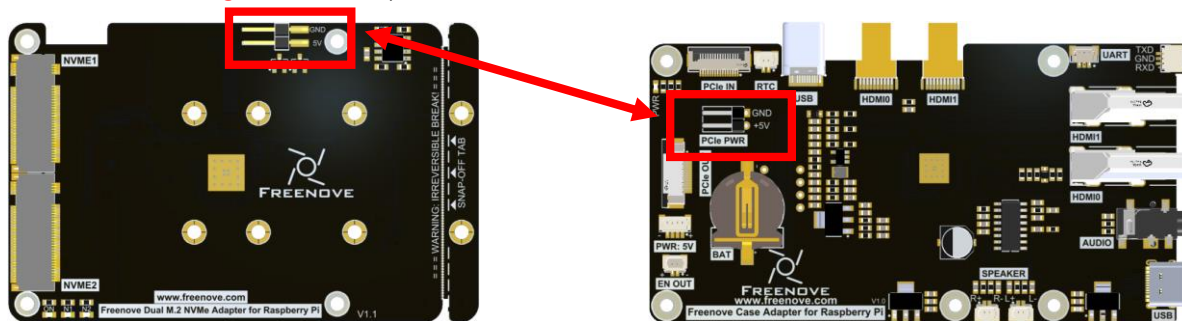
Step 2: Tilt the SSDs to insert them into the two NVMe slots, and then secure them using M2.5x2.5x5 flat-head screws.



Step 3: Connect one end of the NVMe cable to the PCIe OUT interface on the Audio-Video Board (**contacts facing up**), then connect the other end to the FPC interface on the reverse side of the NVMe Adapter Board (**contacts facing toward the board**). **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**

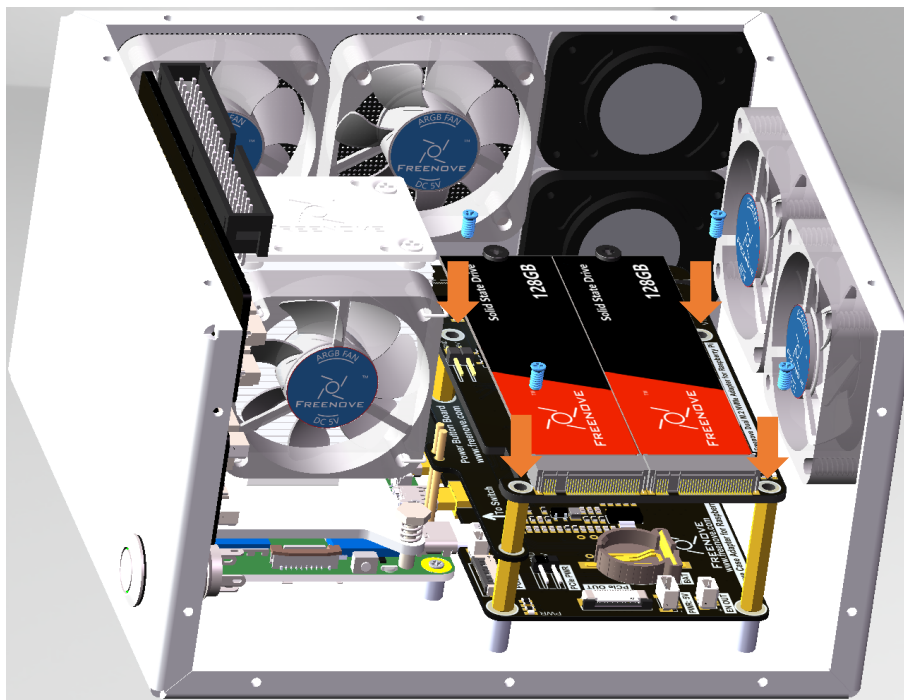


Step 4: Connect two female-to-female jumper wires. Attach one end to the **PCIe PWR interface** on the Audio-Video Board, and the other end to the **pin headers** on the Dual-NVMe Adapter Board. (**Caution:** The red wire must connect to the 5V pin, and the black wire to the GND pin. Incorrect wiring may cause a short circuit and damage the device.)



Generally, connecting the two jumper wires is not required. However, for SSDs with higher power consumption, it is necessary to connect them to ensure adequate power supply and stable operation.

Step 5: Secure the Dual-NVMe Adapter Board with M2.5x5 Countersunk Head Screws.



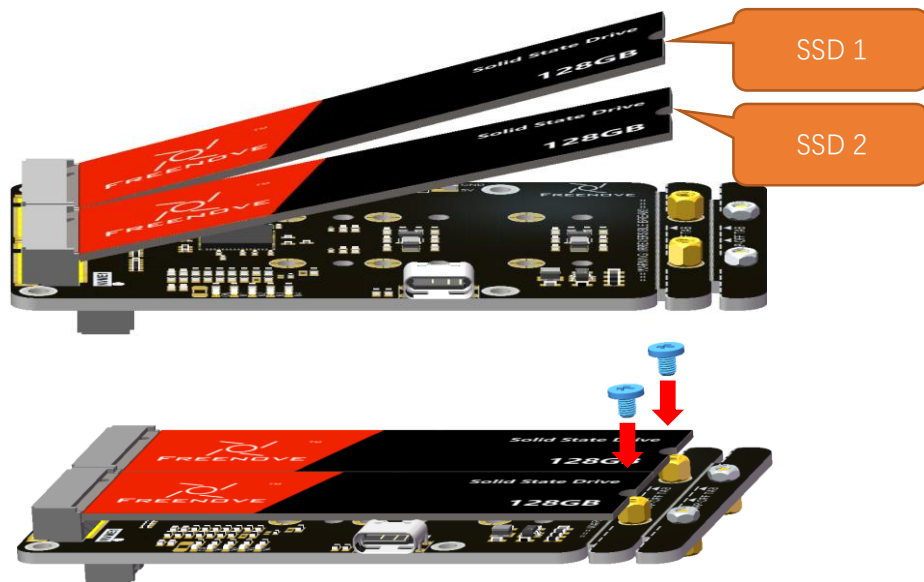
Installing Quad-NVMe Adapter Board

Step 1: Based on the physical dimensions of your SSD, install four M2.5x3+3 brass standoffs into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure them from the reverse side with M2.5 nuts. It is recommended that you install standoffs to all the four mounting holes.

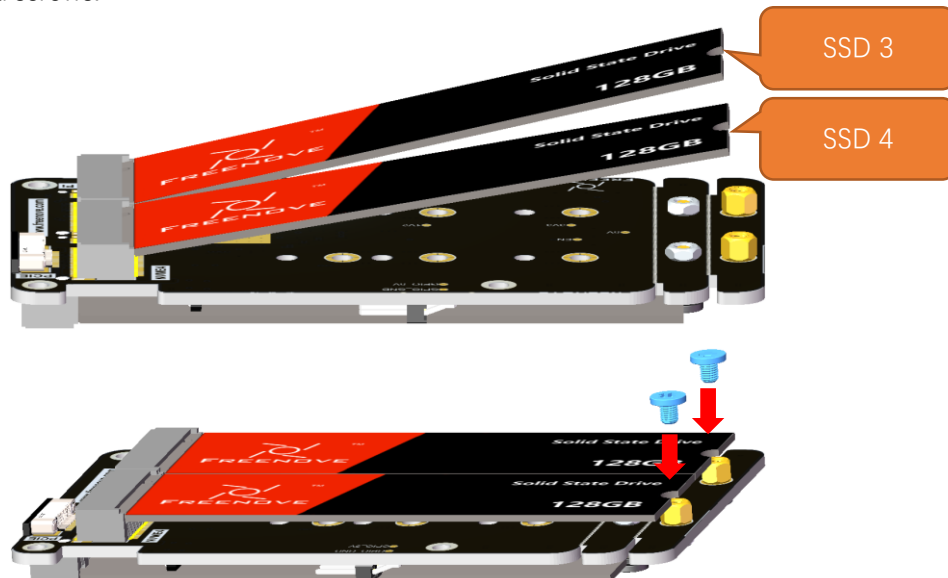


Note: This Adapter Board supports up to four SSDs. Install 0 to 4 SSDs based on your preference.

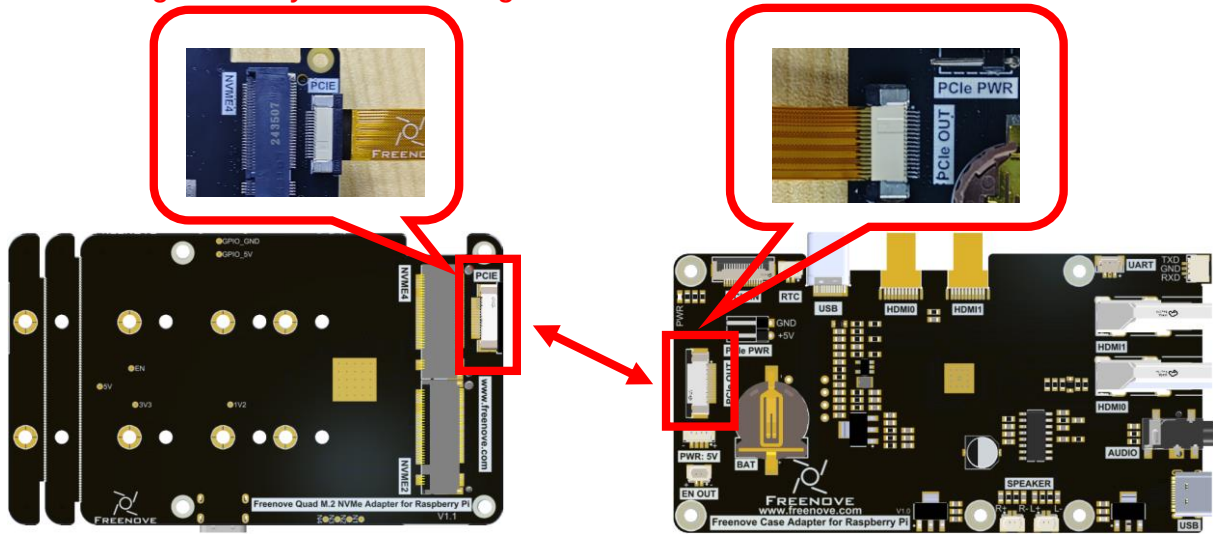
Step 2: Tilt the SSDs to insert them into the two NVMe slots on the front side, and then secure them using M2.5x2.5x5 flat-head screws.



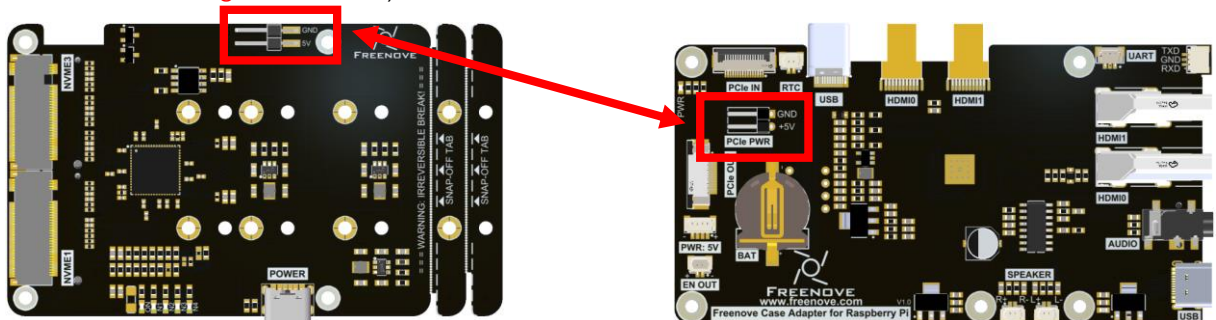
Step 3: Tilt the SSDs to insert them into the two NVMe slots on the back side, and secure them using M2.5x2.5x5 flat-head screws.



Step 4: Connect one end of the NVMe cable to the **PCIe OUT** interface on the Audio-Video Board (**contacts facing up**), and connect the other end to the **PCIe** interface on the reverse side of the Quad-NVMe Adapter Board (**contacts facing toward the board**). **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**



Step 5: Connect two female-to-female jumper wires. Attach one end to the **PCIe PWR** interface on the Audio-Video Board, and the other end to the **pin headers** on the Quad-NVMe Adapter Board. (**Caution: The red wire must connect to the 5V pin, and the black wire to the GND pin. Incorrect wiring may cause a short circuit and damage the device.**)



Generally, connecting the two jumper wires is not required. However, for SSDs with higher power consumption, it is necessary to connect them to ensure adequate power supply and stable operation.

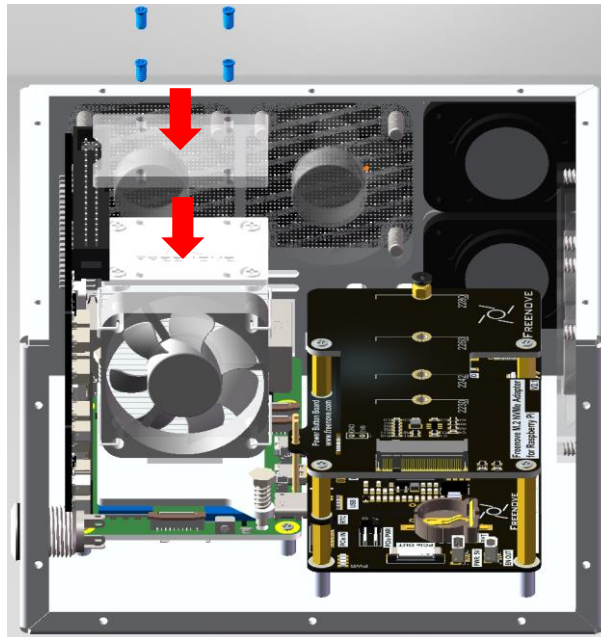
Step 6: Secure the Dual-NVMe Adapter Board with M2.5x5 Countersunk Head Screws.



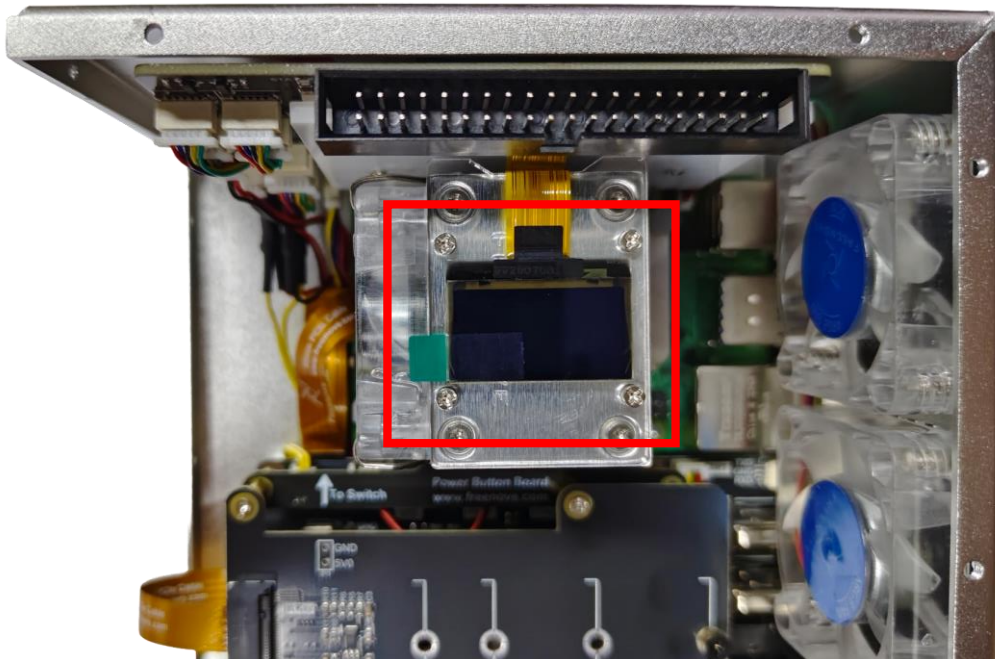
2.3 Installing Acrylic Parts and 4.3" Touchscreen

2.3.1 Installing OLED's Acrylic

Step 1: Mount the acrylic part for the OLED to the top of the tower cooler with M2.5x5 countersunk head screws. Ensure the acrylic notch faces the GPIO Board.



Step 2: Remove the white film from the 3M adhesive on the back of the OLED, align it with the center of the four screw holes, and press it into place.



Note: The glass on the 0.96-inch OLED display is fragile. Ensure it is properly aligned before applying. Repeated attachment and removal may cause the glass to crack.

2.3.2 Installing Case Acrylic

- **For models without a 4.3-inch Screen (Model numbers: FNK0107A/B/C/H/K/L):**

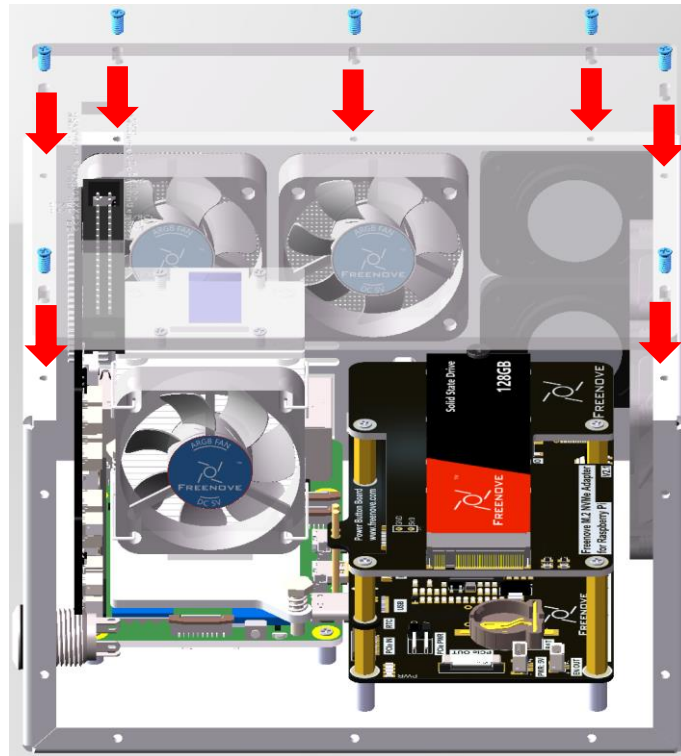
If you have purchased one of the models listed above, your kit does not include a 4.3-inch Screen. To view the installation details for the corresponding acrylic case, please click here: [Installation without Screen](#)

- **For models equipped with a 4.3-inch Screen (Model numbers: FNK0107P\Q\R\U\V\W):**

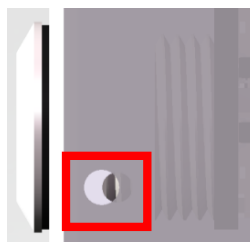
If you have purchased one of the models listed above, your kit includes a 4.3-inch Screen and two sets of acrylic case parts with different structures, supporting both landscape and portrait orientations. Please choose your preferred installation method: [Landscape Mounting](#) or [Portrait Mounting](#).

Installation without Screen

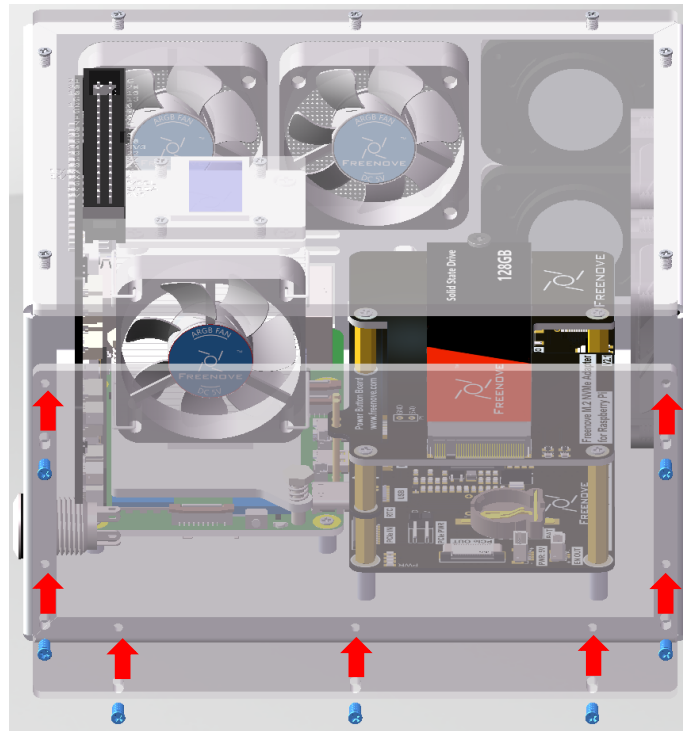
Step 1: Align the top acrylic platel with the GPIO headers, and secure it using M2.5x5 countersunk head screws.



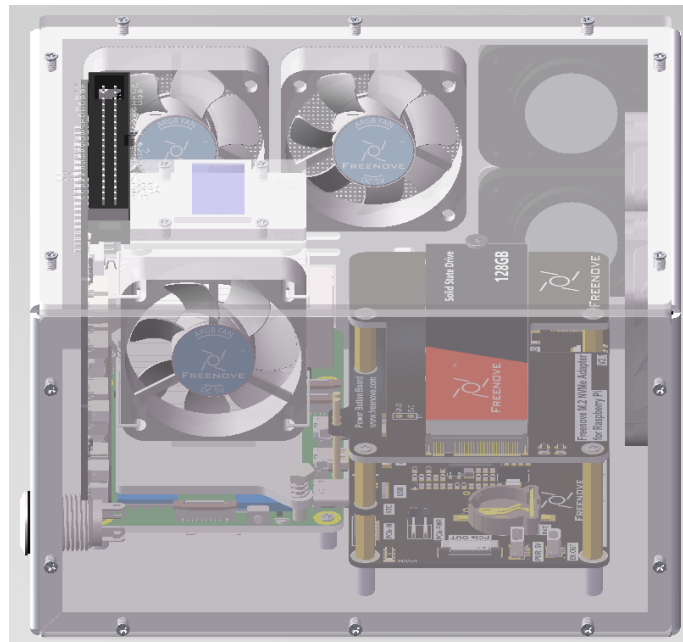
Note: Owing to the machining precision limitations of sheet metal parts, if you notice that certain screw holes do not align properly, this is not an indication of an error. In such cases, you can apply force to bend and align them as needed.



Step 2: Secure the side acrylic plate with M2.5x5 countersunk head screws.

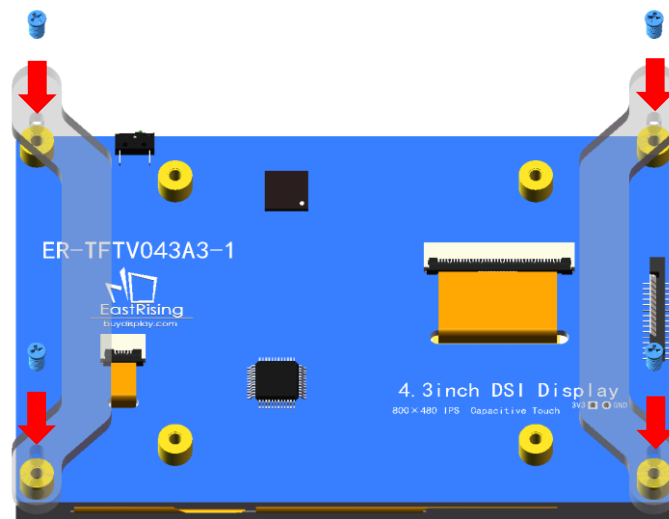


Now you have finished assembling the case without the touchscreen.

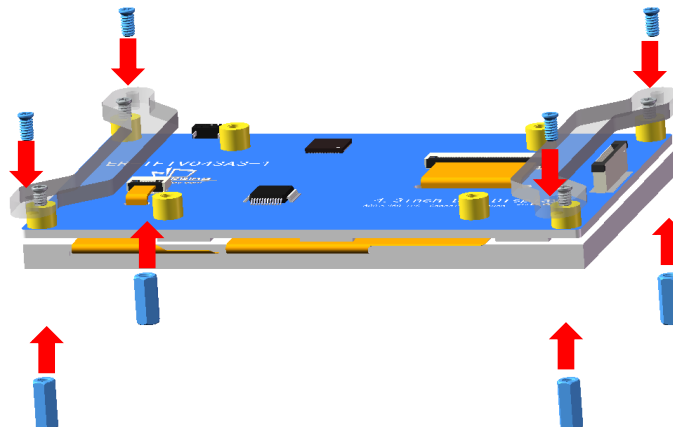


Landscape Mounting

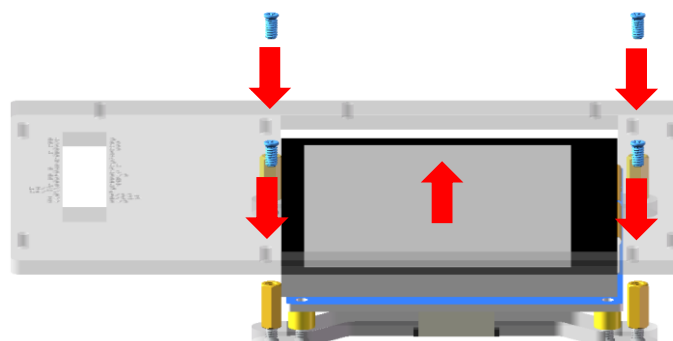
Step 1: Secure the **landscape display bracket** to the back of the 4.3-inch display using M2.5x5 countersunk head screws. The fixing holes are the **inner set of holes** on the bracket.



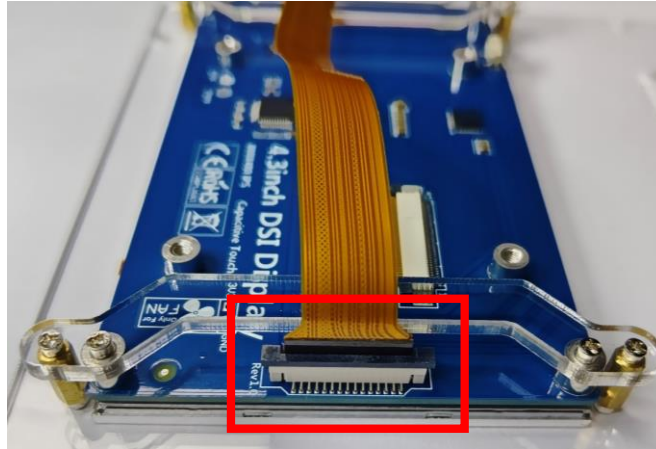
Step 2: Secure the M2.5x9 brass standoff to the landscape display bracket using M2.5x5 countersunk head screws. Use the **outer set of holes** on the bracket.



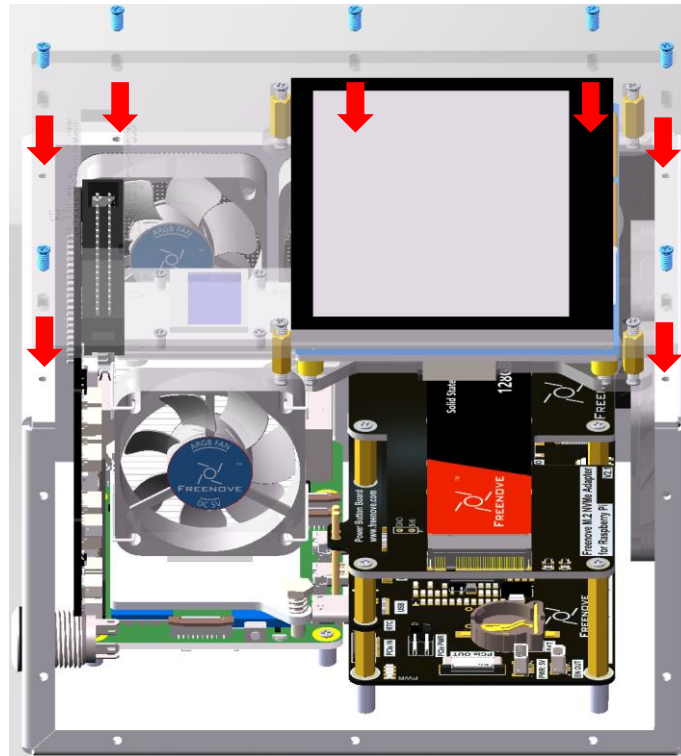
Step 3: Align the pre-assembled 4.3-inch screen with the mounting holes on the top acrylic plate, and secure it using M2.5x5 countersunk head screws.



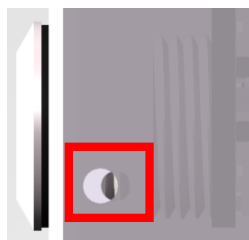
Step 4: Connect the other end of the 160mm DSI FPC cable to the connector on the back of the 4.3-inch screen, ensuring the contacts are facing inward toward the display. **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**



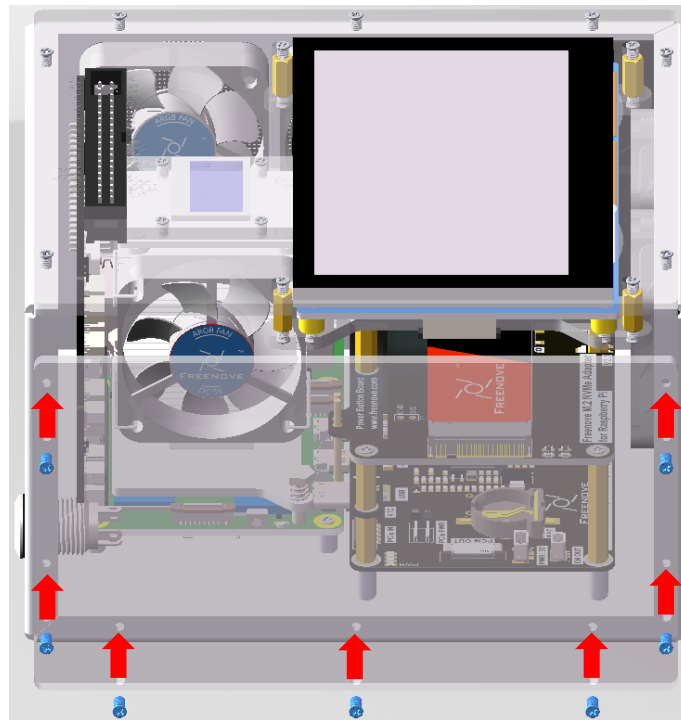
Step 5: Align the top acrylic platel with the GPIO headers, and secure it using M2.5x5 countersunk head screws.



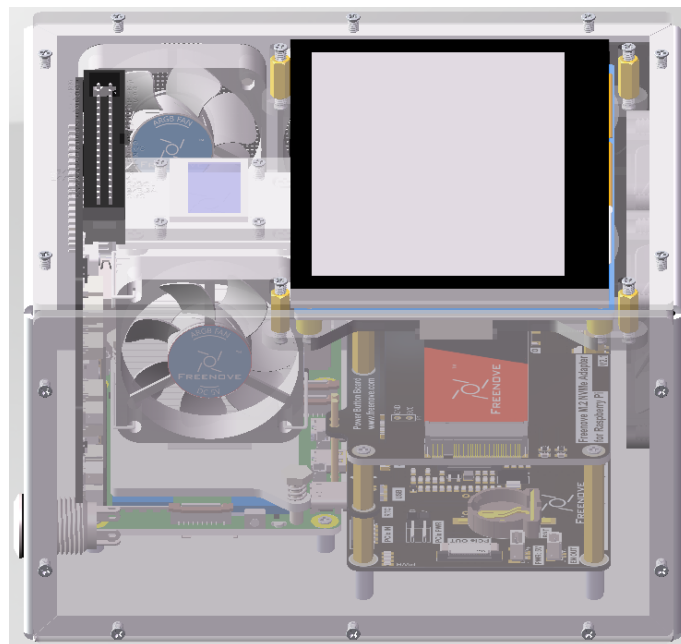
Note: Owing to the machining precision limitations of sheet metal parts, if you notice that certain screw holes do not align properly, this is not an indication of an error. In such cases, you can apply force to bend and align them as needed.



Step 6: Secure the side acrylic plate with M2.5x5 countersunk head screws.

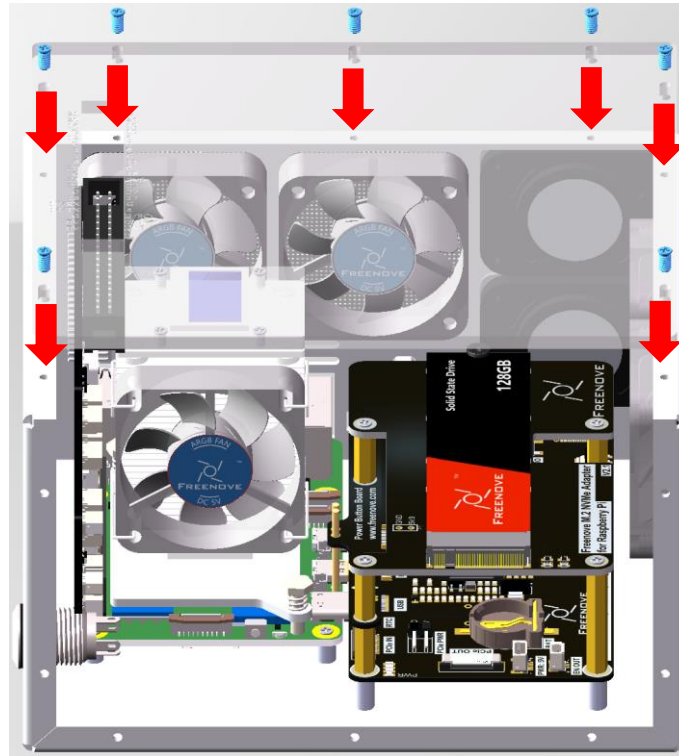


Now, you have finished assembling the case with a landscape-oriented display.

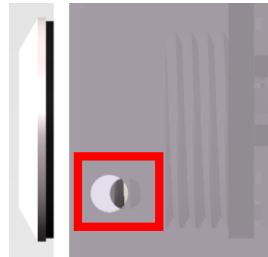


Portrait Mounting

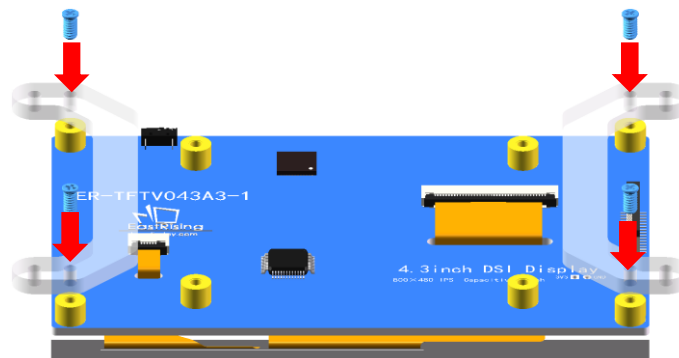
Step 1: Align the top acrylic platel with the GPIO headers, and secure it using M2.5x5 countersunk head screws.



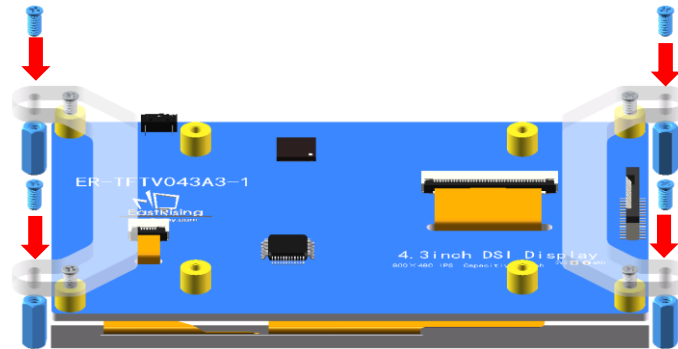
Note: Owing to the machining precision limitations of sheet metal parts, if you notice that certain screw holes do not align properly, this is not an indication of an error. In such cases, you can apply force to bend and align them as needed.



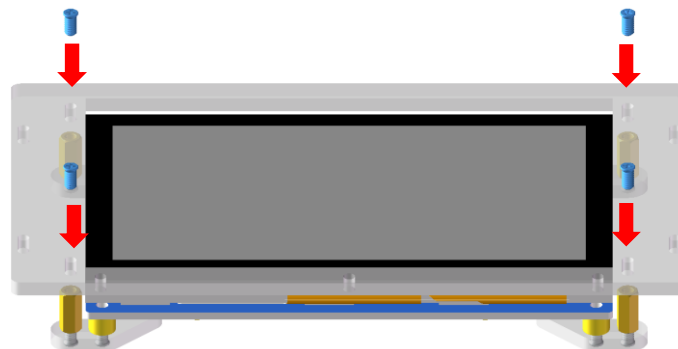
Step 2: Secure the **portrait display bracket** to the back of the 4.3-inch display using M2.5x5 countersunk head screws. The fixing holes are the **inner set of holes** on the bracket.



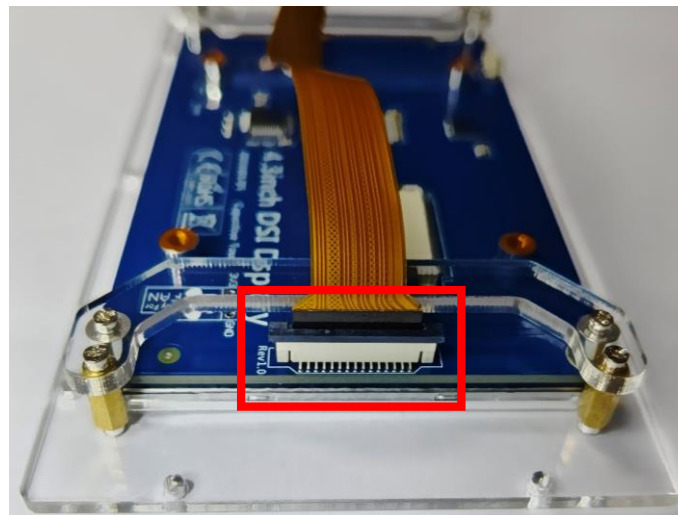
Step 3: Secure the M2.5x9 brass standoff to the portrait display bracket using M2.5x5 countersunk head screws. Use the **outer set of holes** on the bracket.



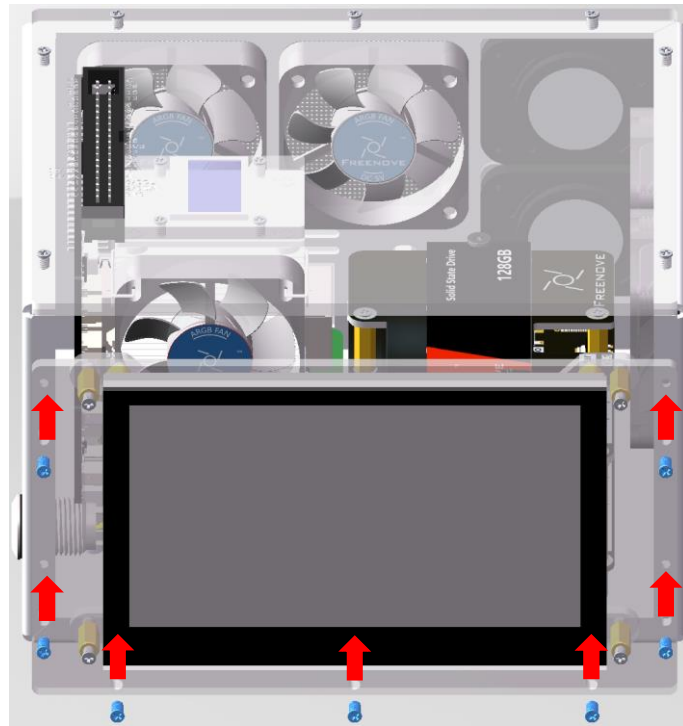
Step 4: Align the pre-assembled 4.3-inch screen with the mounting holes on the top acrylic plate, and secure it using M2.5x5 countersunk head screws.



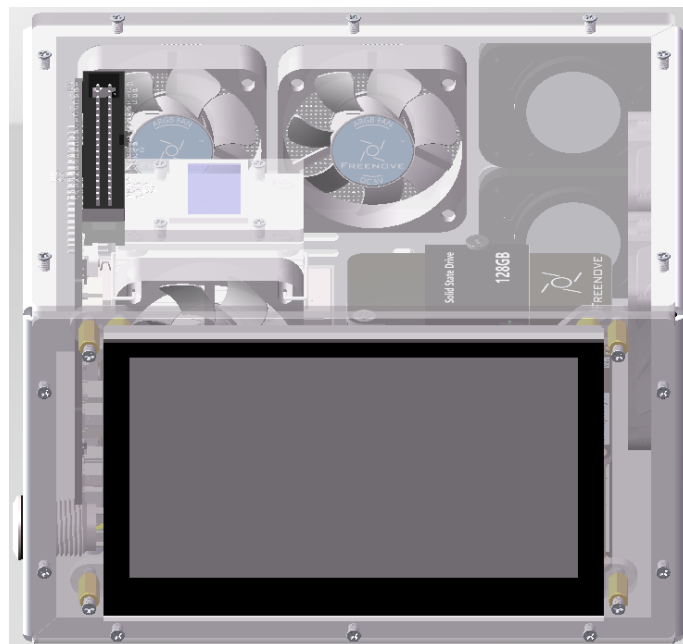
Step 5: Connect the other end of the 160mm DSI FPC cable to the connector on the back of the 4.3-inch screen, ensuring the contacts are facing inward toward the display. **Caution: Open the FPC latch gently to prevent damage that may cause loosening of the FPC cable connection.**



Step 6: Secure the side acrylic plate with M2.5x5 countersunk head screws.

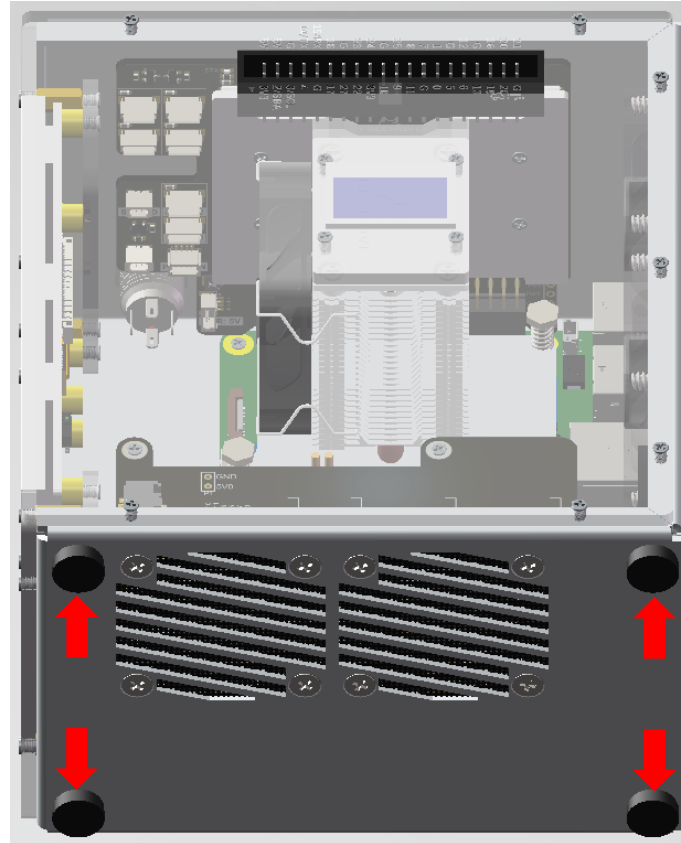


Now, you have finished assembling the case with a portrait-oriented display



2.3.3 Attaching Non-Slip Foot Pad

Peel off the white backing from the 3M adhesive on the black non-slip feet and apply one to each of the four corners on the bottom of the case.

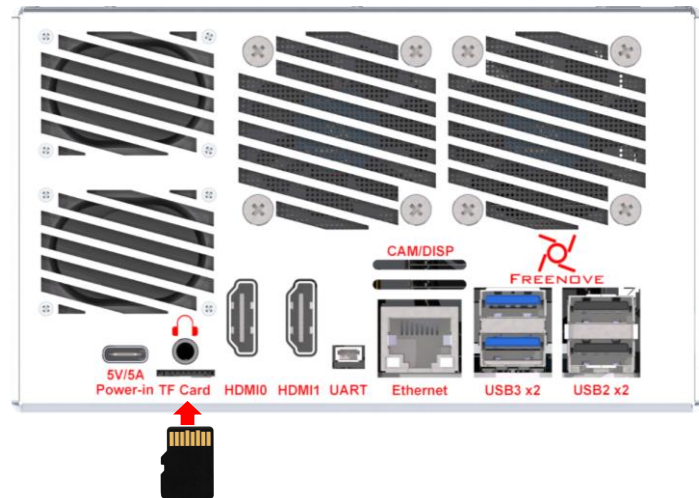


Chapter 3 APP Control

Before powering on the Freenove Computer Case Kit Pro for Raspberry Pi, please make sure that all cable connections are correct.

Due to its multiple functions, this case requires an adequate power supply. We highly recommend using the official Raspberry Pi 5.1V / 5A power adapter (<https://www.raspberrypi.com/products/27w-power-supply>). Failure to do so may result in the Freenove Computer Case Kit Pro for Raspberry Pi being unusable or causing damage to components.

How to Insert and Take out SD Card



Remove the SD card with the written system from the card reader.

Insert it into the self-ejecting TF card slot on the back of the case **with the gold contacts facing up**.

Push the SD card in until you hear a click, indicating it is securely locked in place.

To eject the SD card, press it in again until you hear a click, and the card will pop out.

Installing Raspberry Pi OS

If you have not yet installed the Raspberry Pi OS, please refer to the documentation “[Installing Raspberry Pi OS.pdf](#)” under the directory **Freenove Computer Case Kit Pro for Raspberry Pi** to install the OS to your SD card or SSD.

Important note for users of the **FNK0107B/C/K/L/Q/R/V/W** models: due to hardware limitations of the Dual-NVMe and Quad-NVMe Adapter Boards, please **do not** enable **PCIe 3.0** mode, as this may cause the Raspberry Pi to fail to boot.

If you have a display ready, please continue reading the tutorial below.

3.1 Boot Behavior & Environment Settings

3.1.1 What to Expect on First Startup

When you first power on the assembled chassis, the Raspberry Pi does not talk to the GPIO adapter board. This causes the board to operate on its own in a default mode, and you will observe the following.



1. The case RGB lights will operate in rainbow mode.
2. During initial startup, the fan will run at maximum speed for 3 seconds.
After this period, the system switches to temperature-controlled mode, where fan speed is regulated by the thermal sensor on the GPIO adapter board
3. The screen will remain off during this phase - this is normal and expected.

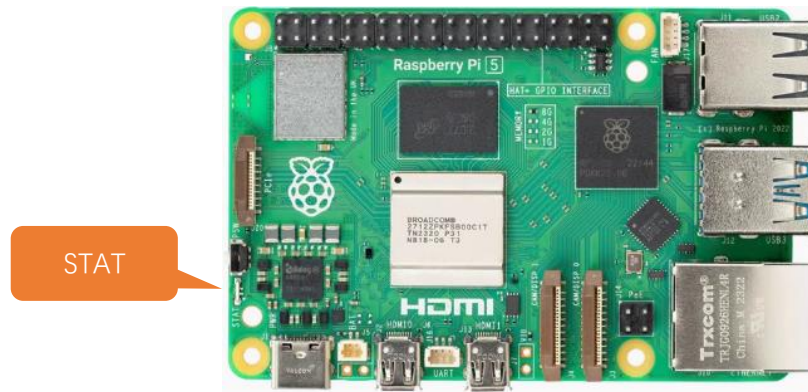
Troubleshooting Guide:

If the RGB lights fail to illuminate, or if the fan doesn't perform the 3-second full-speed initialization, please: Immediately power off the system

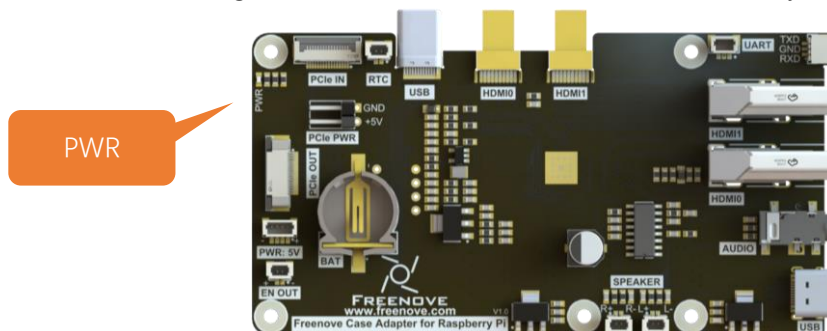
- Verify proper alignment and connection between the GPIO adapter board and Raspberry Pi
- Check all relevant power and data connections

If you have any questions of the above, please contact us at support@freenove.com

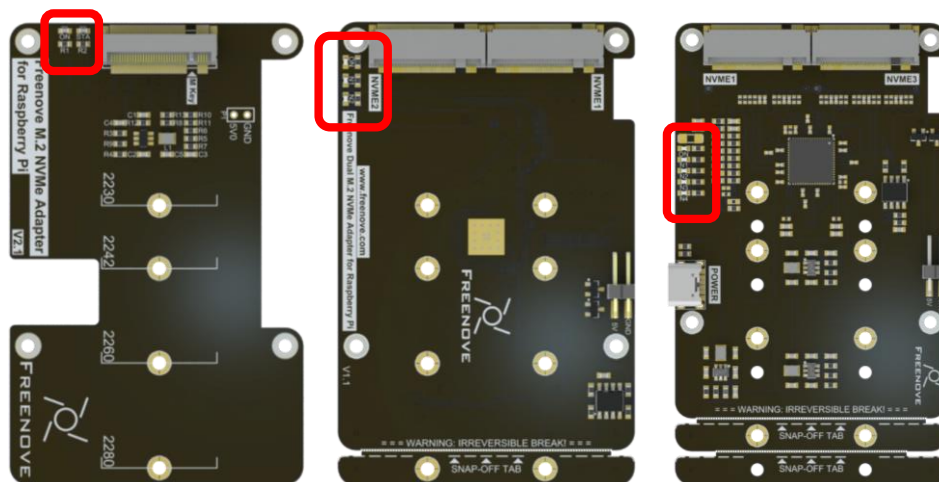
4. RPi 5 Status LED: The green STAT LED will remain steadily illuminated. If this LED is not lit or displays any pattern other than a steady green light, it indicates that the Raspberry Pi operating system has not booted successfully. In this case, please check your Raspberry Pi hardware and OS installation separately.



5. The PWR indicator light on the Audio-Video Board remains steadily lit.



6. NVMe Adapter Board Indicators:
 ON LED: Steadily lit.
 STA LED: Blinks in sync with the SSD's built-in activity light.



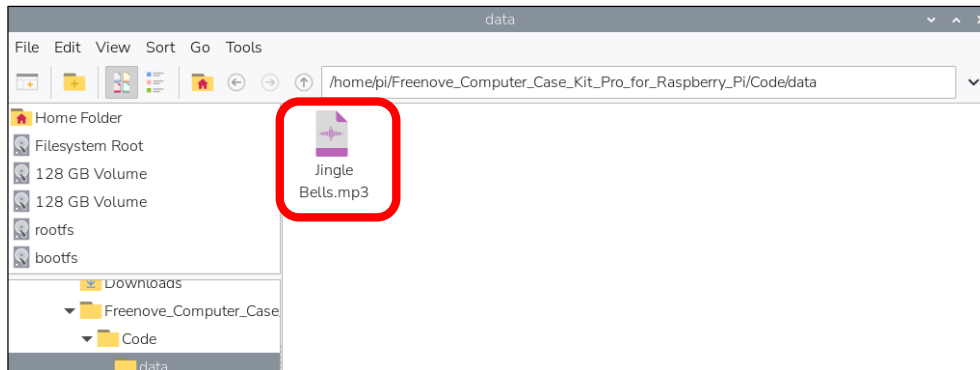
Should the ON indicator fail to illuminate, please check the cable connection.

Important: It is imperative to utilize the onboard pin header for supplemental power when operating multiple solid-state drives to mitigate risks associated with inadequate power supply.

If all components behave as expected, then your computer case is correctly assembled and functioning well. In this case, you can connect a screen to your Raspberry Pi or access it via VNC viewer.

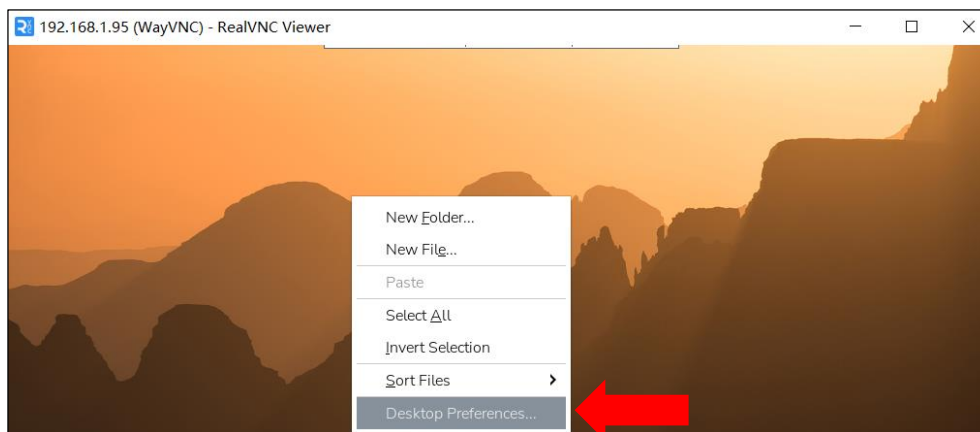
Speaker Test

Open the data folder under the Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/data directory, and click the mp3 file to display.

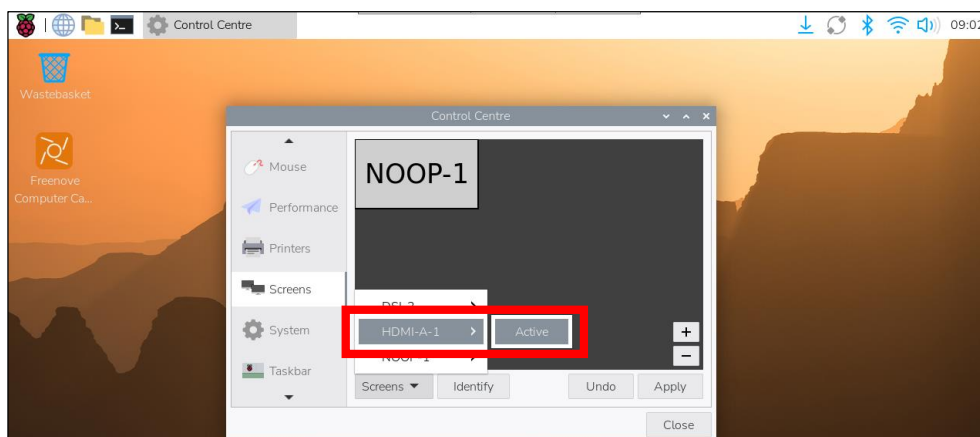


If you have enabled Advanced Options -> A7 Wayland -> W1 X11 in **raspi-config** and experience abnormal audio output, please re-enable HDMI to restore normal audio function.

Right-click in any blank area upon the desktop and choose "**Desktop Preferences...**"



In the pop-up Control Centre window, select "**Screens**" from the left pane. Select Screens → HDMI-A-1 → Active to reactivate HDMI.



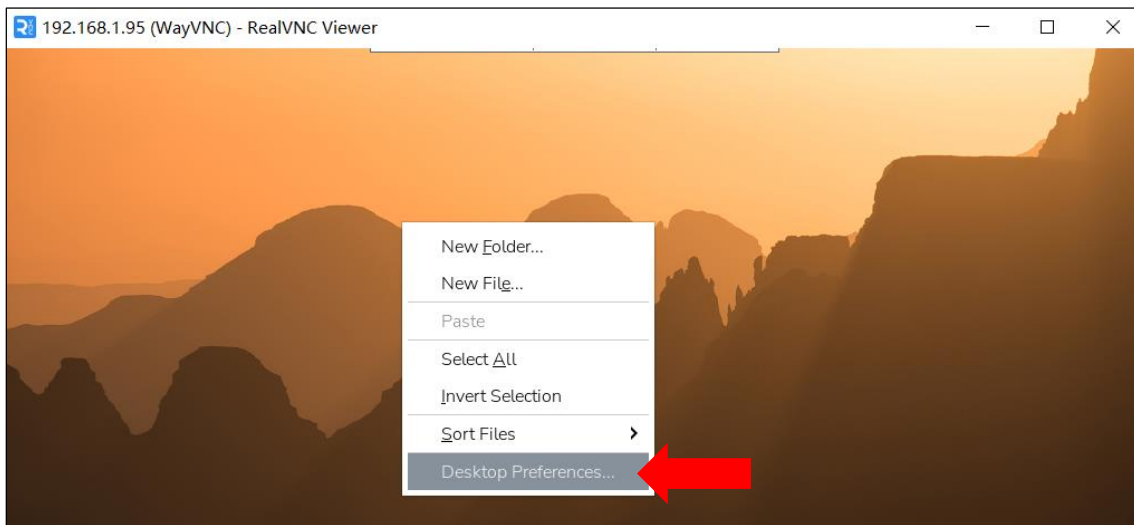
If you have any questions of the above, please contact us at support@freenove.com

3.1.2 Switching Taskbar Display

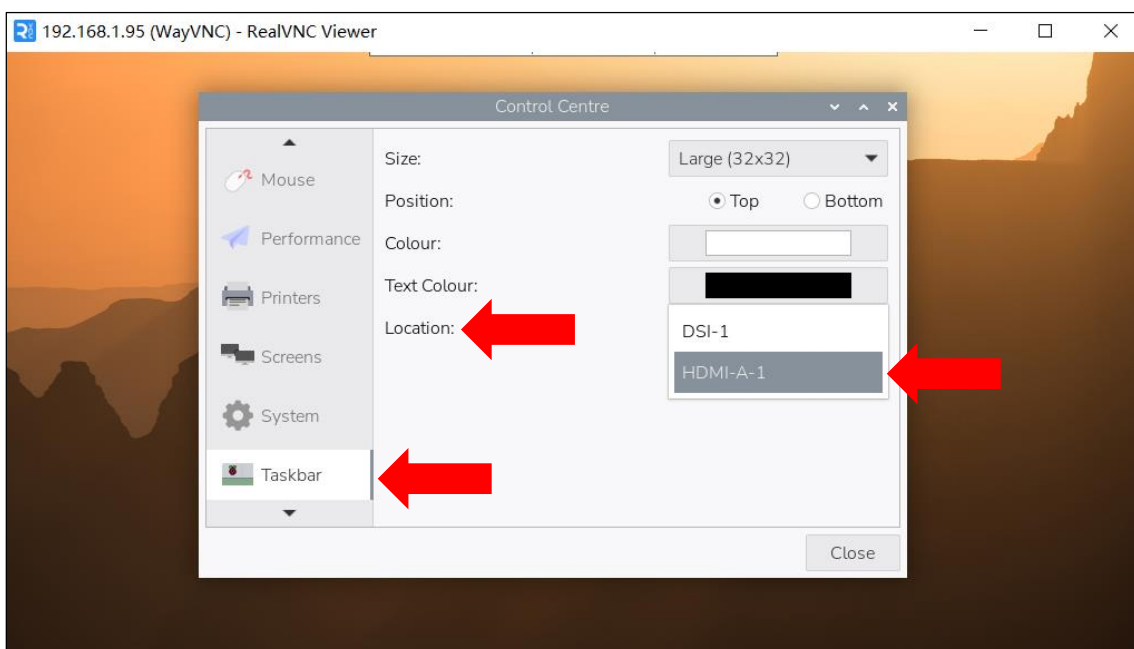
If your kit is a non-display version (**Model numbers: FNK0107A/B/C/H/K/L**), you can [skip this section by clicking here](#).

For versions equipped with a 4.3-inch IPS screen (**Model numbers: FNK0107P/Q/R/U/V/W**), after booting the RPi 5, the system will create two independent display areas: DSI-1 and HDMI-A-1. When connecting via VNC, you are viewing a mirrored display of HDMI-A-1. Since the system status bar is prioritized for display on the DSI screen, it is not included by default in the VNC interface. Please follow the instructions below to display the status bar in the HDMI-A-1 area.

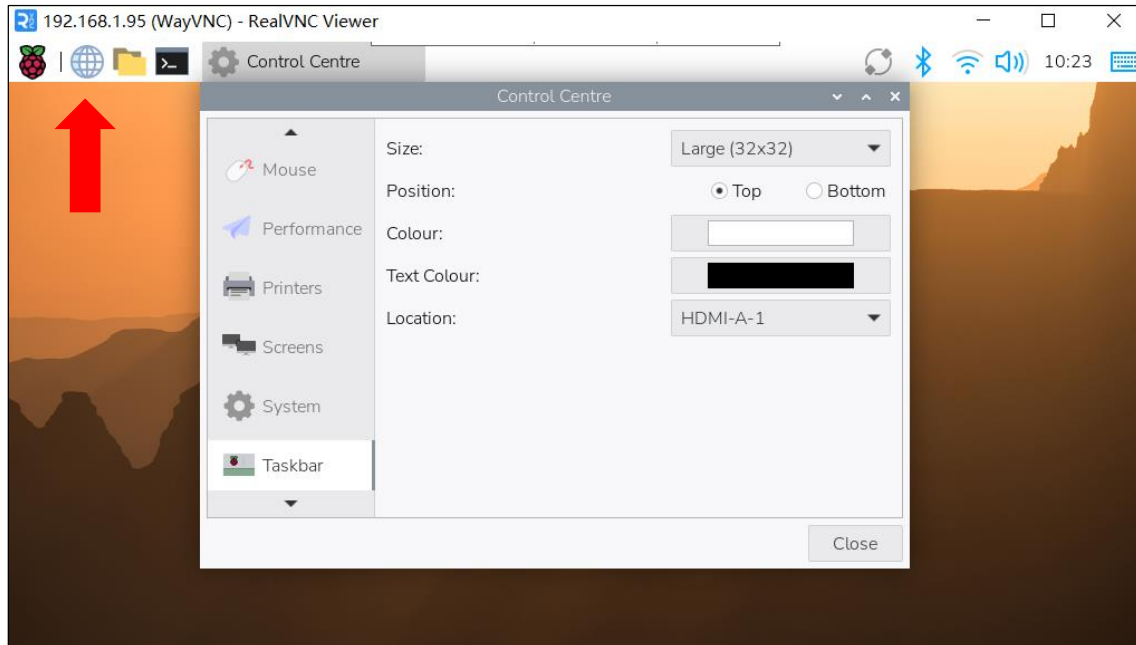
Right-click in any blank area upon the desktop and choose "**Desktop Preferences...**".



In the pop-up Control Centre window, select "**Taskbar**" from the left pane. Find the "**Location**" setting and change it from **DSI-1** to **HDMI-A-1**.



Once configured, the system status bar will switch over from the DSI-1 display area to the HDMI-A-1 display area.



For known system issues in VNC, such as incomplete display or misaligned mouse clicks after resolution switching, please refer to the "[FNK0107-troubleshooting](#)" document located in the same directory.

3.1.3 Software Setup

Code downloading

Open the Raspberry Pi Terminal, type the following two commands one by one to download the code for the case.

```
cd
git clone
https://github.com/Freenove/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi.git

pi@raspberrypi:~ $ cd
pi@raspberrypi:~ $ git clone https://github.com/Freenove/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi.git
Cloning into 'Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi'...
remote: Enumerating objects: 45, done.
remote: Counting objects: 100% (45/45), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 45 (delta 2), reused 45 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (45/45), 9.14 MiB | 3.36 MiB/s, done.
Resolving deltas: 100% (2/2), done.
```

Software Packages Update

Run the following command on the terminal to update your Raspberry Pi's package list to the latest version. In the latest Raspberry Pi 13 (Trixie), passwordless sudo is officially disabled. You will need to enter your user password every time you run a sudo command. Please enter your user password when prompted. (Note: The default password is "raspberrypi", please replace it with your user password.)

```
sudo apt update

pi@raspberrypi:~ $ sudo apt update
[sudo] password for pi:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://archive.raspberrypi.com/debian trixie InRelease
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
127 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

OLED Library Installation (Important)

Run the following command to install the OLED library. Fail to do so will result in software malfunction.

```
sudo apt install python3-luma.oled

pi@raspberrypi:~ $ sudo apt install python3-luma.oled
Installing:
  python3-luma.oled

Installing dependencies:
  python3-cbor2 python3-ftdi python3-luma.core python3-usb

Suggested packages:
  python-cbor2-doc
```

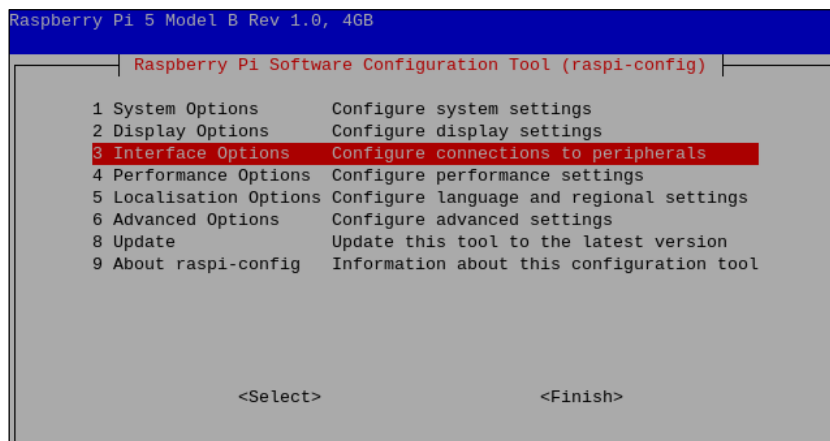
Enable I2C (Important)

The I2C function must be enabled, as it is required for communication between the Raspberry Pi, the OLED display, and the GPIO adapter board. Without it, the chassis software will fail to operate.

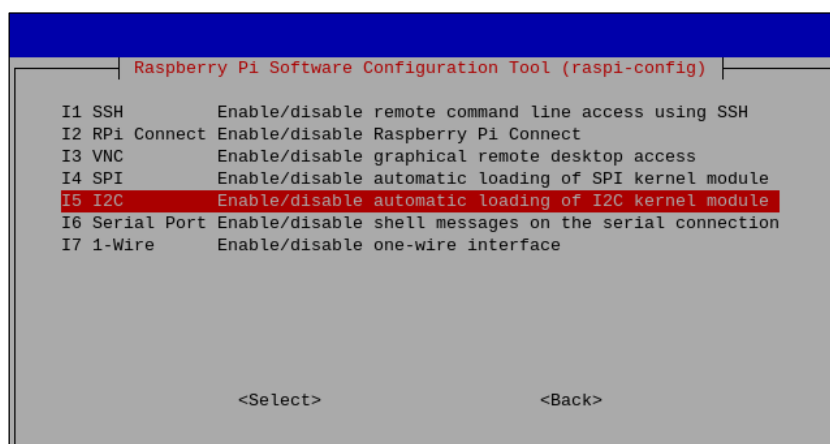
Run the following command to open the Raspberry Pi configuration interface.

```
sudo raspi-config
```

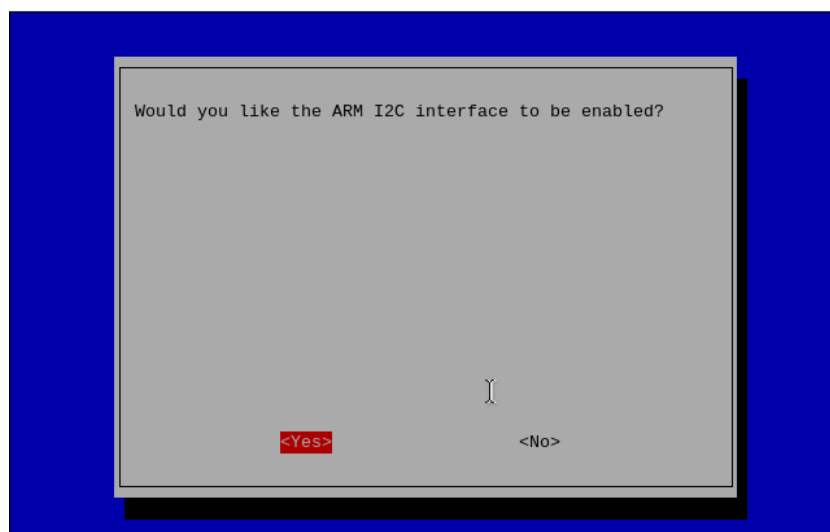
Navigate to Interface Options and press Enter.



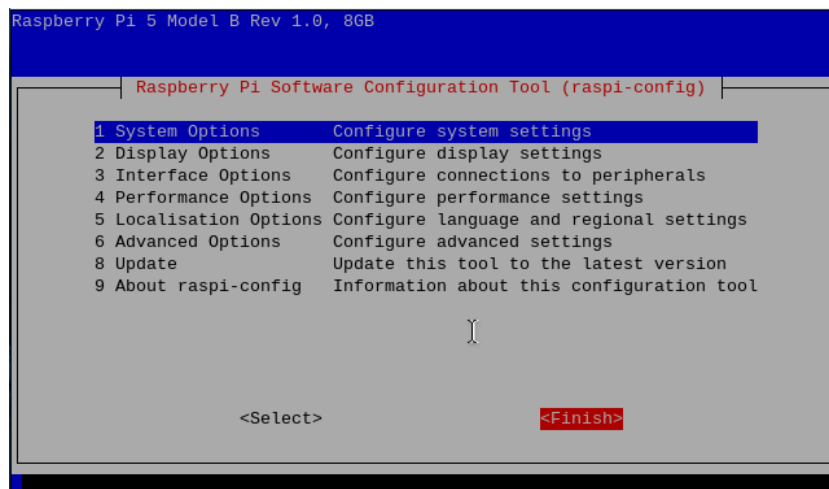
Select I2C and press Enter to enable it.



Select Yes, and press Enter.



Select Finish to save the change and exit the configuration interface.

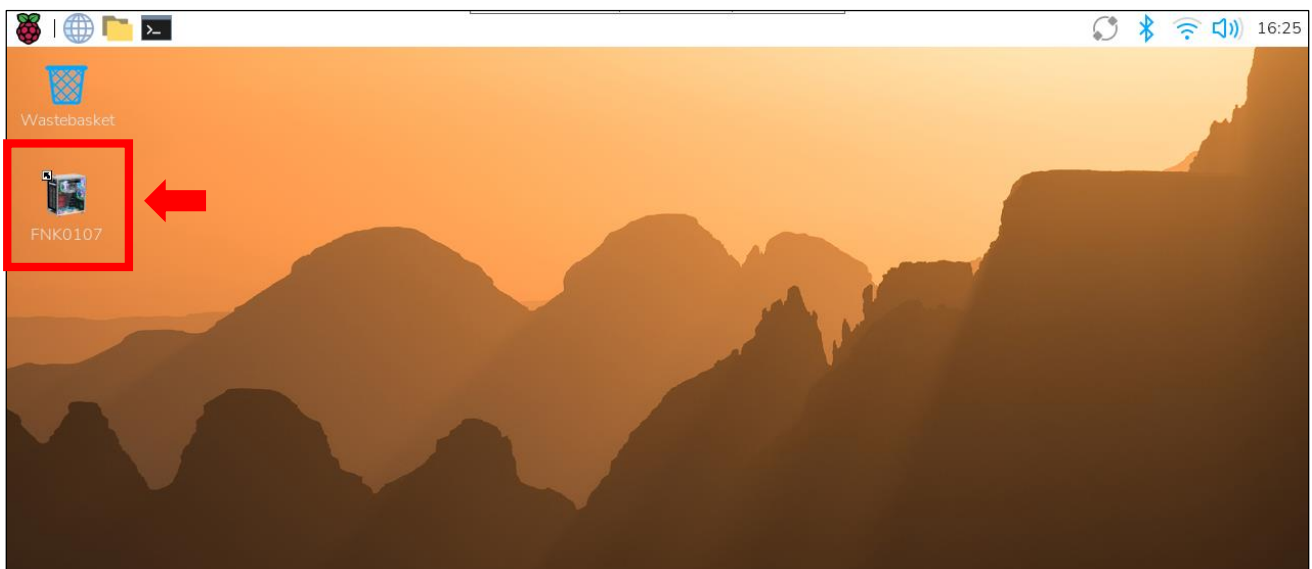


Creating Desktop Shortcut

Run the following two commands one by one in the Raspberry Pi terminal to create a desktop shortcut for the case control software.

```
cd Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/
sudo python create_desktop_shortcut.py
```

```
pi@raspberrypi:~/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code $ sudo python create_desktop_shortcut.py
Current directory: /home/pi/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code
Created new desktop file directly on desktop
Desktop file and script set as executable
Desktop shortcut created successfully!
```

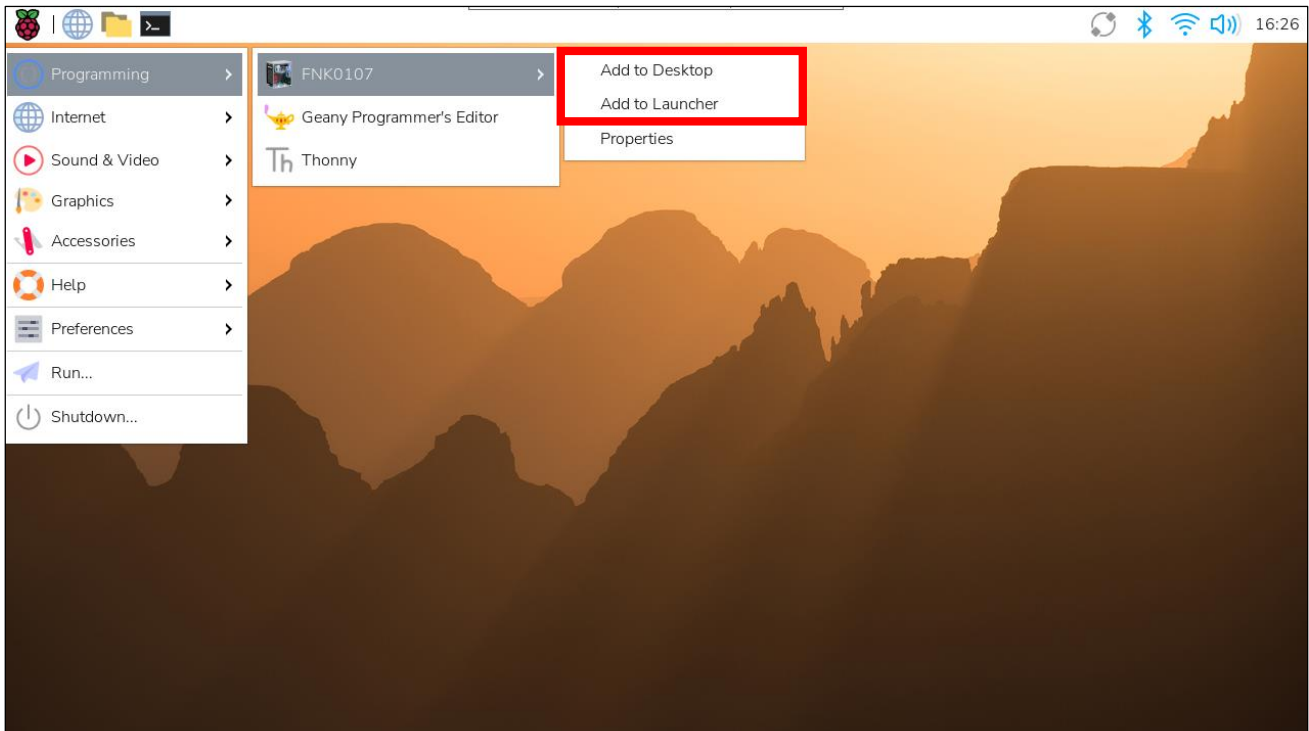


The icon above corresponds to the command shown below.

```
cd Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/
python app_ui.py
```

```
pi@raspberrypi:~ $ cd Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/
pi@raspberrypi:~/Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code $ python app_ui.py
```


After completing this step, you will find the application in the “**Programming**” section of the menu. If you accidentally delete the desktop shortcut or wish to add the app to the top launcher, simply right-click “**FNK0107**” and select “**Add to Desktop**” or “**Add to Launcher**”.



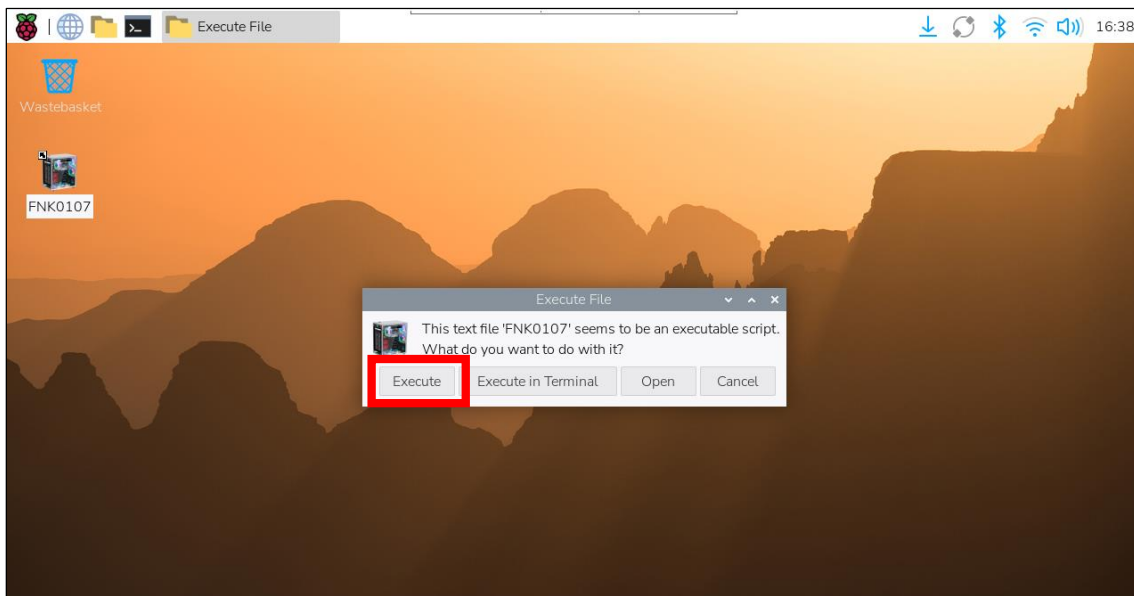
If you are interested in the code implementation, you can explore the files in the [Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code](#) directory. Should you wish to modify the code, please ensure you back it up first to avoid potential software malfunctions caused by unintended changes.

3.2 About the Case Control Software

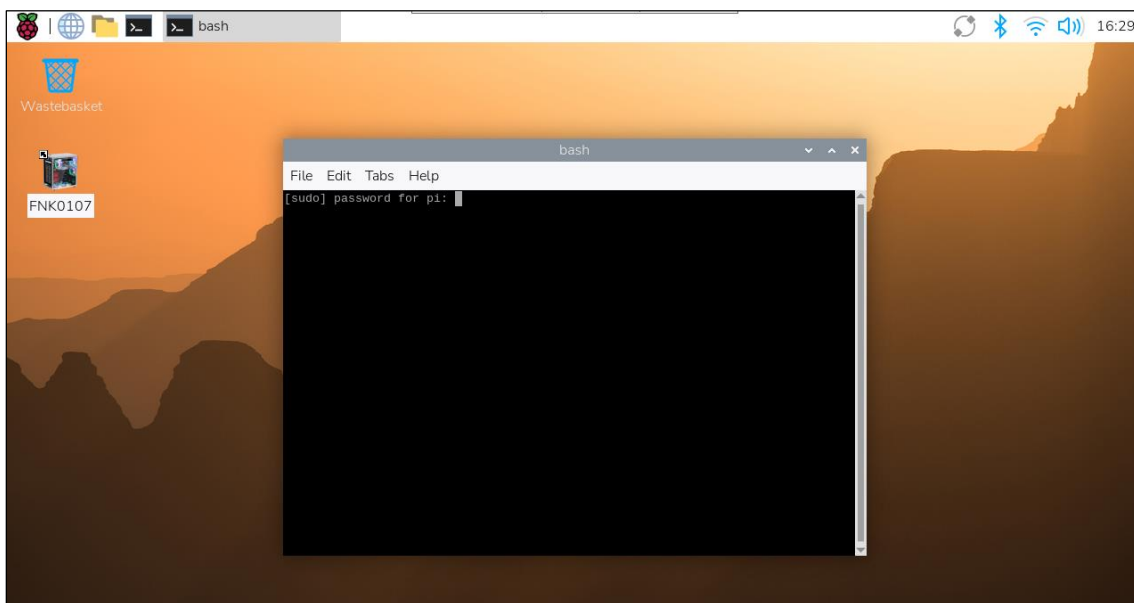
With the environment configured from the previous chapter, the accompanying host software can now be used to manage case functions including ARGB lighting, the OLED display, and fan control.

This chapter provides a detailed guide on the software's usage. For insight into the interface design, the `app_ui.py` file is available in the `Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code` directory.

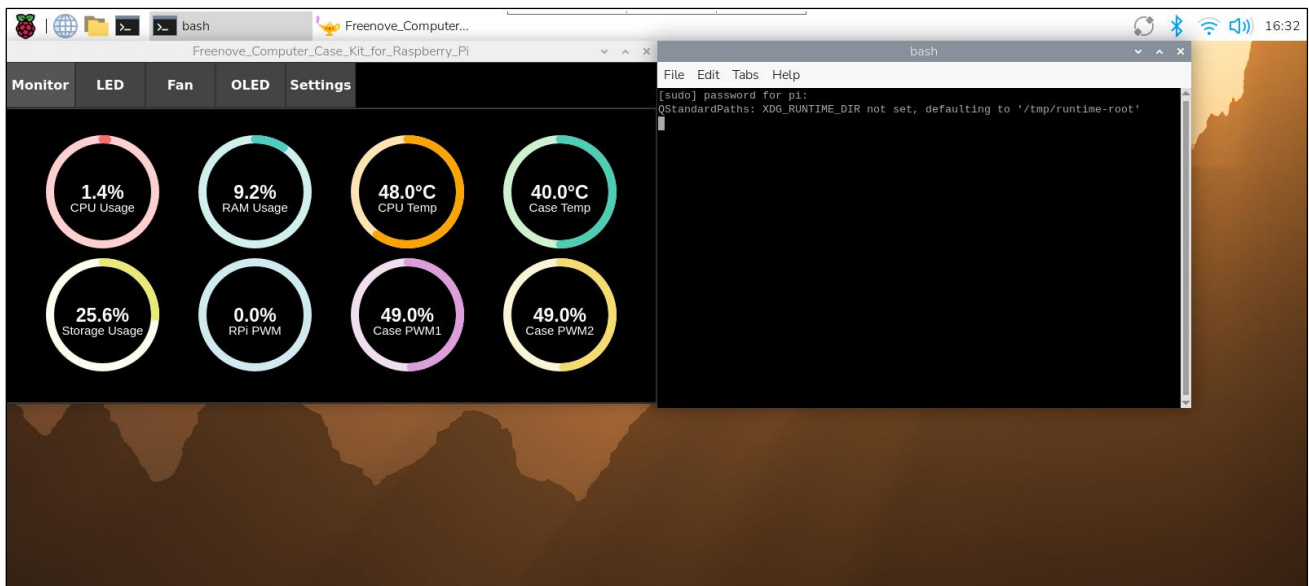
Double click the software with Freenove logo on RPi's desktop, and a window will pop up. Click "**Execute**" to run the program.



Since passwordless sudo is disabled in Trixie, you will also be asked for your password again after reboot. Enter your user password in the pop-up terminal window.

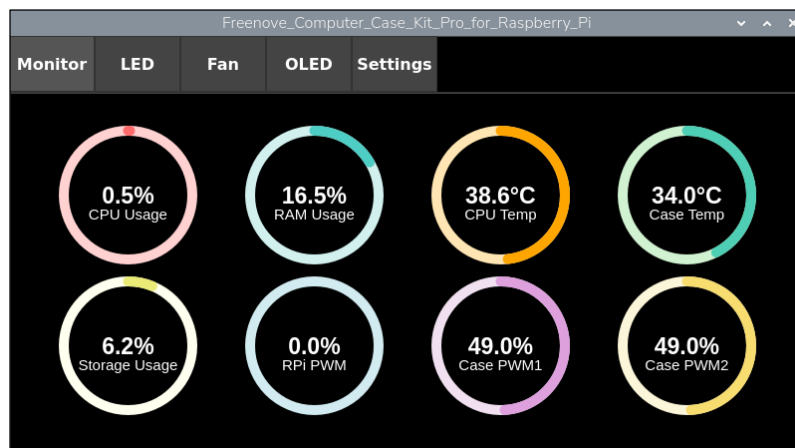


The software interface is as shown below. The left window is the main software interface, and the right window displays debugging information.



3.2.1 Dashboard Monitoring

The dashboard provides live monitoring of key RPi 5 and case component stats, giving you an at-a-glance view of the system status.



Below is a detailed explanation of the parameters displayed on the monitoring interface:

- **CPU Usage:** Current processor utilization percentage of the Raspberry Pi 5
- **RAM Usage:** Current system memory utilization percentage of the Raspberry Pi 5
- **CPU TEMP:** The internal temperature of the Raspberry Pi 5's SoC (System on Chip)
- **Case Temp:** The temperature inside the chassis, provided by the temperature sensor on the GPIO Board
- **Storage Usage:** The total storage space utilization. This value reflects the usage of the SD card if one is used. If single or multiple SSDs are installed, it calculates the aggregate usage across all drives.
- **RPi PWM:** PWM duty cycle for the Raspberry Pi's active cooler fan
- **Case PWM1:** PWM duty cycle for case fan group 1 (FAN1/FAN2)
- **Case PWM2:** PWM duty cycle for case fan group 2 (FAN3/FAN4)

Data Source & Troubleshooting:

CPU Usage, RAM Usage, CPU Temp, Storage Usage, and RPi PWM are retrieved directly from the Raspberry Pi. If any of these values are missing, check the Raspberry Pi for faults.

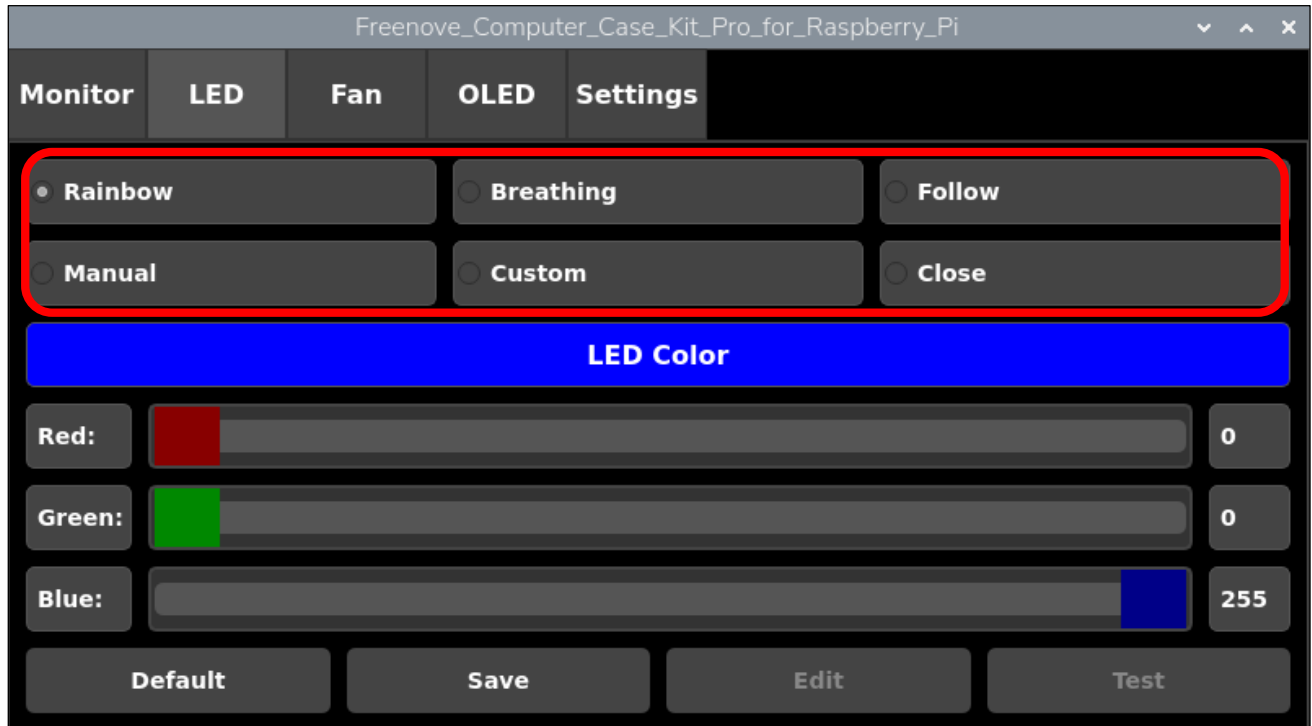
Case Temp, Case PWM1, and Case PWM2 are read from the GPIO Board via I2C communication with Raspberry Pi 5.

If these data cannot be obtained consistently, please contact us by email: support@freenove.com

For those interested in the interface implementation, please refer to the files `api_systemInfo.py` and `api_expansion.py` in the `Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code` directory.

3.2.2 LED Control Interface

This is the control interface for the case ARGB lights. You can select different modes to display various lighting effects.

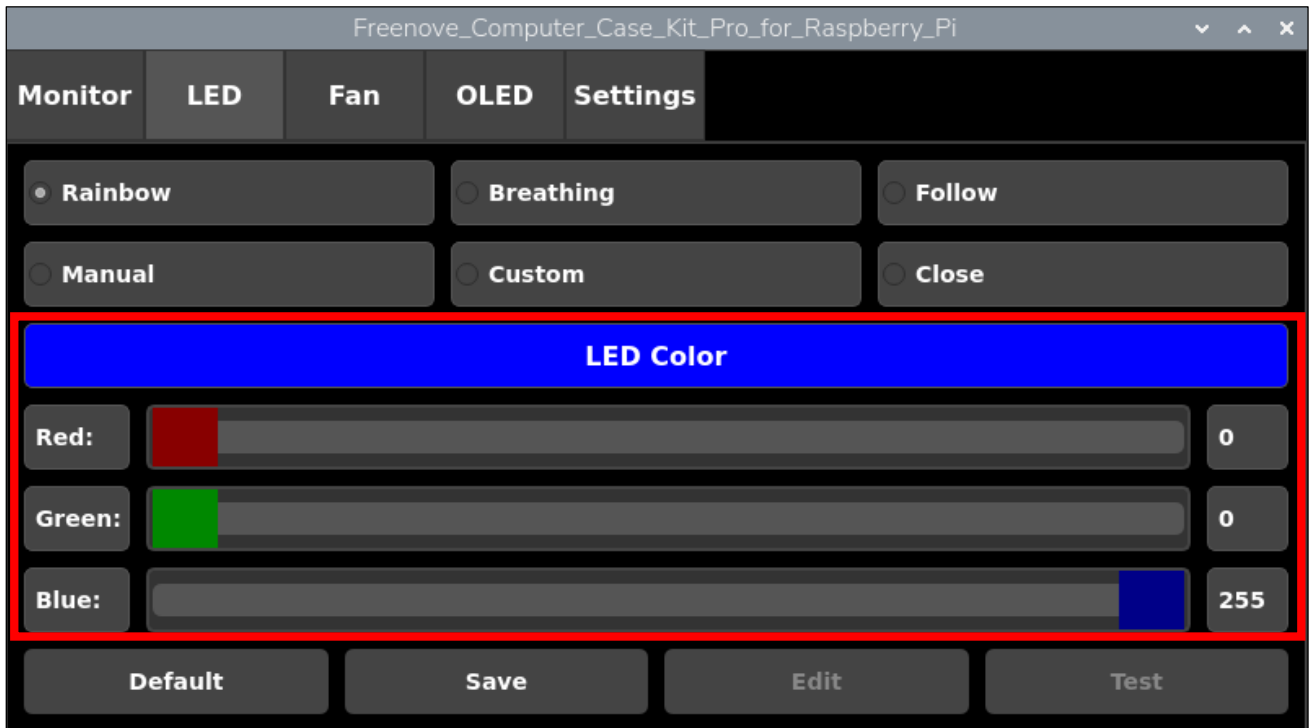


It is set to rainbow mode by default, as shown below.



There are four preset lighting modes available: **Rainbow**、**Breathing**、**Follow**、**Manual**. You can select the corresponding option to switch among the modes.

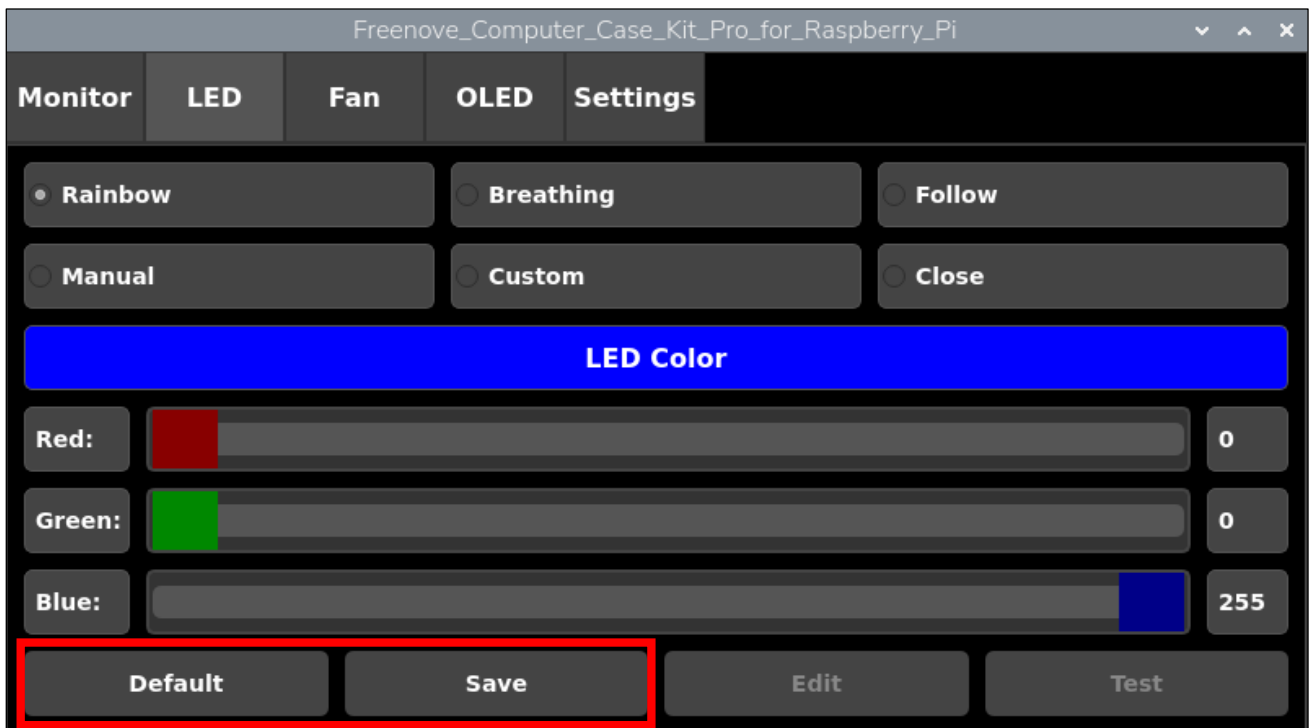
Note: Only in **Breathing**, **Follow**, and **Manual** modes can the color of the RGB lights be controlled using the slider below. In other modes, the color of the RGB lights cannot be adjusted.



If you do not turn ON the RGB lights, you can select the Close mode to turn off them.

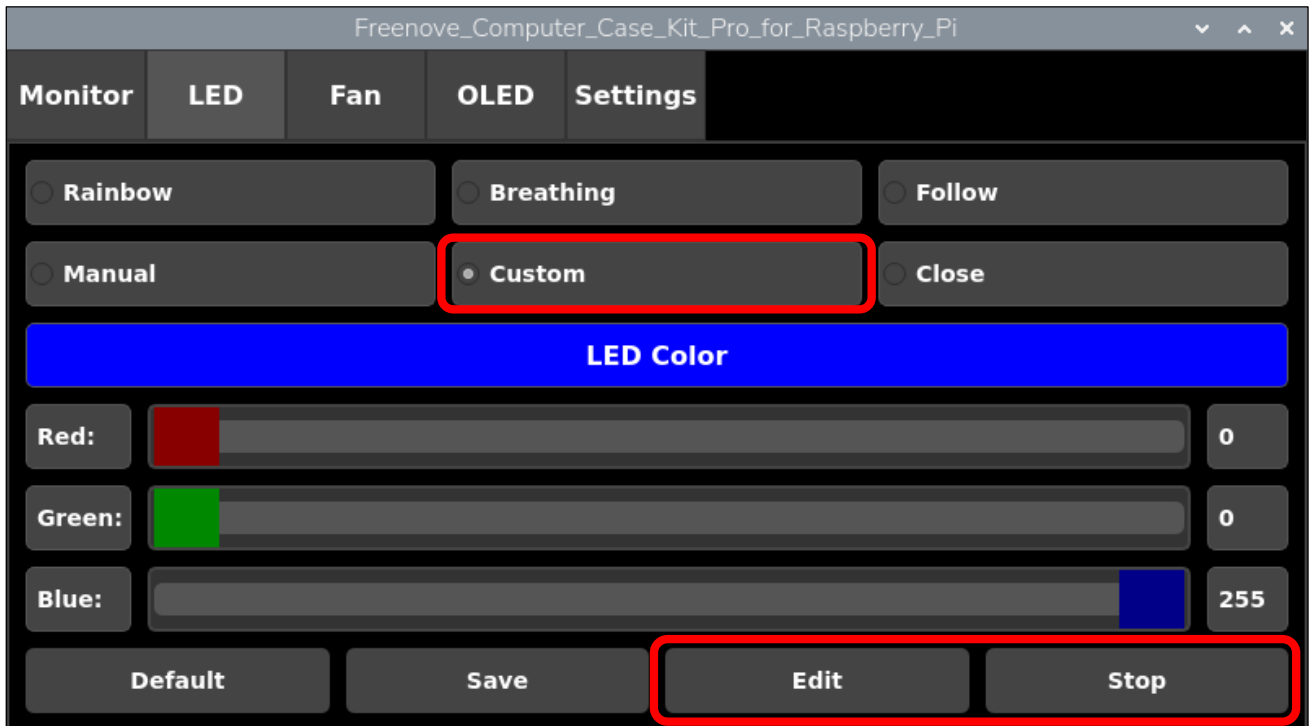
Please note: Any lighting mode you select is temporary. The case will revert to its default mode after a shutdown and restart.

To set a new default mode, select your desired mode and click the "Save" button.



To restore the RGB lights and sliders to their default settings, click the "Default" button once.

If the four preset lighting modes do not meet your needs, you can use the "**Custom**" mode for personalized settings.



Click "**Edit**" will open the "task_led.py" file. You can modify the code in the editor.

Click "**Test**" will create a temporary thread and run task_led.py, allowing you to quickly debug and observe the LED light effects.

You can also view the source code directly:

Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/task_led.py

For more about LED control interface, you can refer to the source file:

Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/api_expansion.py

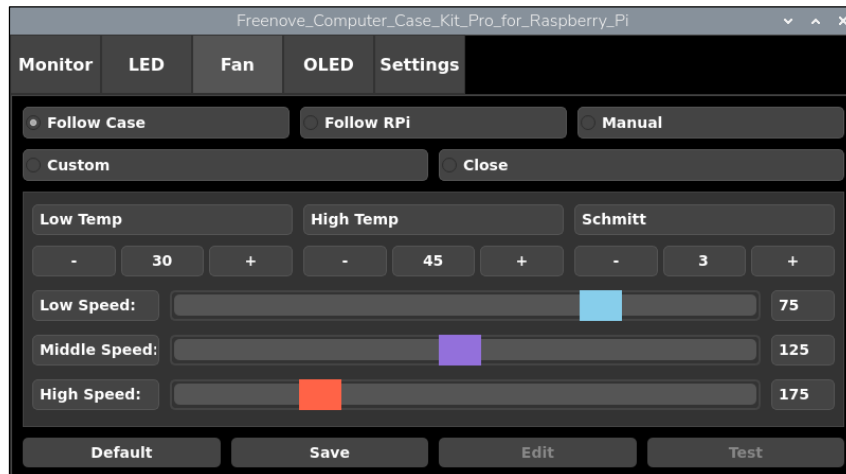
Important Notes:

1. Back up the source code before making any modifications.
2. Custom mode is an advanced feature. Modifying the code in task_led.py requires a basic understanding of Python programming. If you are not familiar with programming, we do not recommend editing this file.
3. The "Test" button only runs the code in the current state. The code will not automatically run after rebooting the Raspberry Pi.
4. If the LED lights show no response after you modify the code and click "Test," it indicates an error in your code preventing it from running properly. Please review your code.
5. If you want the Custom mode code to execute automatically on every startup, please refer to the instructions in the "[3.2.5 Settings Interface](#)" Section.

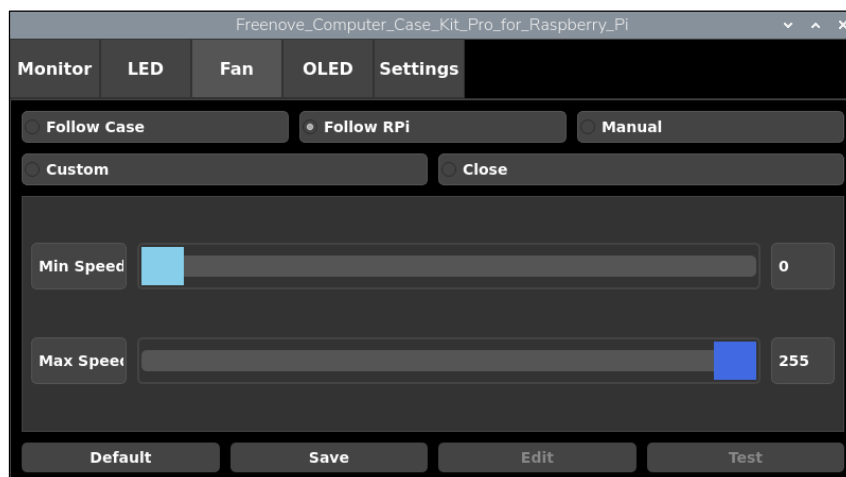
3.2.3 Fan Control Interface

This is the case fan control interface for convenient fan management. The interface will change when you select different modes: Follow Case, Follow RPi, or Manual mode.

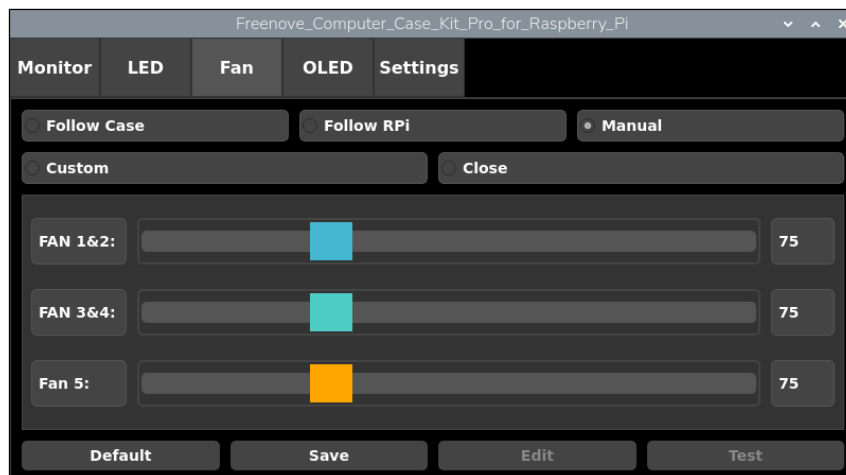
Follow Case Mode



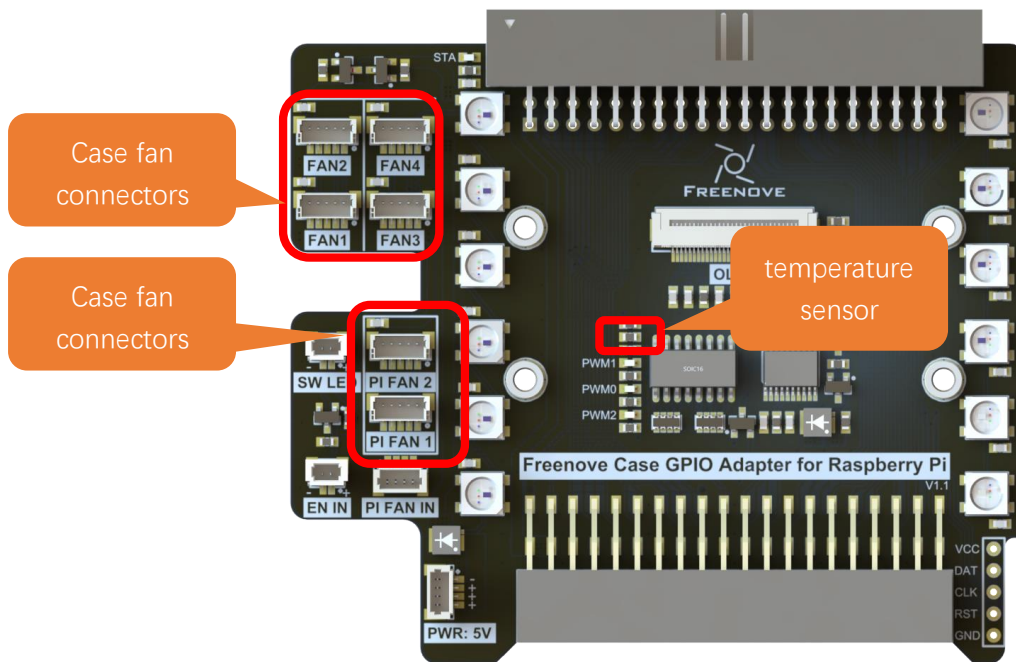
Follow RPi Mode



Manual Mode



The following diagram illustrates the locations of the case fan connectors and the temperature sensor.



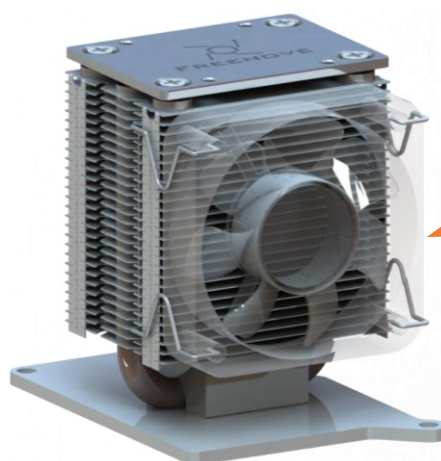
The fan connectors control by GPIO Adapter board are divided into three groups:

- Group 1: FAN1 + FAN2
- Group 2: FAN3 + FAN4
- Group 3: PI FAN 2

For the RPi 5' active cooler, it is recommended that it connect connect to PI FAN 1. In this way, it is controlled by Raspberry Pi.

If connecting to PI FAN 2 connector, it will be controlled by the GPIO Adapter Board.

Note: If the cooling tower fan does not spin, or if all fans fail to spin in Follow RPi mode, please refer to the ["FNK0107-troubleshooting"](#) document located in the same directory.



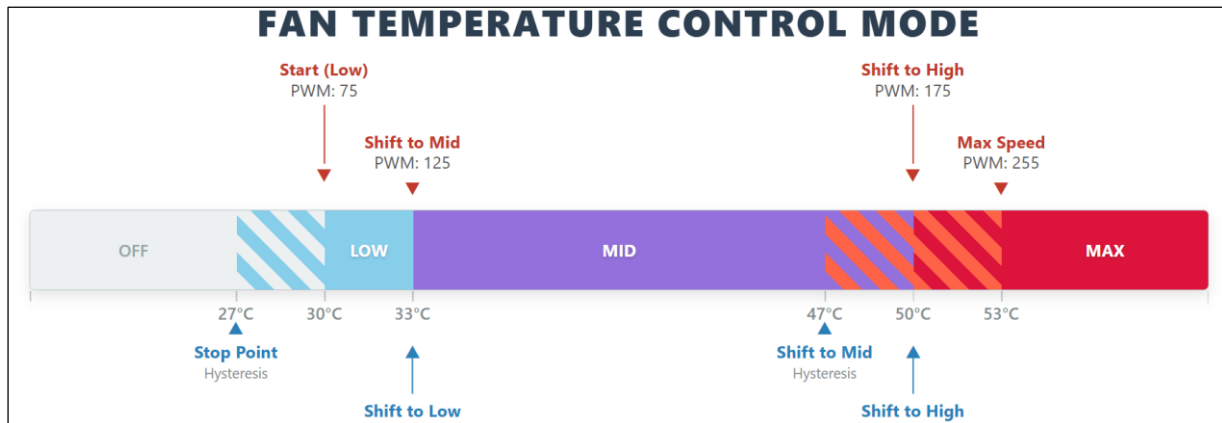
Active Cooler

Next, we will walk you through the different fan modes to help you better understand the case fan control interface.

Follow Case

In Follow Case mode, the GPIO Adapter board samples the temperature sensor at regular intervals and regulates the fan PWM signal based on the readings.

The specific control logic governing this process is depicted in the following figure.



The diagram uses various colors to represent different temperature zones. All values shown are configurable defaults.

- **Heating Phase:**
 - When temperature $\geq$ **Low Temp** (default: 30°C), the fans start at **Low Speed** (default: 75).
 - When temperature $\geq$ **Low Temp + Schmitt** (default: 30 + 3 = 33°C), the fan speed increases to **Middle Speed** (default: 125).
 - When temperature $\geq$ **High Temp** (default: 50°C), the fan speed increases to **High Speed** (default: 175).
 - When temperature $\geq$ **High Temp + Schmitt** (default: 50 + 3 = 53°C), the fan runs at **full speed** (255).
- **Cooling Phase:**
 - When temperature $<$ **High Temp** (default: 50°C), the fan speed drops back to **High Speed** (default: 175).
 - When temperature $<$ **High Temp - Schmitt** (default: 50 - 3 = 47°C), the fan speed decreases to **Middle Speed** (default: 125).
 - When temperature $<$ **Low Temp + Schmitt** (default: 30 + 3 = 33°C), the fan speed decreases to **Low Speed** (default: 75).
 - When temperature $<$ **Low Temp - Schmitt** (default: 30 - 3 = 27°C), the fan stops.

You can configure the following parameters via the software.

Parameter	Value
Low Temp	30
High Temp	50
Schmitt	3
Low Speed	75
Middle Speed	125
High Speed	175

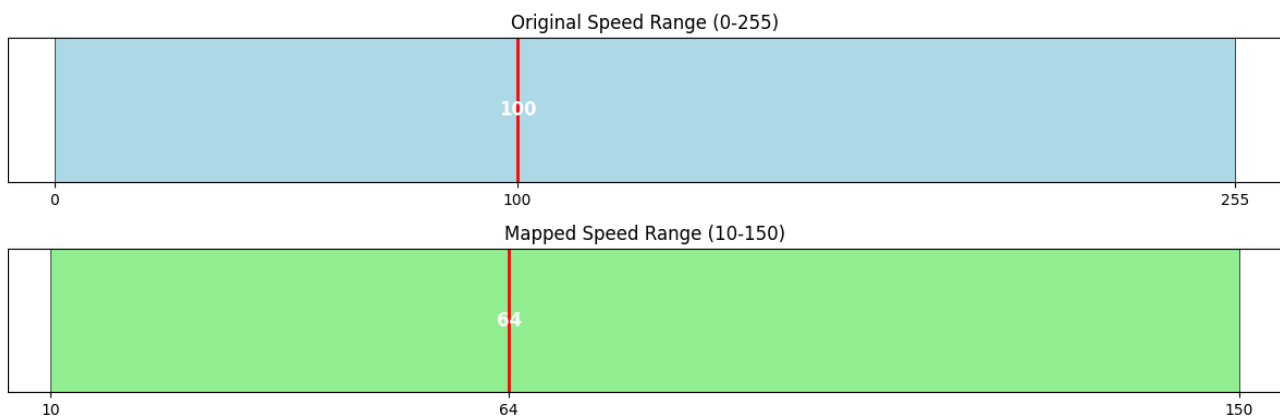
- **Low Temp:** The minimum temperature threshold to activate the case fans. Once reached, the fans begin spinning to accelerate airflow and expel warm air from the case.
- **High Temp:** A higher temperature threshold indicating significant heat buildup. When reached, the fans operate at a higher speed to enable rapid cooling of the system.
- **Schmitt (Hysteresis):** A hysteresis value that prevents the fans from frequently switching on/off or changing speeds near a temperature threshold. It is recommended to keep it at the default value.
- **Low Speed:** The PWM duty cycle applied when the temperature first exceeds the Low Temp threshold. This low speed provides a quiet and power-efficient means of initial cooling.
- **Middle Speed:** The medium PWM duty cycle used when the temperature is between the Low Temp and High Temp thresholds. This speed ensures stable thermal conditions for the Raspberry Pi during sustained workloads, preventing performance fluctuations caused by rapid speed changes.
- **High Speed:** The high PWM duty cycle activated when the temperature exceeds the High Temp threshold. This is critical during heavy computational loads to maximize cooling efficiency, expel heat rapidly, and maintain optimal Raspberry Pi performance.

Follow Rpi

In Follow RPi mode, the GPIO adapter board periodically samples the PWM value from the PI FAN IN interface (which originates from the Raspberry Pi, with a range of 0-255). It then calculates a new PWM value and outputs it to the FAN1-4 and PI FAN 2 connectors.

Calculation Example:

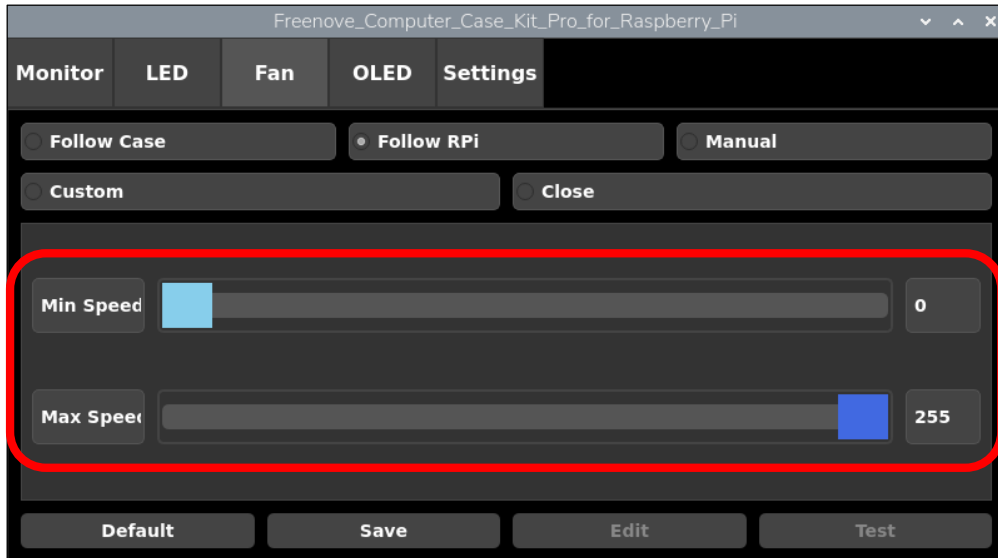
If the sampled PWM value is 100, and the software-configured limits are Min Speed = 10 and Max Speed = 150, the output PWM on the GPIO Adapter will be scaled to 64.



The Calculation formula is shown below:

$$\text{Mapped PWM} = \text{Original PWM} \times \frac{(\text{Max Speed} - \text{Min Speed})}{(255 - 0)} + \text{Min Speed}$$

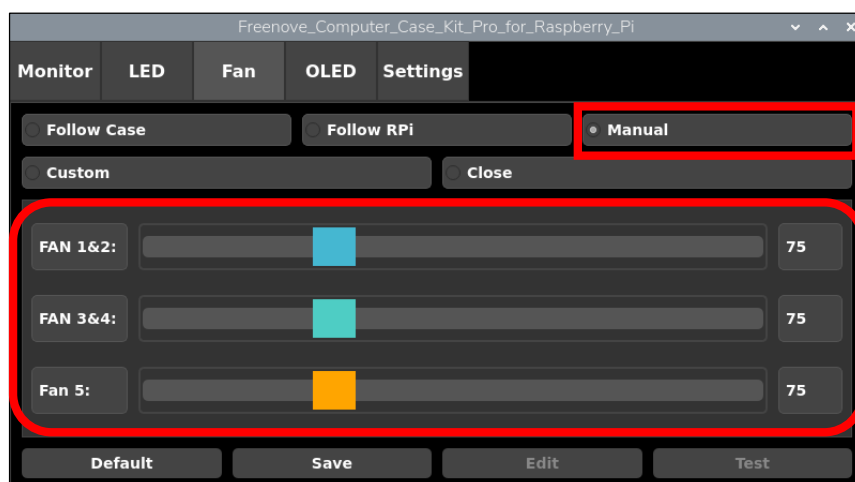
You can configure the Min Speed and Max Speed via the software.



Manual

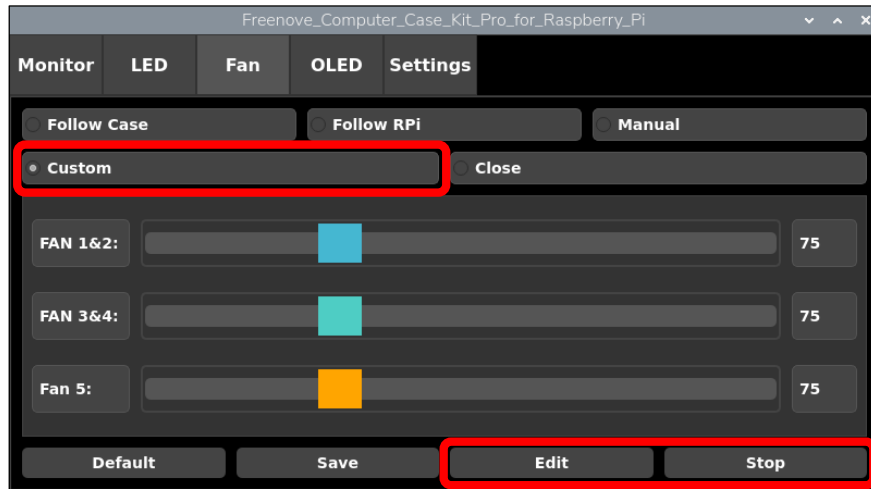
In Manual mode, you can directly set the PWM output for the FAN1-4 and PI FAN 2 connectors via the software.

Note: The slider labeled "**FAN 5**" in the interface controls the **PI FAN 2**. To manually adjust the speed of the Raspberry Pi 5 active cooler, it must be connected to the **PI FAN 2** header.



Custom

If the predefined modes do not meet your needs, you can select the **Custom** mode for advanced configuration.



Click "**Edit**" will open the "task_fan.py" file. You can modify the code in the editor.

Click "**Test**" will create a temporary thread and run task_fan.py, allowing you to quickly debug and observe the fan effects.

You can also view the source code directly:

Freenove_Computer_Case_Kit_Pro_for_Raspberry_Pi/Code/task_fan.py

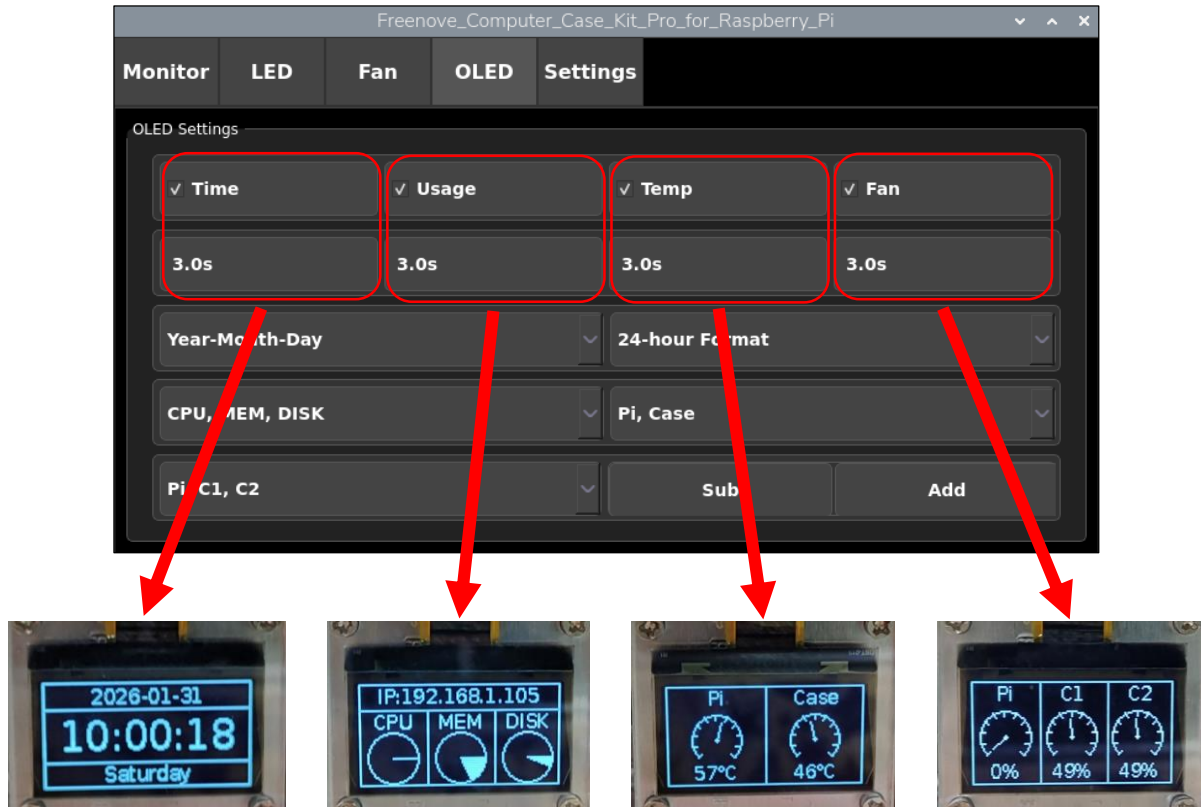
Important Notes:

1. Back up the source code before making any modifications.
2. Custom mode is an advanced feature. Modifying the code in task_fan.py requires a basic understanding of Python programming. If you are not familiar with programming, we do not recommend editing this file.
3. The "Test" button only runs the code in the current state. The code will not automatically run after rebooting the Raspberry Pi.
4. If the fans do no response after you modify the code and click "Test," it indicates an error in your code preventing it from running properly. Please review your code.
5. If you want the Custom mode code to execute automatically on every startup, please refer to the instructions in the ["3.2.5 Settings Interface"](#) Section.

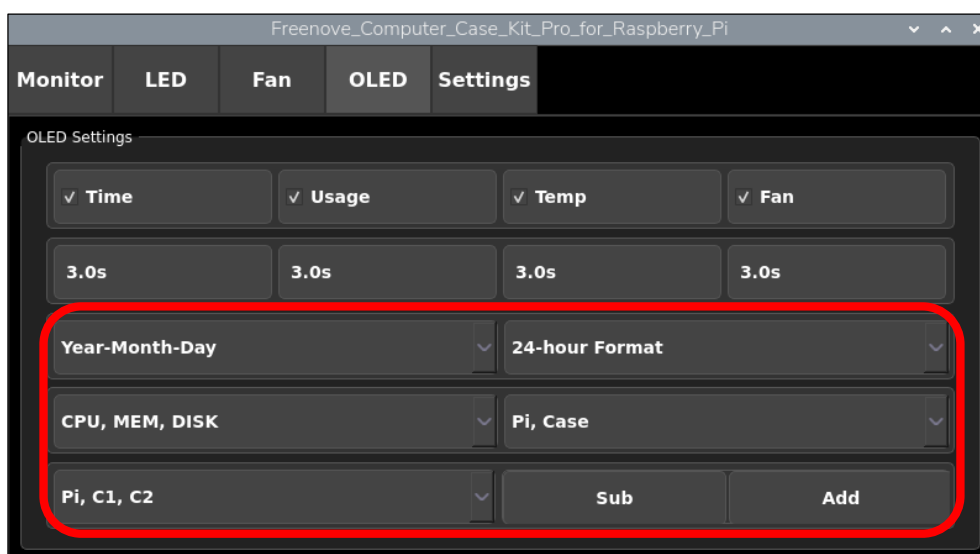
3.2.4 OLED Control Interface

This is the interface for controlling the OLED display content. You can choose from different display layouts according to your preference.

By default, the OLED cycles through four screens: Time, Usage, Temp, and Fan. If you uncheck any of these, that screen will be skipped during the cycle.



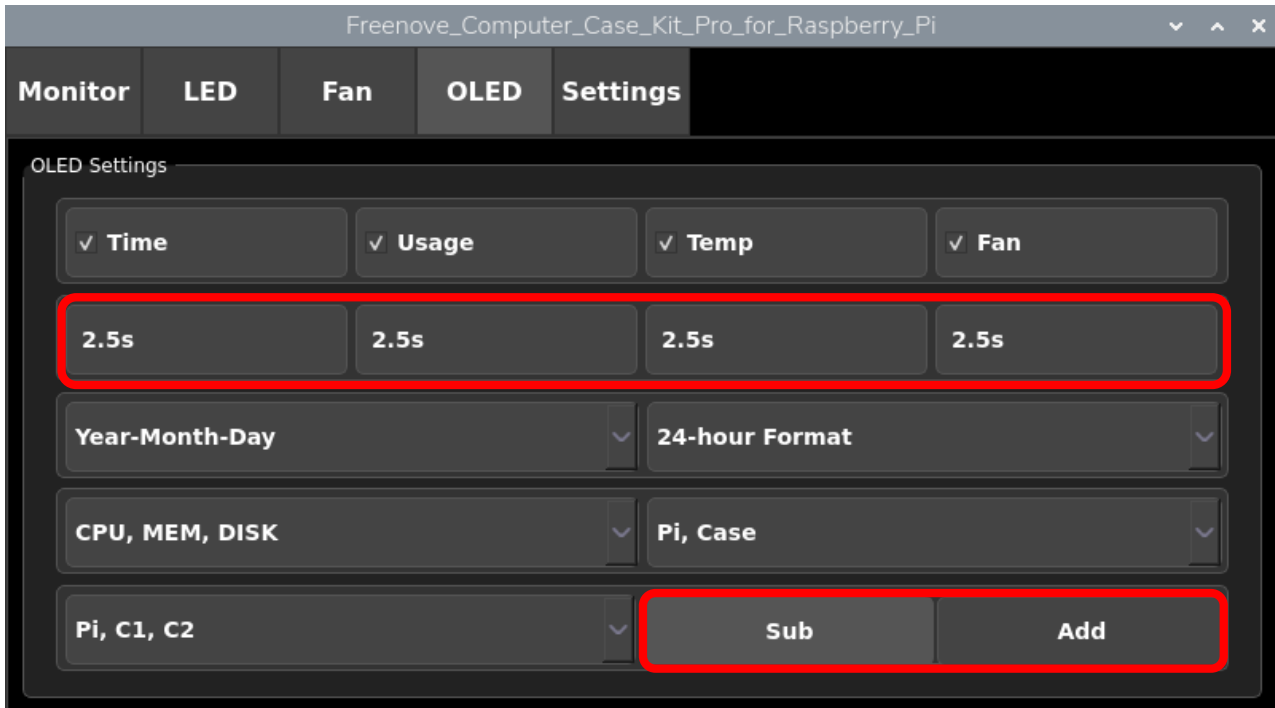
Below are six customizable options. Click the drop-down lists to select different time formats, data display orders, display durations, and more.



Adjusting Display Durations

The duration for each display screen can be adjusted by clicking the "Sub" button to decrease by 0.5s or the "Add" button to increase by 0.5s.

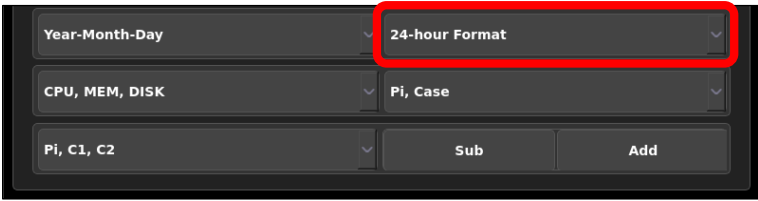
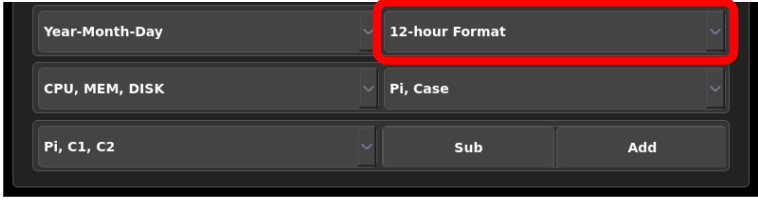
To adjust the display time for a single interface, simply uncheck the other screens so that only the desired one remains active.



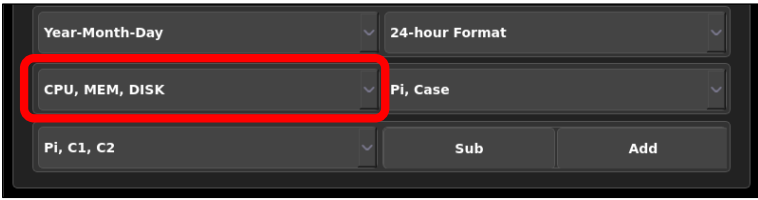
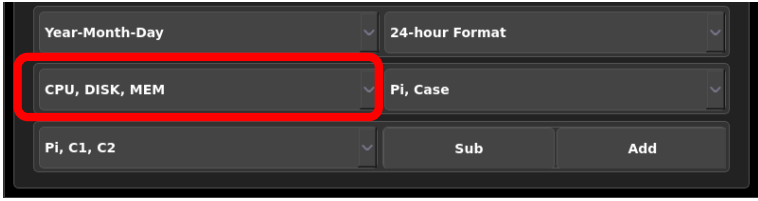
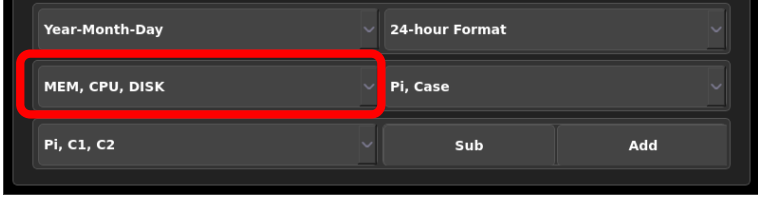
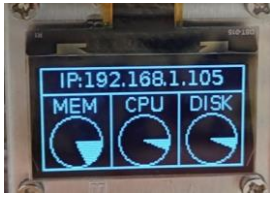
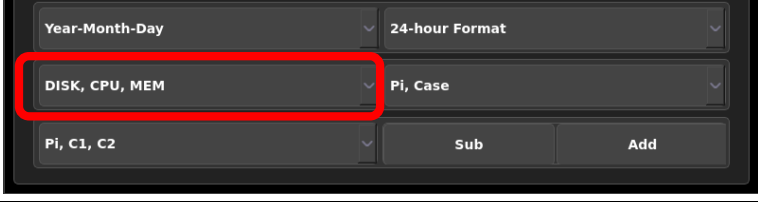
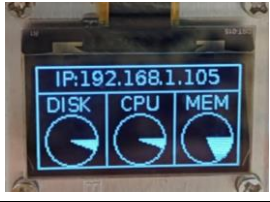
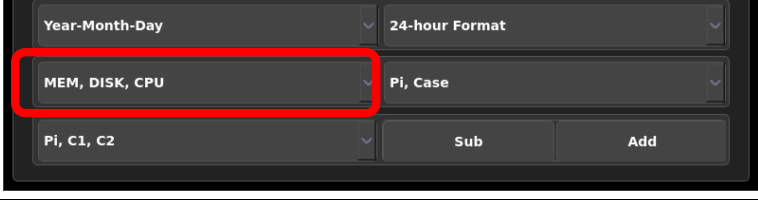
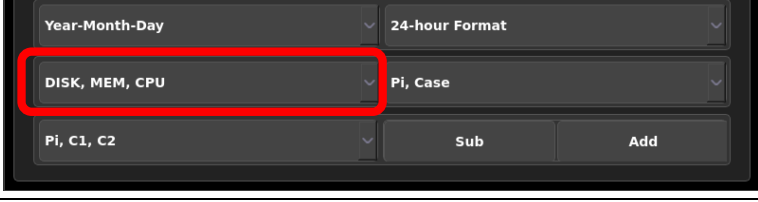
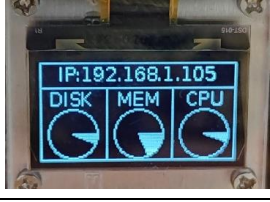
Modifying Date Format

Options	OLED Display

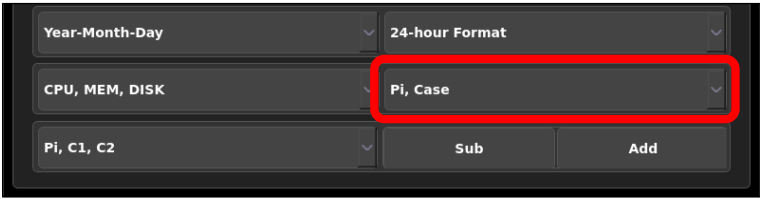
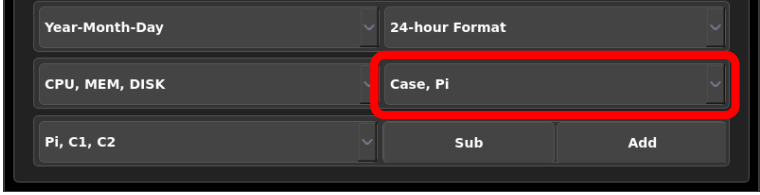
Changing Time Format

Options	OLED Display
	
	

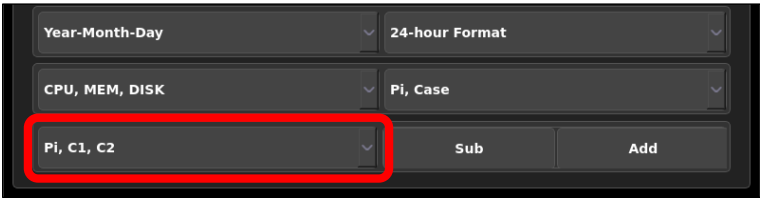
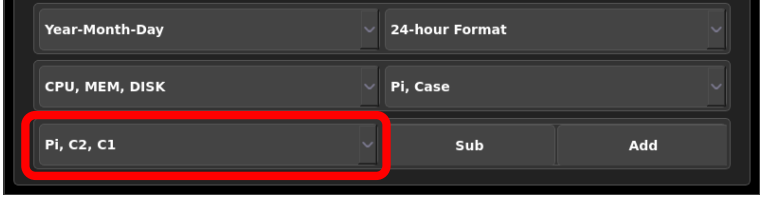
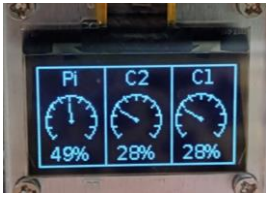
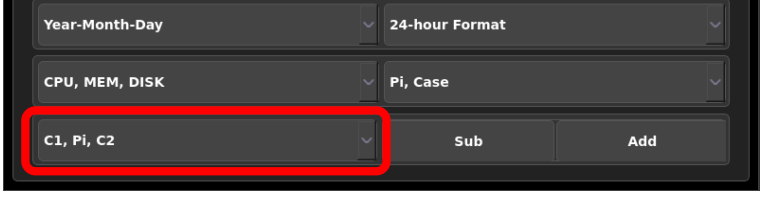
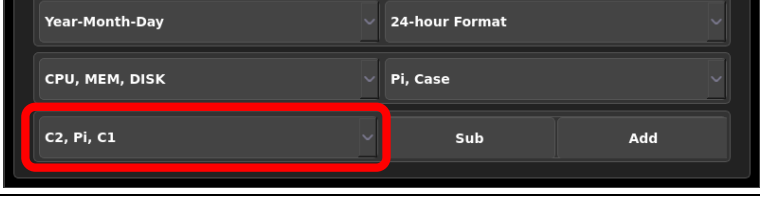
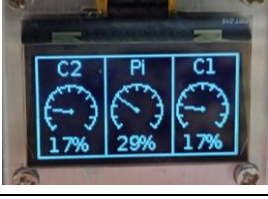
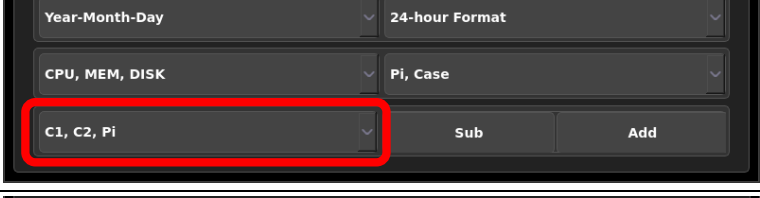
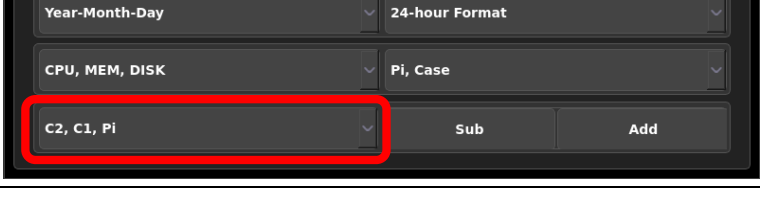
Modifying the Display Order of CPU, MEM, and DISK Usage

Options	OLED Display
	
	
	
	
	
	

Changing the Display Order of Pi and Case Temperature

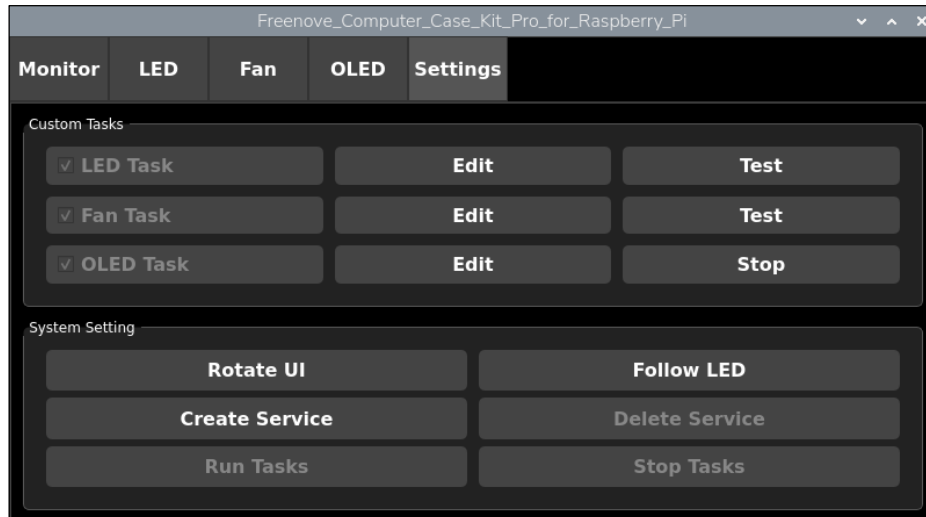
Options	OLED Display
	
	

Adjusting the Display Order of Pi, C1, and C2 Fans PWM

Options	OLED Display
	
	
	
	
	
	

3.2.5 Settings Interface

This interface integrates advanced customization features, allowing you to edit code, debug, and create startup tasks for the case LED lights, fans, and OLED display.



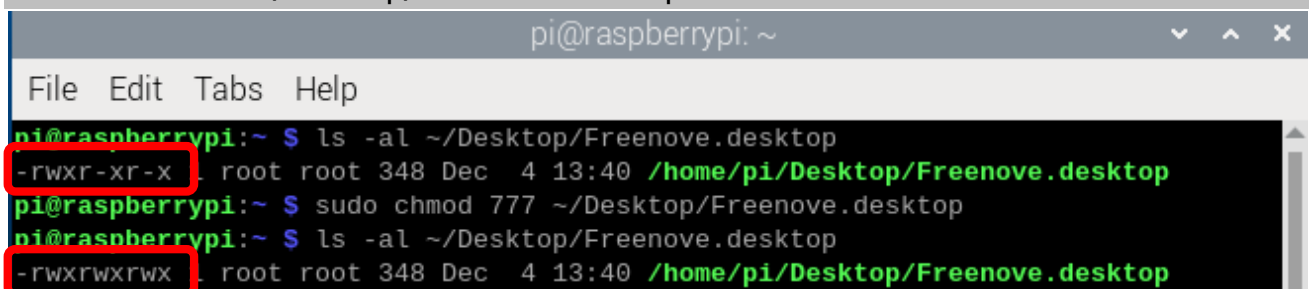
Enable Terminal Display (For Debugging)

If you do not need to view the debugging information and want to close the command terminal window, follow the steps below.

Open the terminal on your Raspberry Pi.

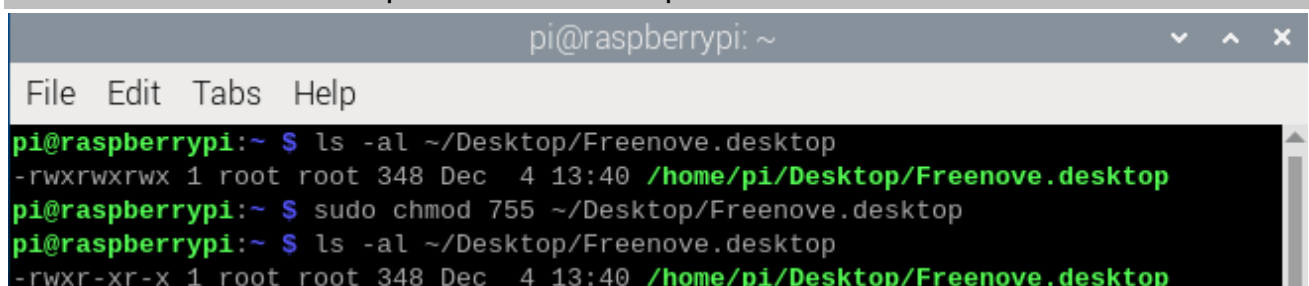
Run the following command to set the correct permissions for the Freenove desktop software:

```
ls -al ~/Desktop/Freenove.desktop
sudo chmod 777 ~/Desktop/Freenove.desktop
```



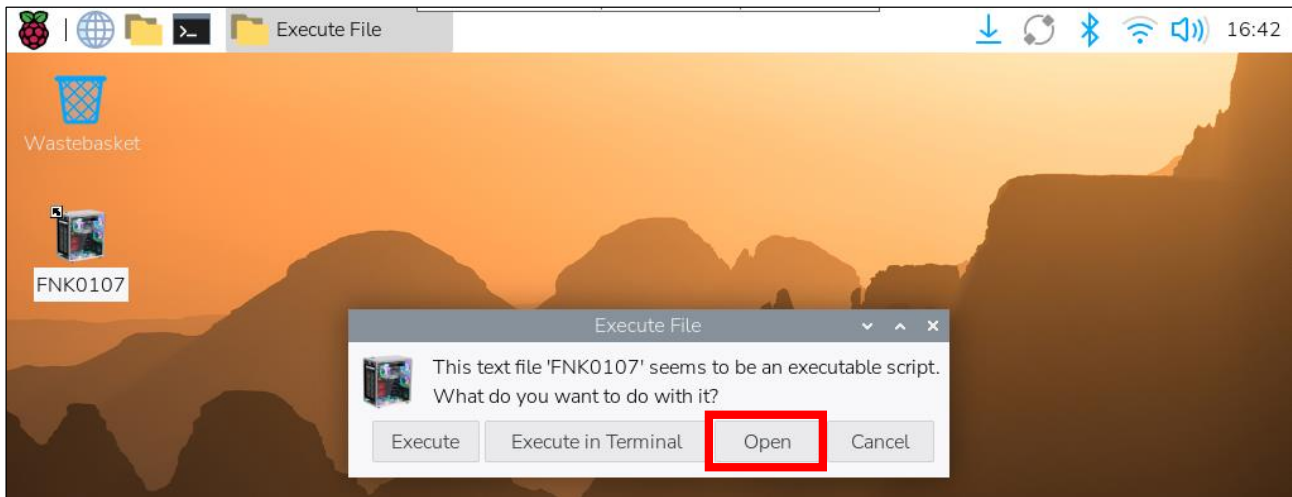
To revert to the default permissions, run the following command:

```
ls -al ~/Desktop/Freenove.desktop
sudo chmod 755 ~/Desktop/Freenove.desktop
```

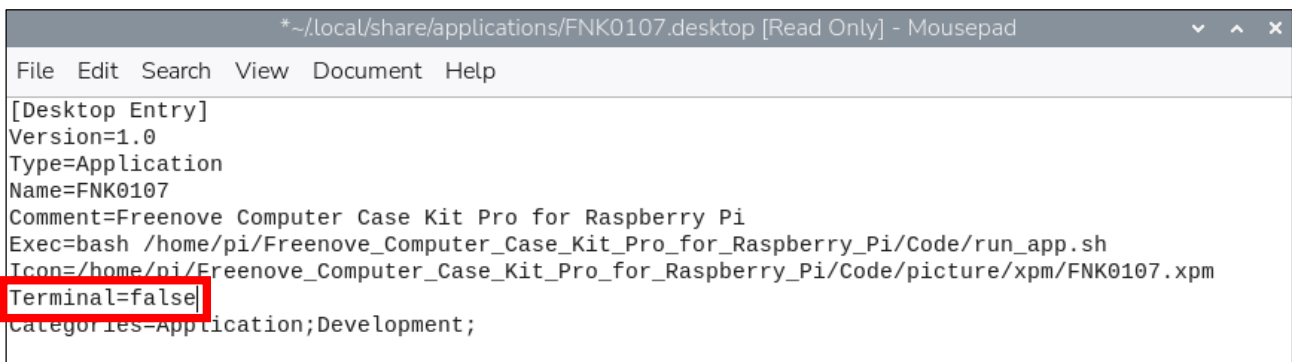


After modifying the permissions for Freenove.desktop, please follow these steps:

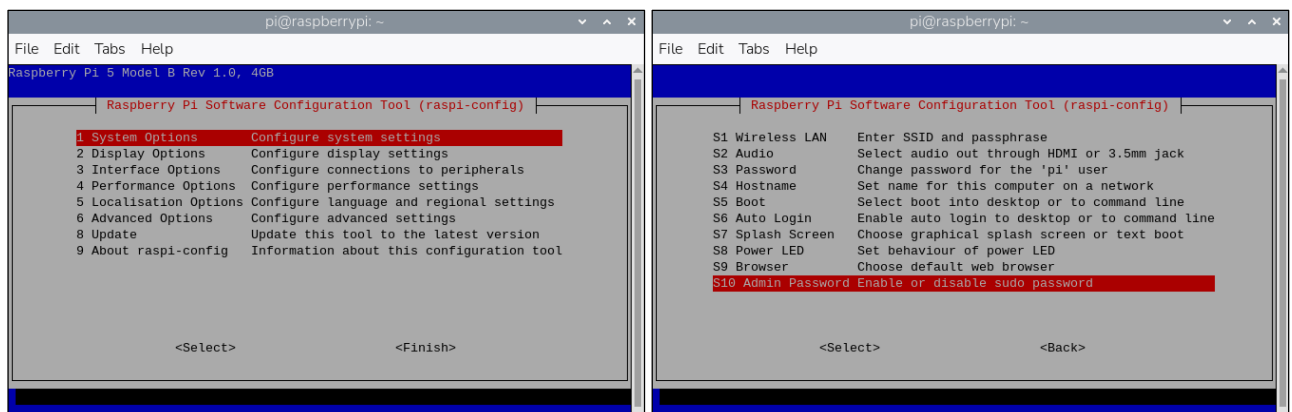
1. Close the current upper-computer software completely.
2. Double-click the Freenove software on the desktop to restart the application.
3. In the pop-up window, click **Open** to launch the software.

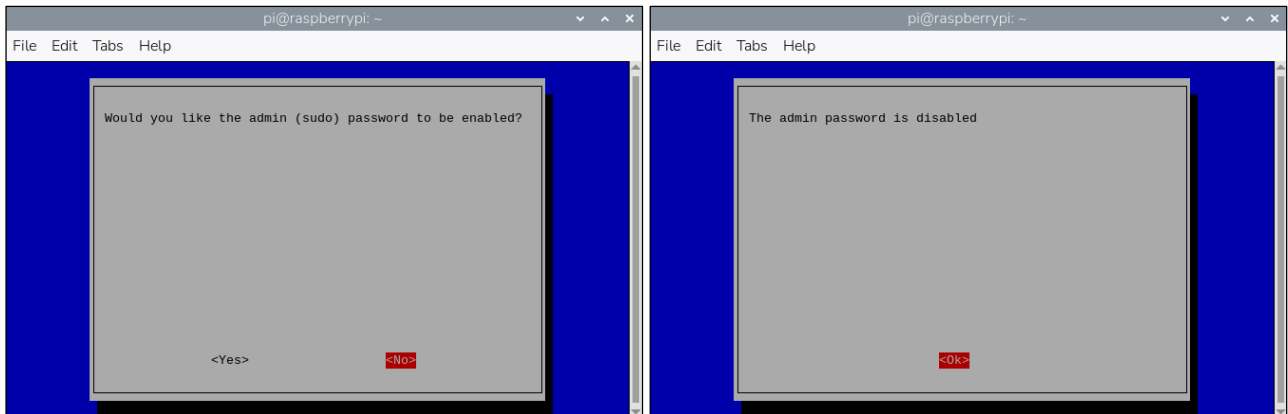


Change the parameter of **Terminal** from true to **false**, save and close the file. Restart the software – the command terminal window will no longer open.



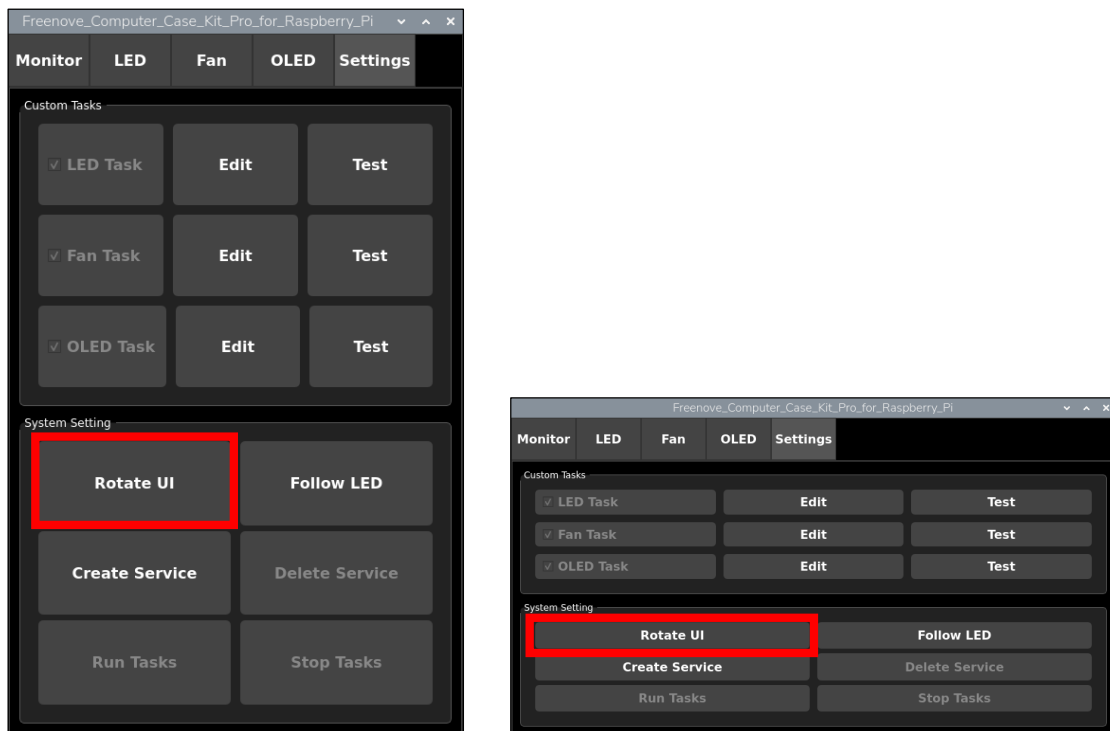
However, with the command terminal closed, you will have nowhere to enter your password when sudo is needed (because passwordless sudo is disabled by default in Trixie). To allow the software to run properly, you must enable passwordless sudo. To do this, open a terminal and run “**sudo raspi-config**”, then select: “1 System Options” → “S10 Admin Password” → “No” → “OK”.



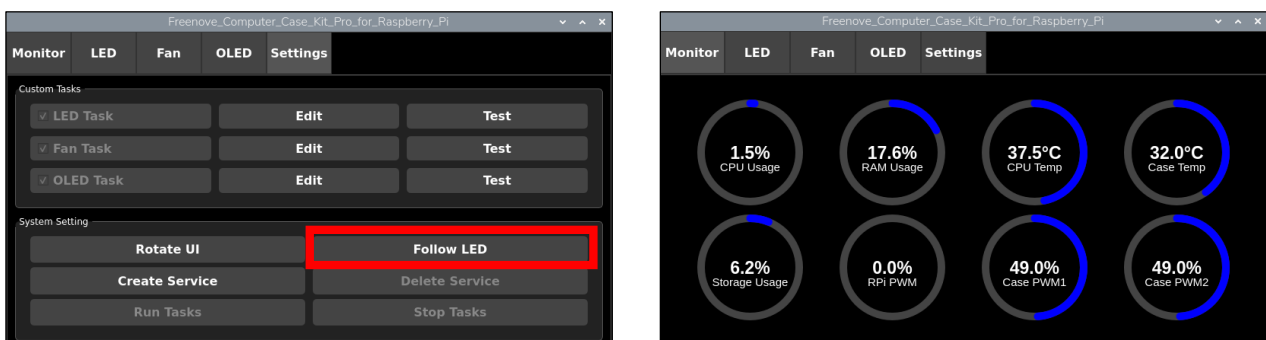


Interface Configuration

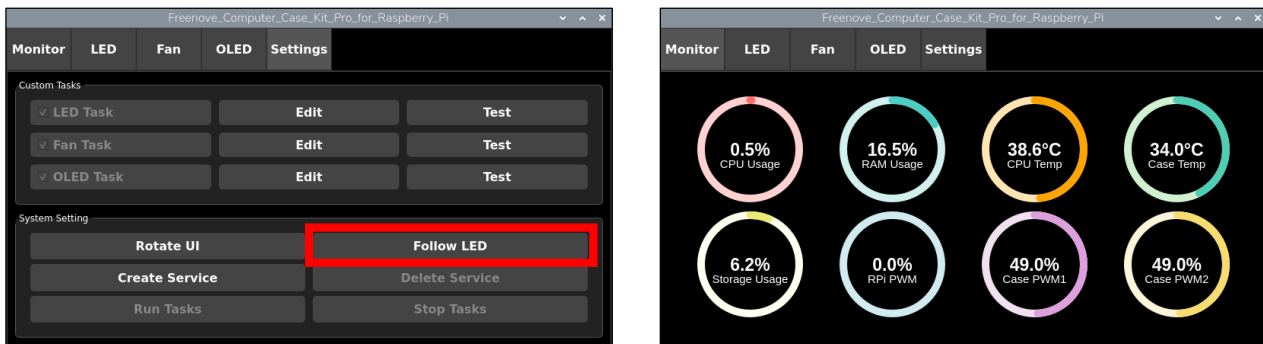
The **"Rotate UI"** button toggles the software interface instantly between landscape and portrait display modes.



Clicking the **"Follow LED"** button activates synchronization between the LED effects and the Monitor interface. Once enabled, the color of the circular progress bar will mirror any real-time changes you make to the LED colors.

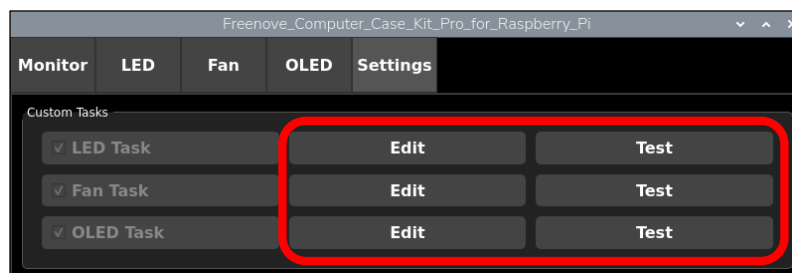


The "**Default Color**" button reverts the circular progress bar (on the Monitor interface) to its default color.

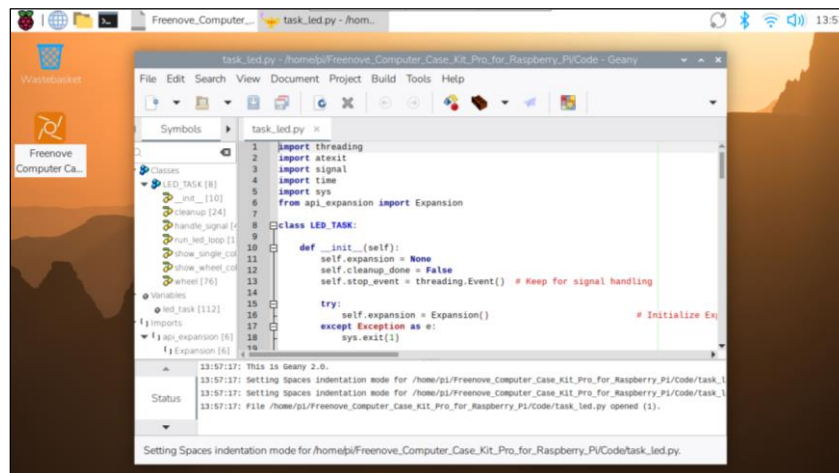


Code Editing and Testing

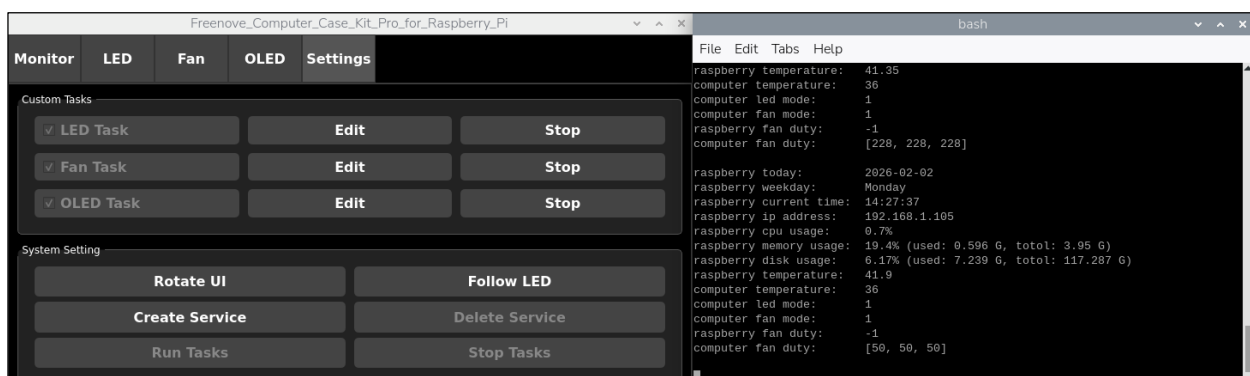
The Edit and Test buttons on the Settings interface allows for quick editing and testing of your code.



You can open and edit the task_led.py, task_fan.py, task_oled.py files by click the corresponding Edit button.

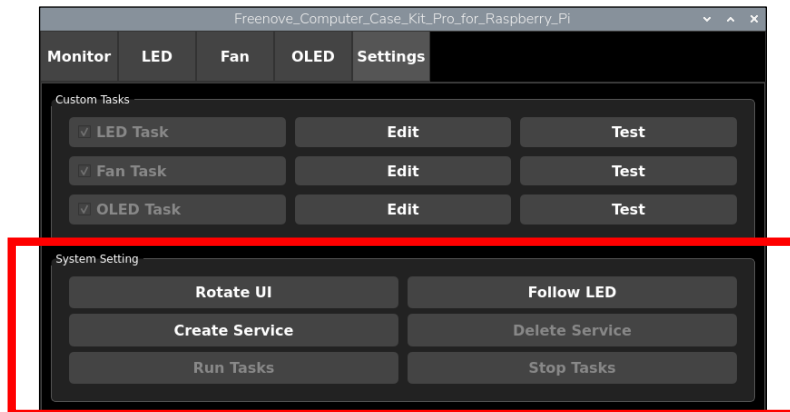


Similarly, you can click Test to quickly verify if the edited code performs as expected. Debug information will typically be printed in the terminal window.



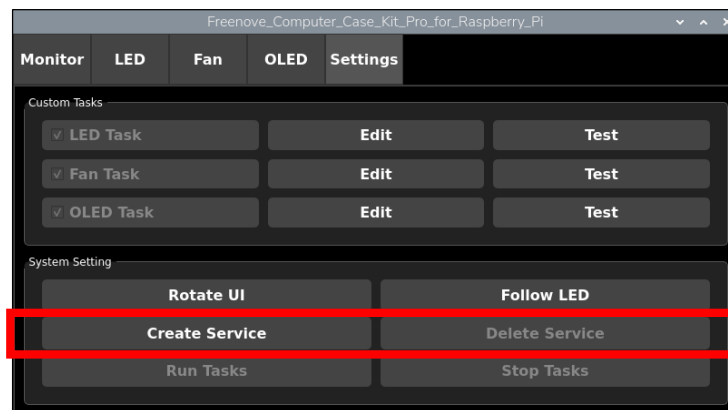
Background Task Management

This section is for comprehensive configuration. Here you can set up the software's interface features and configure background tasks for automatic operation of LEDs, fans, and the OLED display.



Click the **"Create Service"** button creates a background system service on your Raspberry Pi. This service will automatically run every time the case boots up, ensuring that your customized task_led.py, task_fan.py, and task_oled.py scripts are executed after startup.

This service is activated with every boot of the Raspberry Pi 5.

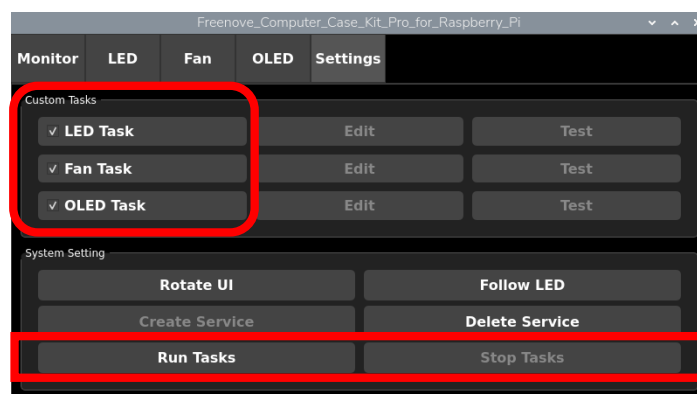


To prevent the background service from running on startup, click the **Delete Service** button to remove it.

Delete Service: Click to permanently remove the service. This prevents it from running automatically upon case startup.

Run Tasks: Click to temporarily start the service for immediate testing.

Stop Tasks: Click to temporarily halt the currently running service.

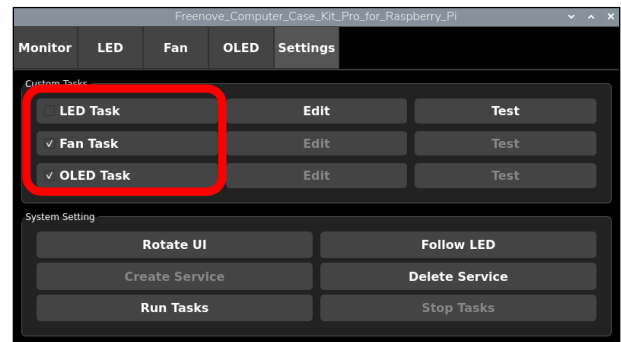
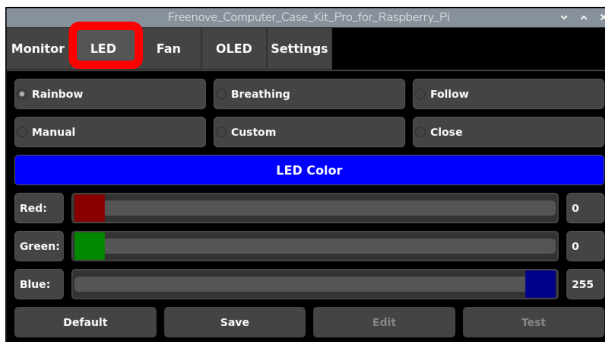


You can enable the execution of one or more custom tasks by checking their corresponding boxes

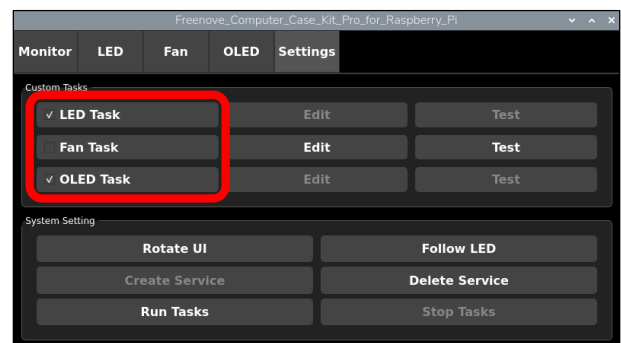
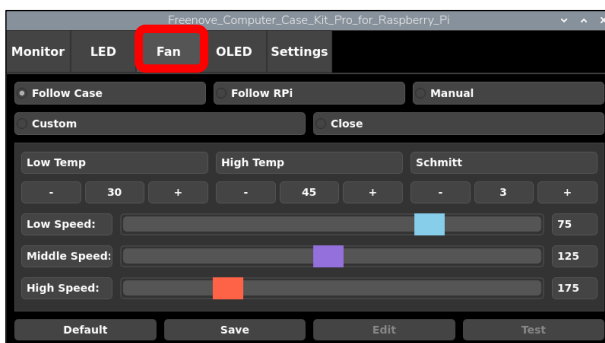
Please Note:

When switching from the Settings interface to the LED or Fan control interface, the corresponding **LED Task** or **Fan Task** will be automatically unchecked, pausing their background operation. You can re-enable them by checking the boxes again.

To stop the custom LED task and use a built-in mode instead, simply switch from the Settings interface to the LED interface and select one of the preset modes (Rainbow, Breathing, Follow, or Manual). This action will automatically uncheck the **LED Task**.



To stop the custom Fan Task and use a preset mode instead, simply switch from the Settings interface to the Fan interface and select one of the standard modes (Follow Case, Follow RPi, or Manual). This action will automatically uncheck the **Fan Task**.



Chapter 4 Speed Test

This is an additional chapter for those who wish to test the read and write speeds of their SSD and SD Card.

4.1 SD Card Speed Test

Raspberry Pi's official microSD cards support DDR50 and SDR104 bus speeds, with a maximum transfer rate of up to 104 MB/s.

<https://www.raspberrypi.com/documentation/accessories/sd-cards.html#content>

<https://www.sdcard.org/developers/sd-standard-overview/bus-speed-default-speed-high-speed-uhs-sd-express/>

Open the terminal and enter the following command:

```
git clone https://github.com/TheRemote/PiBenchmarks
```

```
pi@raspberrypi:~ $ git clone https://github.com/TheRemote/PiBenchmarks
Cloning into 'PiBenchmarks'...
remote: Enumerating objects: 652, done.
remote: Counting objects: 100% (112/112), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 652 (delta 69), reused 65 (delta 31), pack-reused 540 (from 1)
Receiving objects: 100% (652/652), 119.41 KiB | 269.00 KiB/s, done.
Resolving deltas: 100% (386/386), done.
```

Enter the directory:

```
cd PiBenchmarks/
```

Grant executable permissions to the script:

```
chmod +x Storage.sh
```

```
pi@raspberrypi:~ $ cd PiBenchmarks/
pi@raspberrypi:~/PiBenchmarks $ ls
CODE_OF_CONDUCT.md  CONTRIBUTING.md  LICENSE  README.md  Storage.sh
pi@raspberrypi:~/PiBenchmarks $ chmod +x Storage.sh
pi@raspberrypi:~/PiBenchmarks $
```

Start the speed test. Please be aware that the first execution will involve downloading the required dependencies, so the process could take a relatively long time.

```
sudo ./Storage.sh ~/
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh ~/
Trimming and syncing drives ...
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 21.4 GiB (22927368192 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5 Model B Rev 1.0 - Architecture:
aarch64 - OS: Debian GNU/Linux 13 (trixie)
Fetching required components ...
Install lshw
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease
Reading package lists... Done
```


After the speed test is completed, follow the prompts to enter a description and a name for your SSD (you can use any arbitrary characters).

```
RandRead: 8181 - RandWrite: 2405 - Read: 13783 - Write: 2344
Enter a description of your storage and setup (Example: Kingston A400 SSD on Pi 4 using StarTech SATA to USB adapter)
Description: sd1
(Optional) Enter alias to use on benchmark results. Leave blank for completely anonymous.
Alias (leave blank for Anonymous): sd1
Result submitted successfully and will appear live on https://pibenchmarks.com within a couple of minutes.
```

Test result:

Category	Test	Result
HDParm	Disk Read	87.49 MB/sec
HDParm	Cached Disk Read	88.76 MB/sec
DD	Disk Write	21.1 MB/s
FIO	4k random read	2261 IOPS (9047 KB/s)
FIO	4k random write	598 IOPS (2395 KB/s)
IOZone	4k read	13783 KB/s
IOZone	4k write	2344 KB/s
IOZone	4k random read	8181 KB/s
IOZone	4k random write	2405 KB/s
Score: 1095		

4.2 SSD Speed Test

The Raspberry Pi 5 includes a PCIe x1 slot that is certified for PCIe Gen 2.0, providing a theoretical maximum throughput of 5GT/sec, which roughly translates to 500MB/sec for read and write operations. Although this slot is not officially certified for PCIe Gen 3.0, it is possible to force the use of Gen 3.0 for potentially higher speeds. **At the time of this writing, due to the limitation of the expansion chip, both the Dual-NVMe Adapter Board and Quad-NVMe Adapter Board only support the PCIe 2.0 protocol.**

In actual tests, it is found that most SSDs can work stably at PCIe2.0, but are slightly unstable at PCIe3.0, while others are just the opposite. Therefore, please choose PCIe2.0 or PCIe3.0 according to your actual situation.

The PCIe consortium states that the speed of PCIe Gen 3.0 x1 is up to 8GT/sec, which translates to approximately 985MB/sec.

https://en.wikipedia.org/wiki/PCI_Express#Comparison_table

<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#pcie-gen-3-0>

4.2.1 Mounting SSD Devices

Open the terminal, run the following commpad to get the SSD's information.

```
lsblk
```

```
pi@raspberrypi:~ $ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0         7:0    0     2G  0 loop
mmcblk0      179:0    0  29.3G  0 disk
├─mmcblk0p1   179:1    0   512M  0 part /boot/firmware
└─mmcblk0p2   179:2    0  28.8G  0 part /
zram0        254:0    0     2G  0 disk [SWAP]
nvme0n1      259:0    0 119.2G  0 disk
├─nvme0n1p1   259:4    0 119.2G  0 part
nvme1n1      259:1    0 119.2G  0 disk
├─nvme1n1p1   259:6    0 119.2G  0 part
nvme2n1      259:2    0 119.2G  0 disk
├─nvme2n1p1   259:7    0 119.2G  0 part
nvme3n1      259:3    0 119.2G  0 disk
└─nvme3n1p1   259:5    0 119.2G  0 part
```

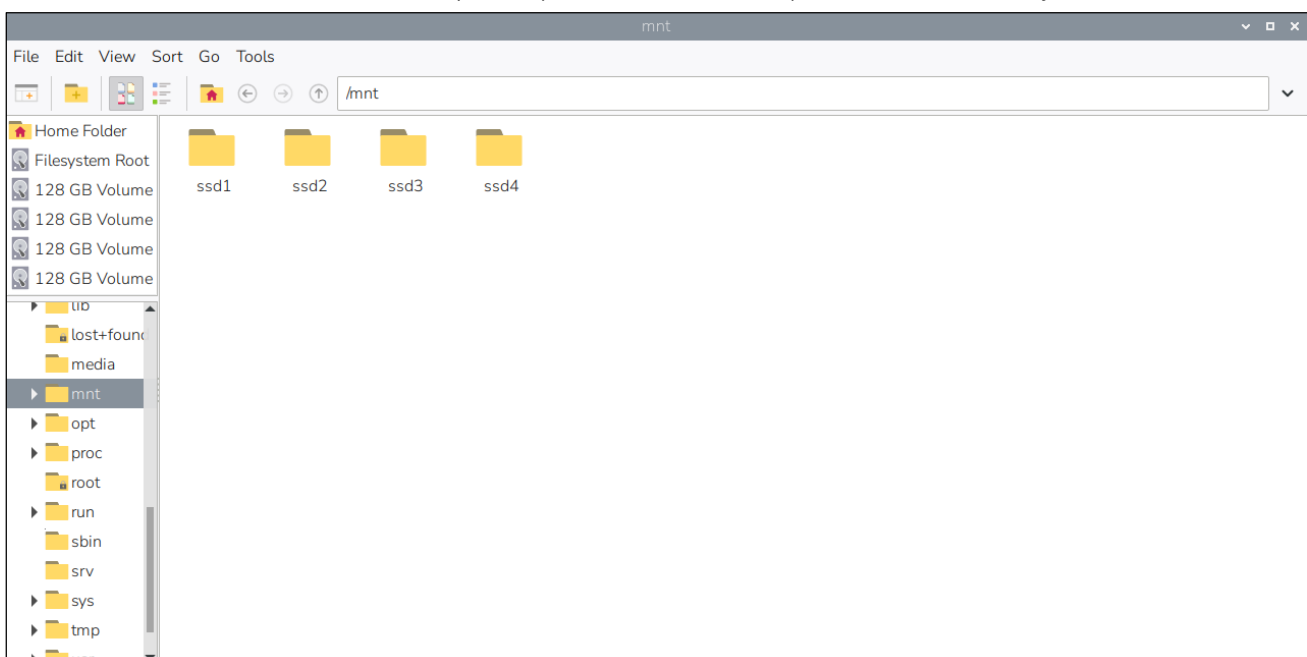
As shown in the image above, the Raspberry Pi 5 has successfully detected four NVMe SSDs, with mmcblk0p1 being the system drive.

Prior to this, we have already partitioned all NVMe SSD devices (If your SSD is not partitioned, please refer to Section 2.3 in the "[Installing Raspberry Pi OS.pdf](#)" file). Next, we need to mount all the SSDs so they can be used by the Raspberry Pi's file system. If you have already mounted them, click [here](#) to proceed to the next step.

```
sudo mkdir -p /mnt/ssd1
sudo mkdir -p /mnt/ssd2
sudo mkdir -p /mnt/ssd3
sudo mkdir -p /mnt/ssd4
```

```
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd1
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd2
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd3
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd4
```

You can view the created SSD mount points (ssd1/ssd2/ssd3/ssd4) in the /mnt directory.



Execute the following commands to complete the mounting process.

```
sudo mount /dev/nvme0n1p1 /mnt/ssd1
sudo mount /dev/nvme1n1p1 /mnt/ssd2
sudo mount /dev/nvme2n1p1 /mnt/ssd3
sudo mount /dev/nvme3n1p1 /mnt/ssd4
```

```
pi@raspberrypi:~ $ sudo mount /dev/nvme0n1p1 /mnt/ssd1
pi@raspberrypi:~ $ sudo mount /dev/nvme1n1p1 /mnt/ssd2
pi@raspberrypi:~ $ sudo mount /dev/nvme2n1p1 /mnt/ssd3
pi@raspberrypi:~ $ sudo mount /dev/nvme3n1p1 /mnt/ssd4
```

4.2.2 Speed Test

After completing the mounting of all NVMe SSD devices, execute the following command to enter the PiBenchmarks directory. We will proceed with individual device speed testing.

```
cd PiBenchmarks/
```

```
pi@raspberrypi:~ $ cd PiBenchmarks/
pi@raspberrypi:~/PiBenchmarks $
```

Run the following command to start the speed test for the first mounted device, **ssd1**.

```
sudo ./Storage.sh /mnt/ssd1
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh /mnt/ssd1
Trimming and syncing drives ...
/mnt/ssd4: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme3n1p1
/mnt/ssd3: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme2n1p1
/mnt/ssd2: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme1n1p1
/mnt/ssd1: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme0n1p1
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 21.3 GiB (22861807616 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5
Model B Rev 1.0 - Architecture: aarch64 - OS: Debian GNU/Linux 13 (trixie)
Fetching required components ...
Clock speeds: CPU: 2400 - Core: 910
Chosen partition (/mnt/ssd1) has been detected as /dev/nvme0n1p1 (nvme0n1p1)
Starting INXI hardware identification...
```

After the speed test is completed, follow the prompts to enter a description and a name for your SSD (you can use any arbitrary characters).

```
RandRead: 46786 - RandWrite: 95069 - Read: 125241 - Write: 96006
Enter a description of your storage and setup (Example: Kingston A400 SSD on Pi 4 using
StarTech SATA to USB adapter)
Description: ssd1
(Optional) Enter alias to use on benchmark results. Leave blank for completely anonymou
s.
Alias (leave blank for Anonymous): ssd1
Result submitted successfully and will appear live on https://pibenchmarks.com within a
couple of minutes.
```

Test result:

Category	Test	Result
HDParm	Disk Read	387.83 MB/sec
HDParm	Cached Disk Read	376.84 MB/sec
DD	Disk Write	339 MB/s
FIO	4k random read	95255 IOPS (381023 KB/s)
FIO	4k random write	76704 IOPS (306816 KB/s)
IOZone	4k read	170906 KB/s
IOZone	4k write	135278 KB/s
IOZone	4k random read	51424 KB/s
IOZone	4k random write	137275 KB/s
Score: 33798		

Next, we will sequentially execute the following commands to perform individual speed tests on each SSD.

```
sudo ./Storage.sh /mnt/ssd2
```

```
sudo ./Storage.sh /mnt/ssd3
```

```
sudo ./Storage.sh /mnt/ssd4
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh /mnt/ssd2
Trimming and syncing drives ...
/mnt/ssd4: 0 B (0 bytes) trimmed on /dev/nvme3n1p1
/mnt/ssd3: 0 B (0 bytes) trimmed on /dev/nvme2n1p1
/mnt/ssd2: 0 B (0 bytes) trimmed on /dev/nvme1n1p1
/mnt/ssd1: 624.2 MiB (654471168 bytes) trimmed on /dev/nvme0n1p1
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 0 B (0 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5 Model B
```

The performance varies among different SSDs, and each test may have certain error, which is normal. The following figures show the speed test results of the four SSDs.

Category	Test	Result
HDParm	Disk Read	387.87 MB/sec
HDParm	Cached Disk Read	373.89 MB/sec
DD	Disk Write	341 MB/s
FIO	4k random read	95255 IOPS (381023 KB/s)
FIO	4k random write	77283 IOPS (309132 KB/s)
IOZone	4k read	173486 KB/s
IOZone	4k write	135975 KB/s
IOZone	4k random read	52014 KB/s
IOZone	4k random write	142343 KB/s
Score: 34439		
Compare with previous benchmark results at: https://pibenchmarks.com/		
pi@raspberrypi:~/PiBenchmarks \$		

Category	Test	Result
HDParm	Disk Read	389.68 MB/sec
HDParm	Cached Disk Read	322.94 MB/sec
DD	Disk Write	304 MB/s
FIO	4k random read	55501 IOPS (222005 KB/s)
FIO	4k random write	70620 IOPS (282482 KB/s)
IOZone	4k read	132950 KB/s
IOZone	4k write	115905 KB/s
IOZone	4k random read	50319 KB/s
IOZone	4k random write	117567 KB/s
Score: 28653		
Compare with previous benchmark results at: https://pibenchmarks.com/		
pi@raspberrypi:~/PiBenchmarks \$		

Category	Test	Result
HDParm	Disk Read	384.62 MB/sec
HDParm	Cached Disk Read	371.46 MB/sec
DD	Disk Write	337 MB/s
FIO	4k random read	91838 IOPS (367354 KB/s)
FIO	4k random write	71859 IOPS (287438 KB/s)
IOZone	4k read	159767 KB/s
IOZone	4k write	127351 KB/s
IOZone	4k random read	49789 KB/s
IOZone	4k random write	132234 KB/s
Score: 32394		
Compare with previous benchmark results at: https://pibenchmarks.com/		
pi@raspberrypi:~/PiBenchmarks \$		

Category	Test	Result
HDParm	Disk Read	387.03 MB/sec
HDParm	Cached Disk Read	359.10 MB/sec
DD	Disk Write	310 MB/s
FIO	4k random read	94814 IOPS (379259 KB/s)
FIO	4k random write	63405 IOPS (253622 KB/s)
IOZone	4k read	145952 KB/s
IOZone	4k write	115302 KB/s
IOZone	4k random read	44552 KB/s
IOZone	4k random write	116463 KB/s
Score: 29348		
Compare with previous benchmark results at: https://pibenchmarks.com/		
pi@raspberrypi:~/PiBenchmarks \$		

4.2.3 PCIe Gen3.0 Speed Test

In the Preface, it is mentioned that the Raspberry Pi's PCIe Gen 3.0 has not been officially certified. While it is functional, its performance is not as reliable as desired. This chapter is presented as an exploratory section for assessing the speed capabilities of SSDs when used with PCIe Gen 3.0. **For practical applications, it is advised to opt for PCIe Gen 2.0 to ensure greater stability and dependability.**

If your kit includes an NVMe Adapter Board (Model: **FNK0107A/H/P/U**), you can enable PCIe Gen 3.0 mode. If your adapter is a different model or no adjustment is needed, please [skip this step](#).

Enable PCIe Gen 3.0

Add the line `dtparam=pciex1_gen=3` to `/boot/firmware/config.txt` to enable PCIe Gen3.0.

As shown below, enter the command to open the file.

```
sudo nano /boot/firmware/config.txt
```

Add the line `dtparam=pciex1_gen=3` to the end of the file, as shown below:

```
GNU nano 7.2 /boot/firmware/config.txt *
otg_mode=1
[all]
dtparam=pciex1_gen=3
```

Press Ctrl-O to save the file, Enter to confirm, and Ctrl-X to exit.

Reboot your Raspberry Pi.

```
sudo reboot
```

After rebooting, test the speed again.

Category	Test	Result
HDParm	Disk Read	755.51 MB/sec
HDParm	Cached Disk Read	687.95 MB/sec
DD	Disk Write	487 MB/s
FIO	4k random read	93944 IOPS (375779 KB/s)
FIO	4k random write	89824 IOPS (359298 KB/s)
IOZone	4k read	281334 KB/s
IOZone	4k write	199418 KB/s
IOZone	4k random read	60363 KB/s
IOZone	4k random write	223924 KB/s
Score: 47341		

The speed of another SSD.

Category	Test	Result
HDParm	Disk Read	804.45 MB/sec
HDParm	Cached Disk Read	670.21 MB/sec
DD	Disk Write	404 MB/s
FIO	4k random read	206868 IOPS (827474 KB/s)
FIO	4k random write	92252 IOPS (369009 KB/s)
IOZone	4k read	224411 KB/s
IOZone	4k write	167052 KB/s
IOZone	4k random read	61980 KB/s
IOZone	4k random write	190269 KB/s
Score: 47057		

Disable PCIe Gen 3.0

Delete the line added with the previous step to disable PCIe Gen3.0.

Delete the line `dtparam=pciex1_gen=3` in the boot/firmware/config.txt file.

After the line is removed, it will change to PCIe Gen2.0.

What's next?

Thank you again for choosing Freenove products.

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions, or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us:

support@freenove.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

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